$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **8.1.0**

Lecture : **Eigen and Singular Values**

Topic : **Overview of this chapter**

Concept :

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: 3rd July 2020

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We will use Lay's (4th edition) chapter 5.1-5.4 for the slides
Ch 5.5 and 5.6 for future offerings

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **8.1.1**

Lecture : **Eigen and Singular Values**

Topic : **Introducing Eigenvectors and Eigenvalues**

Concept :

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: 3rd July 2020

Eigenvectors and Eigenvalues

DEFINITION

An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$; such an $\mathbf{x}$ is called an *eigenvector corresponding to* λ.[1]

[1] Note that an eigenvector must be *nonzero*, by definition, but an eigenvalue may be zero

The requirement that an eigenvector be nonzero is imposed to avoid the unimportant case $A\mathbf{0} = \lambda\mathbf{0}$, which holds for every A and λ.

Important note:
Eigenvalues and **eigenvectors** are only for **square** matrices.

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Lay, Linear Algebra and its Applications (4th Edition)

Eigenvectors and Eigenvalues

In general, the image of a vector $\mathbf{x}$ under multiplication by a square matrix A differs from $\mathbf{x}$ in both magnitude and direction. However, in the special case where $\mathbf{x}$ is an eigenvector of A, multiplication by A leaves the direction unchanged. For example,

EXAMPLE 1 Let $A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. The images of $\mathbf{u}$ and $\mathbf{v}$ under multiplication by A are shown in Fig. 1. In fact, $A\mathbf{v}$ is just $2\mathbf{v}$. So A only "stretches," or dilates, $\mathbf{v}$. ■

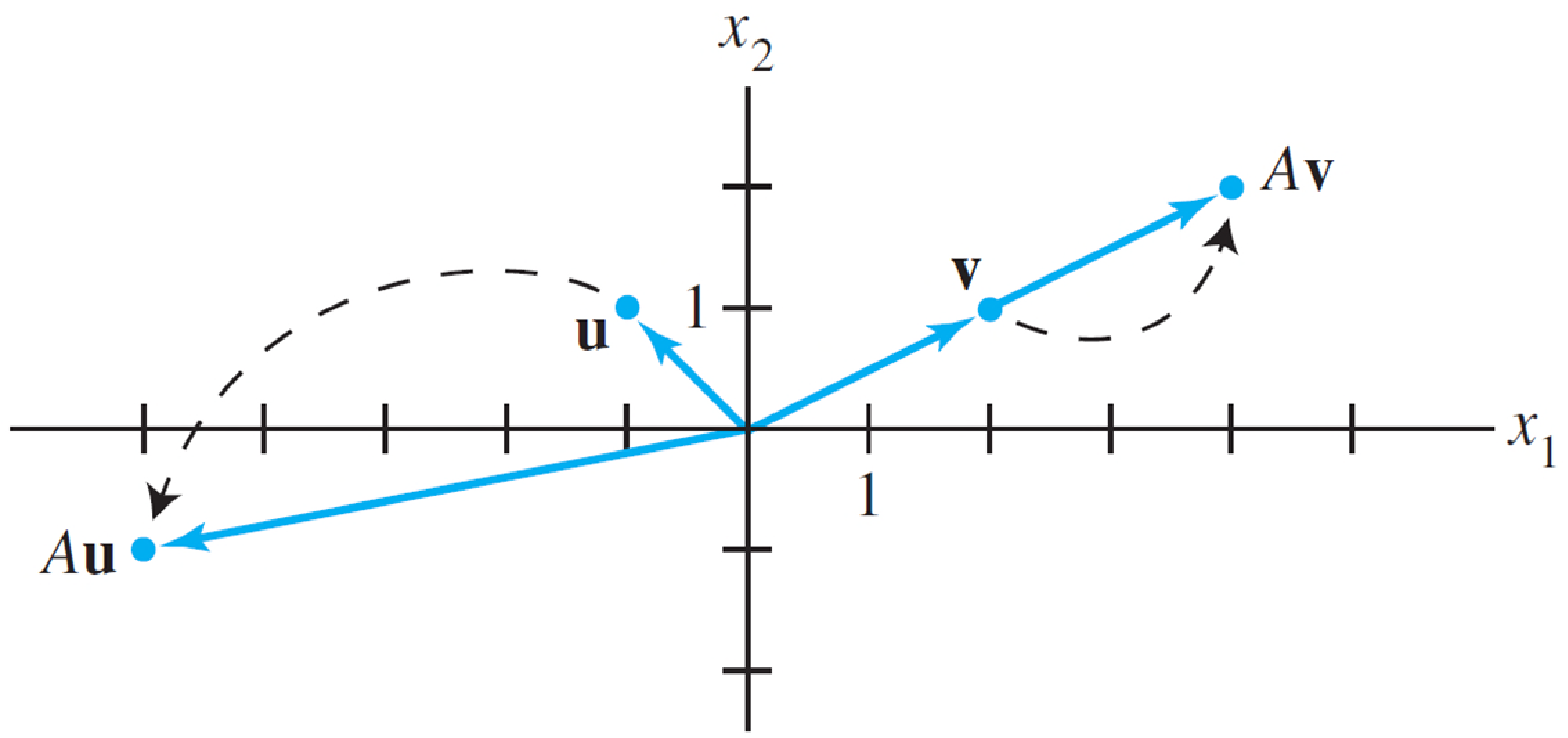


FIGURE 1 Effects of multiplication by A.

EigenVectors and Values

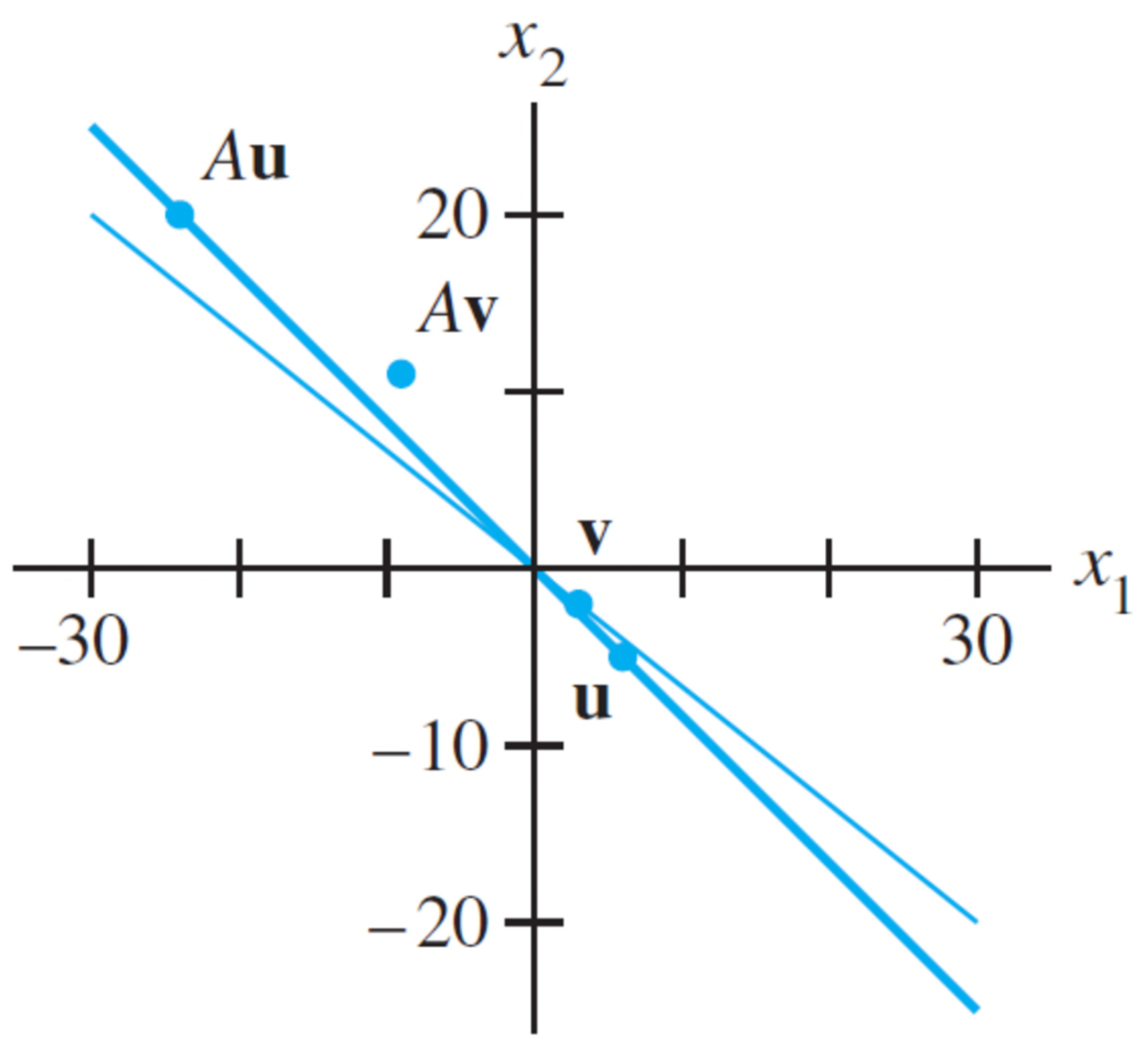
EXAMPLE 2 Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$, and $\mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$. Are $\mathbf{u}$ and $\mathbf{v}$ eigenvectors of A?

SOLUTION

$$A\mathbf{u} = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}\begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4\begin{bmatrix} 6 \\ -5 \end{bmatrix} = -4\mathbf{u}$$

$$A\mathbf{v} = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}\begin{bmatrix} 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -9 \\ 11 \end{bmatrix} \neq \lambda\begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

Thus $\mathbf{u}$ is an eigenvector corresponding to an eigenvalue (-4), but $\mathbf{v}$ is not an eigenvector of A, because $A\mathbf{v}$ is not a multiple of $\mathbf{v}$. ■



$A\mathbf{u} = -4\mathbf{u}$, but $A\mathbf{v} \neq \lambda\mathbf{v}$.

Example: How to find EigenVectors

EXAMPLE 3 Show that 7 is an eigenvalue of matrix A in Example 2, and find the corresponding eigenvectors.

SOLUTION The scalar 7 is an eigenvalue of A if and only if the equation

$$A\mathbf{x} = 7\mathbf{x} \tag{1}$$

has a nontrivial solution. But (1) is equivalent to $A\mathbf{x} - 7\mathbf{x} = \mathbf{0}$, or

$$(A - 7I)\mathbf{x} = \mathbf{0} \tag{2}$$

To solve this homogeneous equation, form the matrix

$$A - 7I = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix}$$

The columns of $A - 7I$ are obviously linearly dependent, so (2) has nontrivial solutions. Thus 7 *is* an eigenvalue of A. To find the corresponding eigenvectors, use row operations:

$$\begin{bmatrix} -6 & 6 & 0 \\ 5 & -5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The general solution has the form $x_2\begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Each vector of this form with $x_2 \neq 0$ is an eigenvector corresponding to $\lambda = 7$. ■

Example: How to find EigenVectors

EigenSpace

The equivalence of equations (1) and (2) obviously holds for any λ in place of $\lambda = 7$. Thus λ is an eigenvalue of an $n \times n$ matrix A if and only if the equation

$$(A - \lambda I)\mathbf{x} = \mathbf{0} \tag{3}$$

has a nontrivial solution. The set of *all* solutions of (3) is just the null space of the matrix $A - \lambda I$. So this set is a *subspace* of $\mathbb{R}^n$ and is called the **eigenspace** of A corresponding to λ. The eigenspace consists of the zero vector and all the eigenvectors corresponding to λ.

Note:
The $\lambda, and\ x$ that are solutions of equation (3) are the eigenvalue and vector respectively.

The eigenvector x that is a solution of the above is tied to the λ eigenvalue.

The eigenvalue could be repeated and
there could be more than 1 independent eigenvector.
(See Example 4)

Example: EigenSpace

EigenSpace

Example 3 shows that for matrix A in Example 2, the eigenspace corresponding to $\lambda = 7$ consists of *all* multiples of (1, 1), which is the line through (1, 1) and the origin. From Example 2, you can check that the eigenspace corresponding to $\lambda = -4$ is the line through $(6, -5)$. These eigenspaces are shown in Fig. 2, along with eigenvectors (1, 1) and $(3/2, -5/4)$ and the geometric action of the transformation $\mathbf{x} \mapsto A\mathbf{x}$ on each eigenspace.

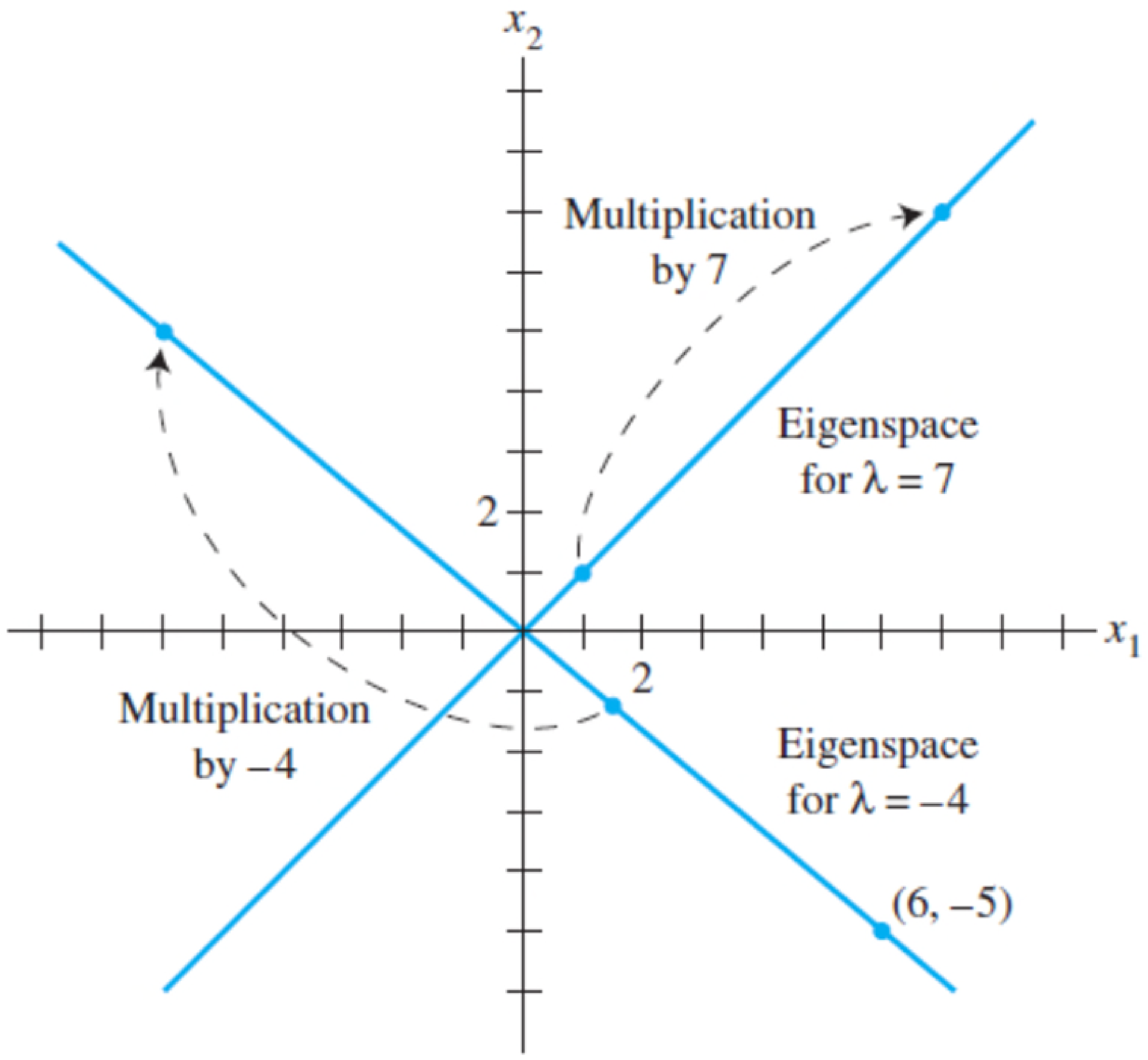


FIGURE 2 Eigenspaces for $\lambda = -4$ and $\lambda = 7$.

EigenSpace: Example two-dimensional subspace of R^3

Example:

In this example, there is repeated eigenvalue.

When this occur, it is possible to have more than 1 eigenvector for that eigenvalue.
I.e, the eigenspace's dimension is greater than 1.

EXAMPLE 4 Let $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$. An eigenvalue of A is 2. Find a basis for the corresponding eigenspace.

SOLUTION Form

$$A - 2I = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

and row reduce the augmented matrix for $(A - 2I)\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Lay,4thEd, pg 269

EigenSpace: Example two-dimensional subspace of R^3

At this point, it is clear that 2 is indeed an eigenvalue of A because the equation $(A - 2I)\mathbf{x} = \mathbf{0}$ has free variables. The general solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}, \quad x_2 \text{ and } x_3 \text{ free}$$

The eigenspace, shown in Fig. 3, is a two-dimensional subspace of $\mathbb{R}^3$. A basis is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

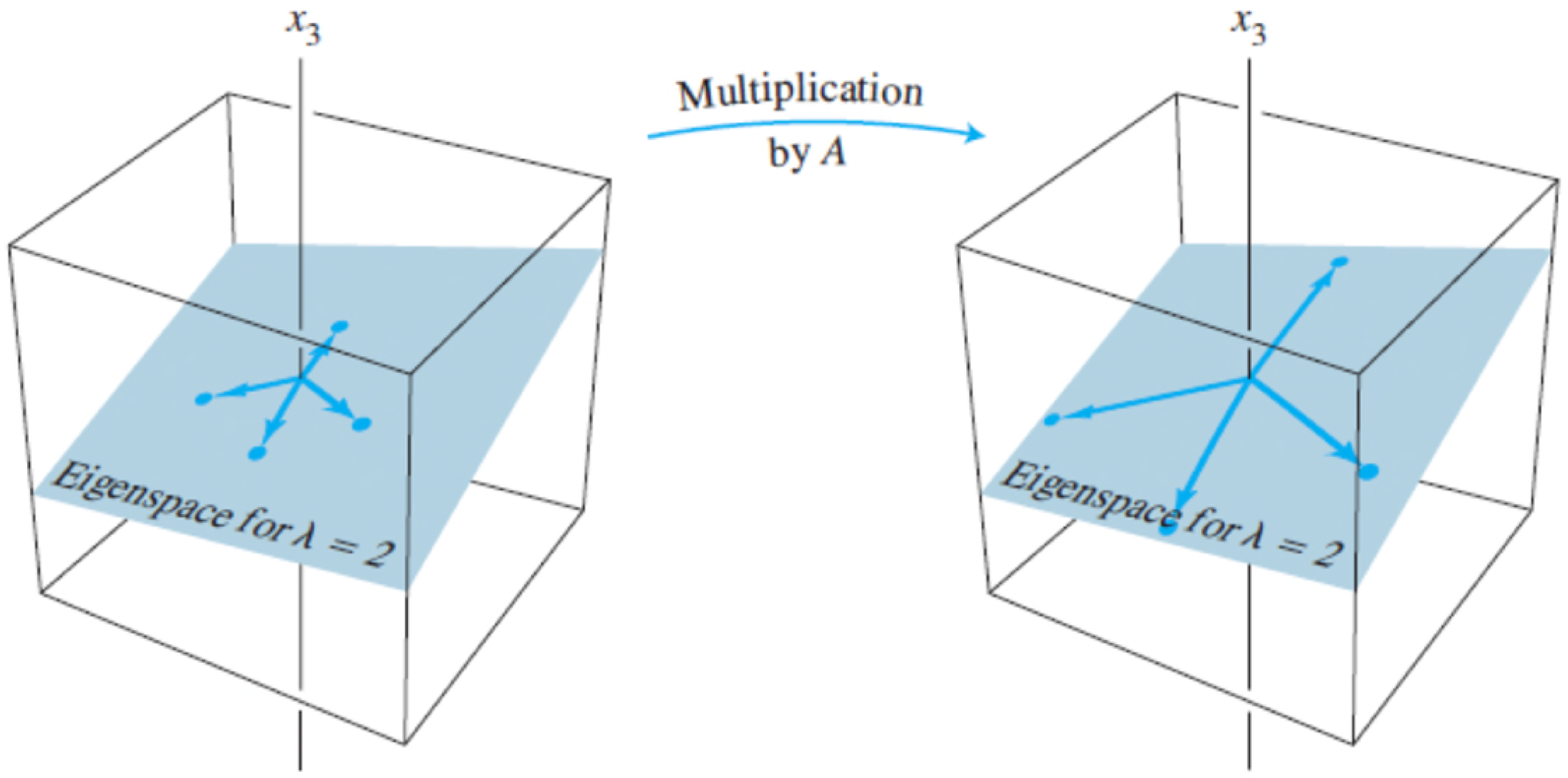


FIGURE 3 A acts as a dilation on the eigenspace.

Sanity check: Matlab

```
% Lay 4th edition, pg 268, Example 4,
% a matrix with repeated roots = -2

A = [4 -1 6; 2 1 6; 2 -1 8]
[P,D] = eig(A)
A_tst = P*D*inv(P)
```

```
A =

     4    -1     6
     2     1     6
     2    -1     8
```

```
A_tst =

    4.0000   -1.0000    6.0000
    2.0000    1.0000    6.0000
    2.0000   -1.0000    8.0000
```

```
P =

   -0.5774   -0.6122    0.3205
   -0.5774   -0.7873   -0.9112
   -0.5774    0.0728   -0.2587


D =

    9.0000         0         0
         0    2.0000         0
         0         0    2.0000
```

Matlab’s EIG function

eig

Eigenvalues and eigenvectors

Syntax

```
e = eig(A)
[V,D] = eig(A)
[V,D,W] = eig(A)
```

Description

`e = eig(A)` returns a column vector containing the eigenvalues of square matrix `A`. example

`[V,D] = eig(A)` returns diagonal matrix `D` of eigenvalues and matrix `V` whose columns are the corresponding right eigenvectors, so that `A*V = V*D`. example

Warning:

1) D is the eigenvalue BUT they are typically not sorted.
2) V are columns of the right eigenvectors AND they (the columsn) are normalized to norm 1. Hence they may be different to Lay’s example BUT their direction will be correct.

Matlab – sanity check

Note: Here, $\lambda_2 = \lambda_3 = 2$. Hence, there are two eigenvectors corresponding to the eigenvalue of 2. The eigen space can be formed by any set of basis vectors that span it. Hence, the eigen space spanned by the basis on LHS is the same eigen space spanned by the 2nd and 3rd columns of matrix U.

MATLAB

```
A =

     4    -1     6
     2     1     6
     2    -1     8

>> [U,D] = eig(A)

U =

   -0.5774   -0.6122    0.3205
   -0.5774   -0.7873   -0.9112
   -0.5774    0.0728   -0.2587
```

Eigenvector corresponding to λ_1 | Eigenvector corresponding to λ_2 | Eigenvector corresponding to λ_3

```
D =

    9.0000         0         0
         0    2.0000         0
         0         0    2.0000
```

λ_1 = 9.0000, λ_2 = 2.0000, λ_3 = 2.0000

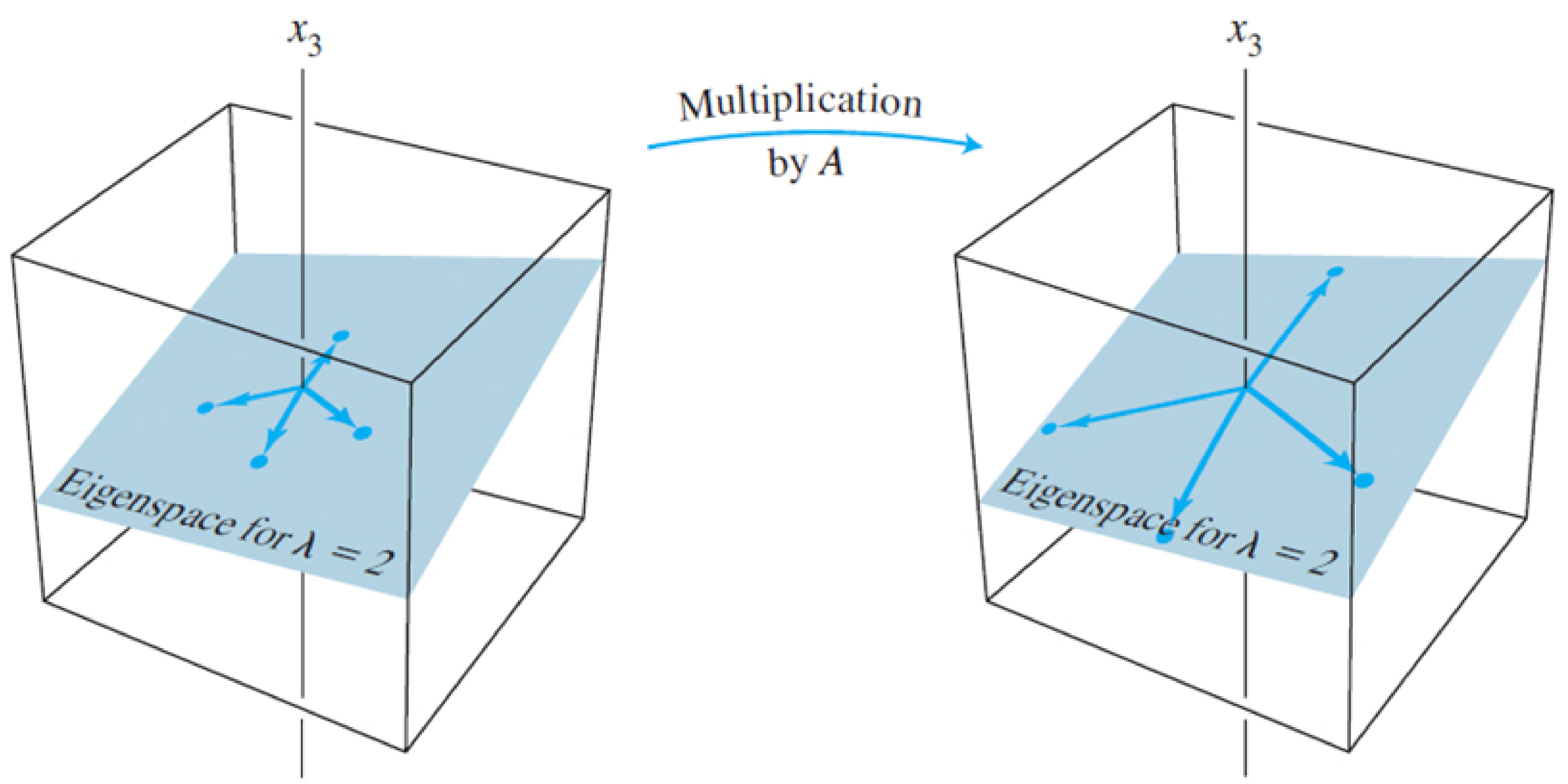


FIGURE 3 A acts as a dilation on the eigenspace.

Matlab provide solutions of eigenvectors such that the norm of the eigenvectors == 1.
When the matrix A is not symmetrical, the eigenvectors will not be orthogonal to each other.
In the special case when matrix is symmetric, eigen vectors will be orthogonal to each other.

Summary

To find the eigenvalue and vectors of a matrix, 2 steps:

a) First find the eigen value of the matrix.

- slide 8.1.2 shows how to find the eigen values.

b) Second, use Gaussian Elimination (row reduction) to find the solution of the homogenous equation

$$(A - \lambda I)x = 0$$

for each eigenvalue λ.

Example:

1) https://www.scss.tcd.ie/Rozenn.Dahyot/CS1BA1/SolutionEigen.pdf
2) https://lpsa.swarthmore.edu/MtrxVibe/EigMat/MatrixEigen.html

References Videos

Ritvik Math
- overview https://youtu.be/KTKAp9Q3yWg
- how to compute 2x2 eigvalue and vector https://youtu.be/glaiP222JWA

Eigen vector 33 mins: Brunton: https://youtu.be/ZSGrJBS_qtc

MIT Strang,
https://youtu.be/cdZnhQjJu4I lecture 21: (51 mins)
https://youtu.be/U8R54zOTVLw (11 mins)

Trevor Bazett:
https://youtu.be/4wTHFmZPhT0 (9. Min - geometric)
Examples -
https://youtu.be/LsZ-nNy0ZRs (4.5min example)
https://youtu.be/4u55V_Yp-QI (7 min)
https://youtu.be/EZkDtcyPP6Q (9 mins - repeated eigenvalues, complex eigenvalues,)

Zach Star: applications of eigen vectors values
https://youtu.be/i8FukKfMKCI

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **8.1.2**

Lecture : **Eigen and Singular Values**

Topic : **Characteristic Equation**

Concept : **How to find eigenvalue and vectors**

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: 3rd July 2020

How to find eigenValues ?

Our next objective is to obtain a general procedure for finding eigenvalues and eigenvectors of an $n \times n$ matrix A. We will begin with the problem of finding the eigenvalues of A. Note first that the equation $A\mathbf{x} = \lambda\mathbf{x}$ can be rewritten as $A\mathbf{x} = \lambda I\mathbf{x}$, or equivalently, as

$$(\lambda I - A)\mathbf{x} = \mathbf{0}$$

For λ to be an eigenvalue of A this equation must have a nonzero solution for $\mathbf{x}$.

$$\begin{aligned} Ax &= \lambda x \\ Ax - \lambda x &= 0 \\ (A - \lambda I)x &= 0 \\ (\lambda I - A)x &= 0 \end{aligned}$$

For the above equation to be true, and x not a zero vector (condition for eigen vector),

then $(\lambda I - A)$ must be a singular matrix (not invertible).

See: invertible matrix theorem (corollary of (d))

How to find eigenValues ?

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A, the statements are either all true or all false.

a. A is an invertible matrix.

b. A is row equivalent to the $n \times n$ identity matrix.

c. A has n pivot positions.

d. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

e. The columns of A form a linearly independent set.

f. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.

g. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.

h. The columns of A span $\mathbb{R}^n$.

i. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.

j. There is an $n \times n$ matrix C such that $CA = I$.

k. There is an $n \times n$ matrix D such that $AD = I$.

l. A^T is an invertible matrix.

Lay,4th, pg112 (Ch2.3)

The Invertible Matrix Theorem divides the set of all $n \times n$ matrices into two disjoint classes: the invertible (nonsingular) matrices, and the noninvertible (singular) matrices. Each statement in the theorem describes a property of every $n \times n$ invertible matrix. The *negation* of a statement in the theorem describes a property of every $n \times n$ singular matrix. For instance, an $n \times n$ singular matrix is *not* row equivalent to I_n, does *not* have n pivot positions, and has linearly *dependent* columns.

Determinant and Invertibility

THEOREM 4

A square matrix A is invertible if and only if $\det A \neq 0$.

Theorem 4 adds the statement "$\det A \neq 0$" to the Invertible Matrix Theorem. A useful corollary is that $\det A = 0$ when the columns of A are linearly dependent. Also, $\det A = 0$ when the *rows* of A are linearly dependent. (Rows of A are columns of A^T, and linearly dependent columns of A^T make A^T singular. When A^T is singular, so is A, by the Invertible Matrix Theorem.) In practice, linear dependence is obvious when two columns or two rows are the same or a column or a row is zero.

Lay, pg171 (Ch3.2)

Clever idea:
The determinant transforms the original problem of $(A - \lambda I)x = 0$,
an equation with two unknowns λ, x into a polynomial with only 1 unknown λ.
Allowing us to solve first for the eigen value λ, the roots of the polynomial.
The polynomial is called the "characteristic equation"of A.

Since $(A - \lambda I)$ is a singular matrix (NOT invertible), therefore

$$\det(A - \lambda I) = 0$$

Characteristic Eqn and Polynomial

EXAMPLE 1 Find the eigenvalues of $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$.

SOLUTION We must find all scalars λ such that the matrix equation

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

has a nontrivial solution. By the Invertible Matrix Theorem in Section 2.3, this problem is equivalent to finding all λ such that the matrix $A - \lambda I$ is *not* invertible, where

$$A - \lambda I = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix}$$

By Theorem 4 in Section 2.2, this matrix fails to be invertible precisely when its determinant is zero. So the eigenvalues of A are the solutions of the equation

$$\det(A - \lambda I) = \det \begin{bmatrix} 2-\lambda & 3 \\ 3 & -6-\lambda \end{bmatrix} = 0$$

Determinant of 2x2 matrix

THEOREM 4

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible.

$$\det A = ad - bc$$

Theorem 4 says that a 2×2 matrix A is invertible if and only if $\det A \neq 0$.

Lay,4th,pg103,sec 2.2

Characteristic Eqn and Polynomial

Recall that

$$\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

So

$$\begin{aligned} \boxed{\det(A - \lambda I)} &= (2-\lambda)(-6-\lambda) - (3)(3) \\ &= -12 + 6\lambda - 2\lambda + \lambda^2 - 9 \\ &\boxed{= \lambda^2 + 4\lambda - 21} \\ &= (\lambda - 3)(\lambda + 7) \end{aligned}$$

If $\det(A - \lambda I) = 0$, then $\lambda = 3$ or $\lambda = -7$. So the eigenvalues of A are 3 and -7. ■

Characteristic Eqn

Characteristic Polynomial

Equivalent statements regarding eigenvalue λ

Given $Ax = \lambda x$, the following theorem applies:

Theorem Given a square matrix A and a scalar λ, the following statements are equivalent:

- λ is an eigenvalue of A,
- $N(A - \lambda I) \neq \{\mathbf{0}\}$,
- the matrix $A - \lambda I$ is singular,
- $\det(A - \lambda I) = 0$.

Ref: https://textbooks.math.gatech.edu/ila/characteristic-polynomial.html

Example: EigenValues of Triangular Matrixes

THEOREM 1 The eigenvalues of a triangular matrix are the entries on its main diagonal.

PROOF For simplicity, consider the 3×3 case. If A is upper triangular, then $A - \lambda I$ has the form

$$A - \lambda I = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ 0 & a_{22} - \lambda & a_{23} \\ 0 & 0 & a_{33} - \lambda \end{bmatrix}$$

The scalar λ is an eigenvalue of A if and only if the equation $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, that is, if and only if the equation has a free variable. Because of the zero entries in $A - \lambda I$, it is easy to see that $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a free variable if and only if at least one of the entries on the diagonal of $A - \lambda I$ is zero. This happens if and only if λ equals one of the entries a_{11}, a_{22}, a_{33} in A. For the case in which A is lower triangular, see Exercise 28. ■

THEOREM 3.2 Determinant of a Triangular Matrix

If A is a triangular matrix of order n, then its determinant is the product of the entries on the main diagonal. That is,

$$\det(A) = |A| = a_{11}a_{22}a_{33} \cdot \cdot \cdot a_{nn}.$$

Fact 7. The determinant of a lower triangular matrix (or an upper triangular matrix) is the product of the diagonal entries. In particular, the determinant of a diagonal matrix is the product of the diagonal entries.

Proof: Khan's academy determinant of triangular matrix.

https://www.khanacademy.org/math/linear-algebra/matrix-transformations/determinant-depth/v/linear-algebra-upper-triangular-determinant

Example: Meaning of eigenvalue == 0 => not invertible matrix

EXAMPLE 5 Let $A = \begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & 3 & 4 \end{bmatrix}$. The eigenvalues of A are 3, 0, and 2. The eigenvalues of B are 4 and 1. ■

What does it mean for a matrix A to have an eigenvalue of 0, such as in Example 5? This happens if and only if the equation

$$A\mathbf{x} = 0\mathbf{x} \tag{4}$$

has a nontrivial solution. But (4) is equivalent to $A\mathbf{x} = \mathbf{0}$, which has a nontrivial solution if and only if A is not invertible. Thus 0 *is an eigenvalue of A if and only if A is not invertible*. This fact will be added to the Invertible Matrix Theorem in Section 5.2.

THEOREM

The Invertible Matrix Theorem (continued)

Let A be an $n \times n$ matrix. Then A is invertible if and only if:

s. The number 0 is *not* an eigenvalue of A.

t. The determinant of A is *not* zero.

Example: eigenvalue and algebraic multiplicity

EXAMPLE 3

Find the characteristic equation of

$$A = \begin{bmatrix} 5 & -2 & 6 & -1 \\ 0 & 3 & -8 & 0 \\ 0 & 0 & 5 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

SOLUTION Form $A - \lambda I$, and use Theorem 3(d):

$$\det(A - \lambda I) = \det \begin{bmatrix} 5-\lambda & -2 & 6 & -1 \\ 0 & 3-\lambda & -8 & 0 \\ 0 & 0 & 5-\lambda & 4 \\ 0 & 0 & 0 & 1-\lambda \end{bmatrix}$$

$$= (5-\lambda)(3-\lambda)(5-\lambda)(1-\lambda)$$

The characteristic equation is

$$(5-\lambda)^2(3-\lambda)(1-\lambda) = 0$$

or

$$(\lambda - 5)^2(\lambda - 3)(\lambda - 1) = 0$$

Expanding the product, we can also write

$$\lambda^4 - 14\lambda^3 + 68\lambda^2 - 130\lambda + 75 = 0$$

The eigenvalue 5 in Example 3 is said to have *multiplicity* 2 because $(\lambda - 5)$ occurs two times as a factor of the characteristic polynomial. In general, the (**algebraic**) **multiplicity** of an eigenvalue λ is its multiplicity as a root of the characteristic equation.

Lay, 4th, pg276

EigenValues and Algebraic multiplicity

EXAMPLE 4 The characteristic polynomial of a 6×6 matrix is $\lambda^6 - 4\lambda^5 - 12\lambda^4$. Find the eigenvalues and their multiplicities.

SOLUTION Factor the polynomial

$$\lambda^6 - 4\lambda^5 - 12\lambda^4 = \lambda^4(\lambda^2 - 4\lambda - 12) = \lambda^4(\lambda - 6)(\lambda + 2)$$

The eigenvalues are 0 (multiplicity 4), 6 (multiplicity 1), and -2 (multiplicity 1). ■

The **set of solutions** $(\lambda_1, \lambda_2, \dots, \lambda_{N_\lambda})$,that is, the **eigenvalues**, is called the **spectrum** of A. The charaacteristic polynomial can be factored as follows:

$$p(\lambda) = (\lambda - \lambda_1)^{n_1}(\lambda - \lambda_2)^{n_2} \cdots (\lambda - \lambda_{N_\lambda})^{n_{N_\lambda}} = 0.$$

The integer n_i is termed the **algebraic multiplicity** of eigenvalue λ_i .It is the number of times an eigenvalue appears as a root of the characteristic polynomial.

The algebraic multiplicities sum to N (the number of rows in A):

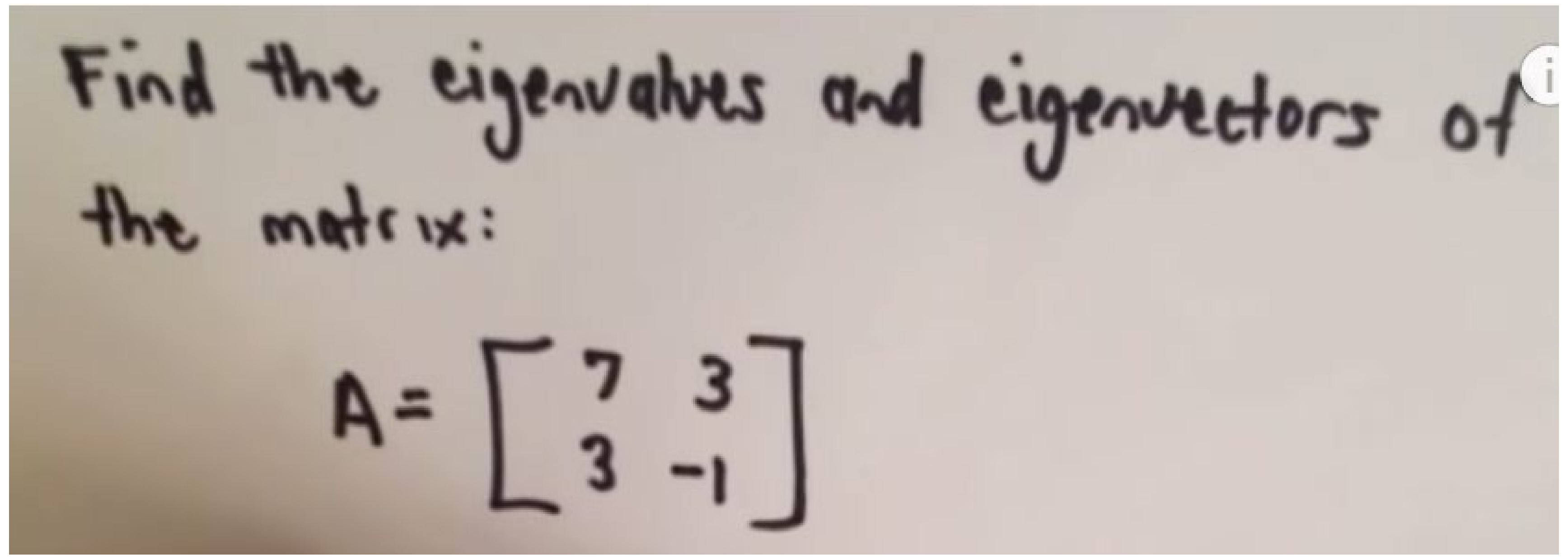
$$\sum_{i=1}^{N_\lambda} n_i = N.$$

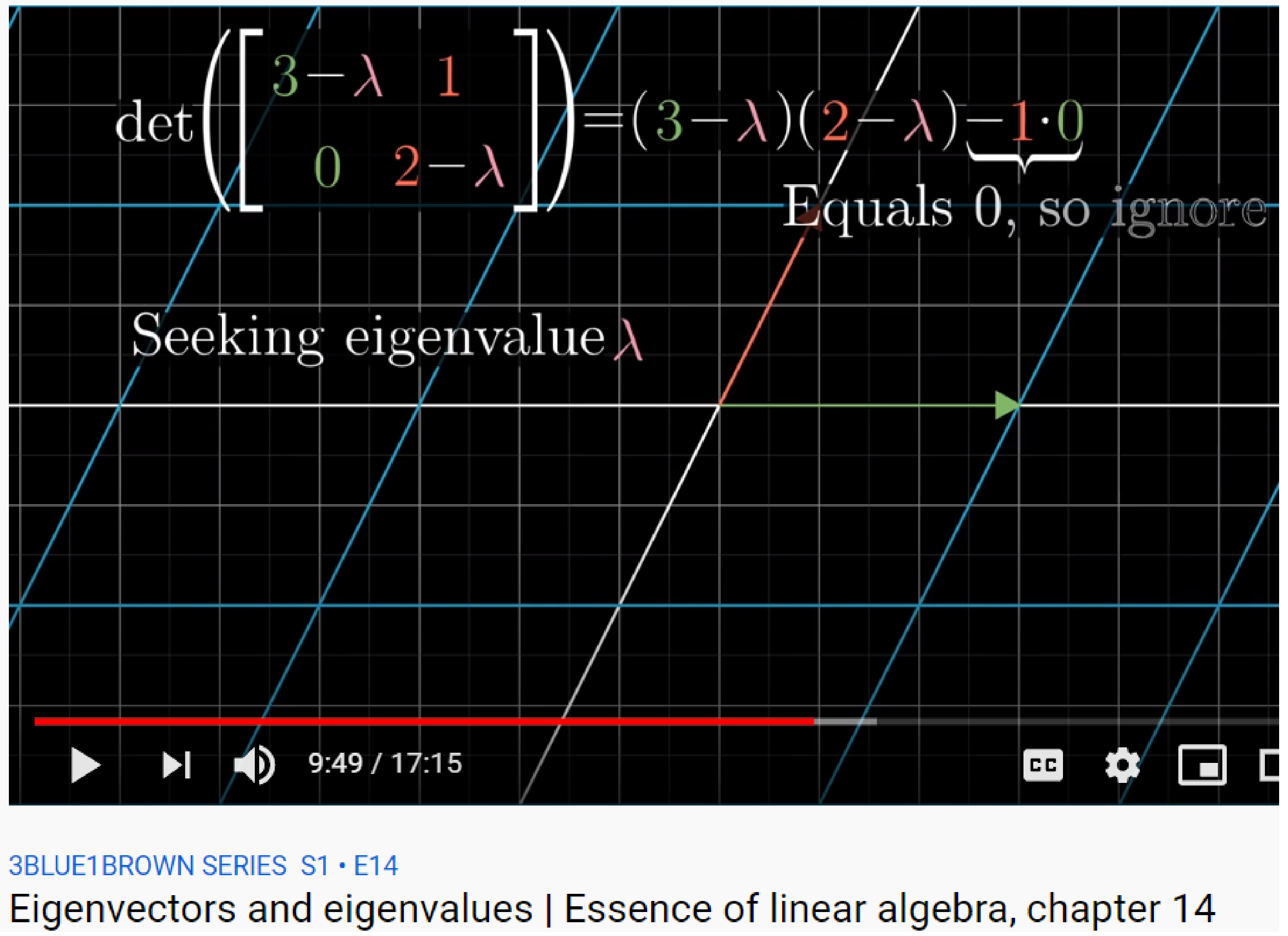
For each eigenvalue λ_i, there is a corresponding EigenSpace $E(\lambda_i)$

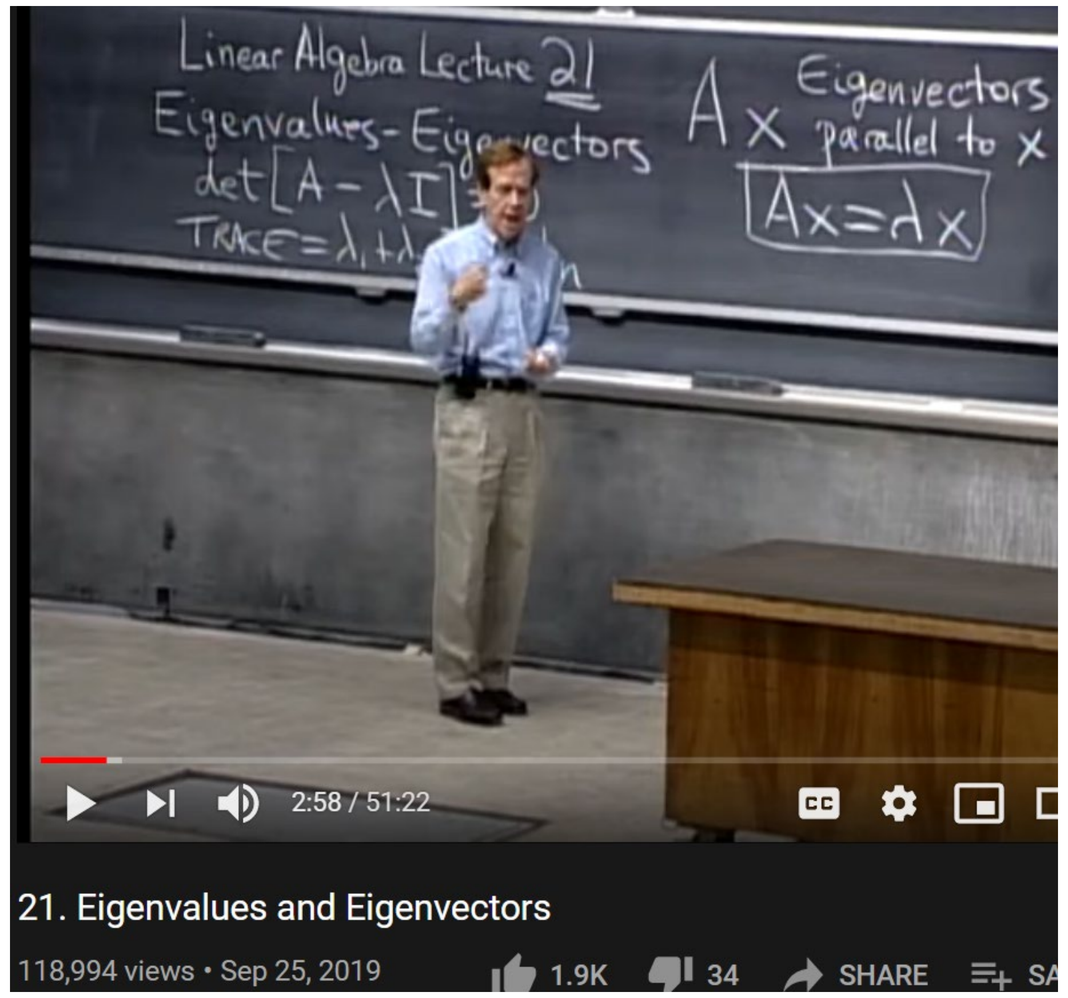
References online

1) PatrickJMT: https://www.youtube.com/watch?v=IdsV0RaC9jM

Work examples:







2) 3Blue1Brown: Ch14, https://www.youtube.com/watch?v=PFDu9oVAE-g

Understanding and perspective

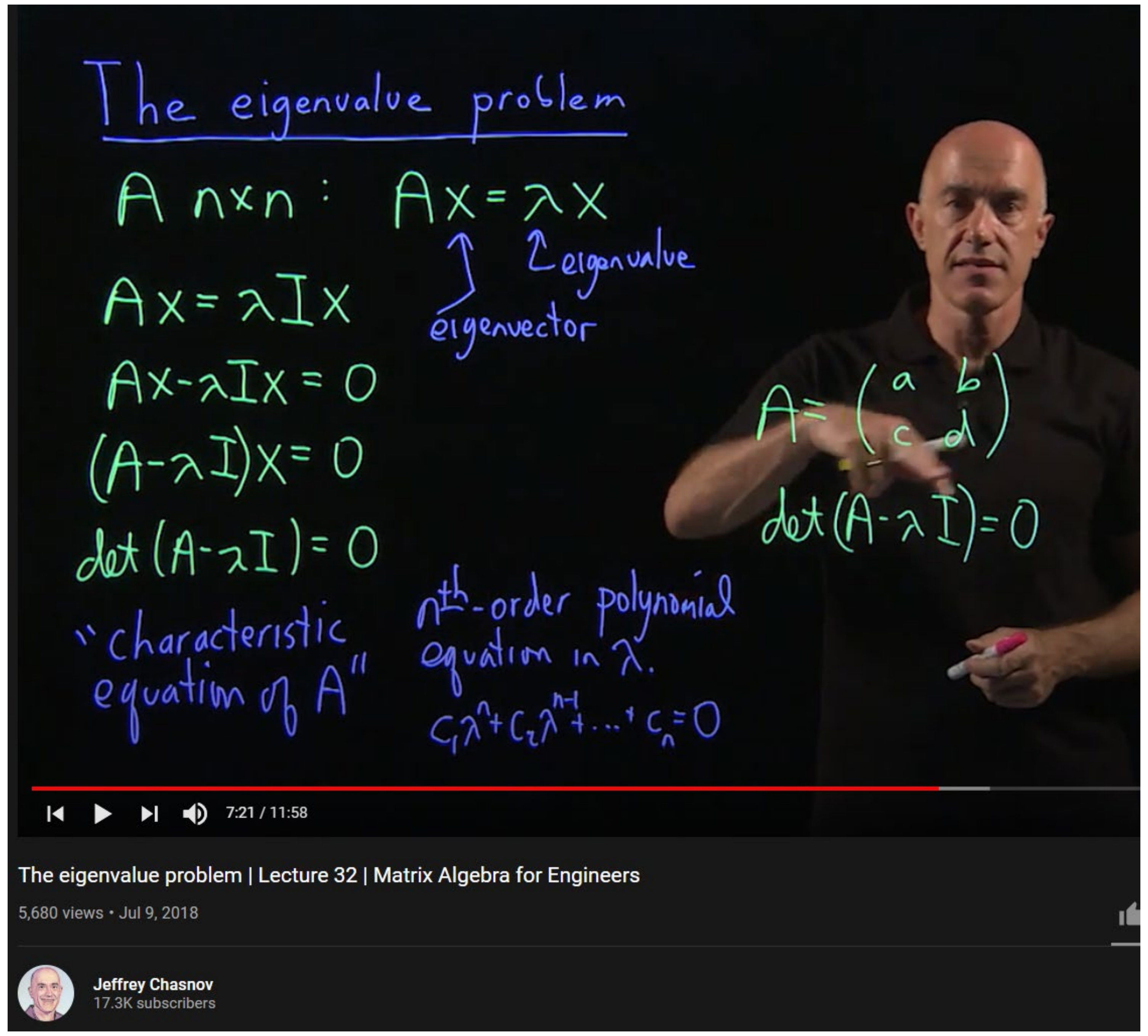
3) Strang introducing EigenValues & vectors

https://www.youtube.com/watch?v=cdZnhQjJu4I

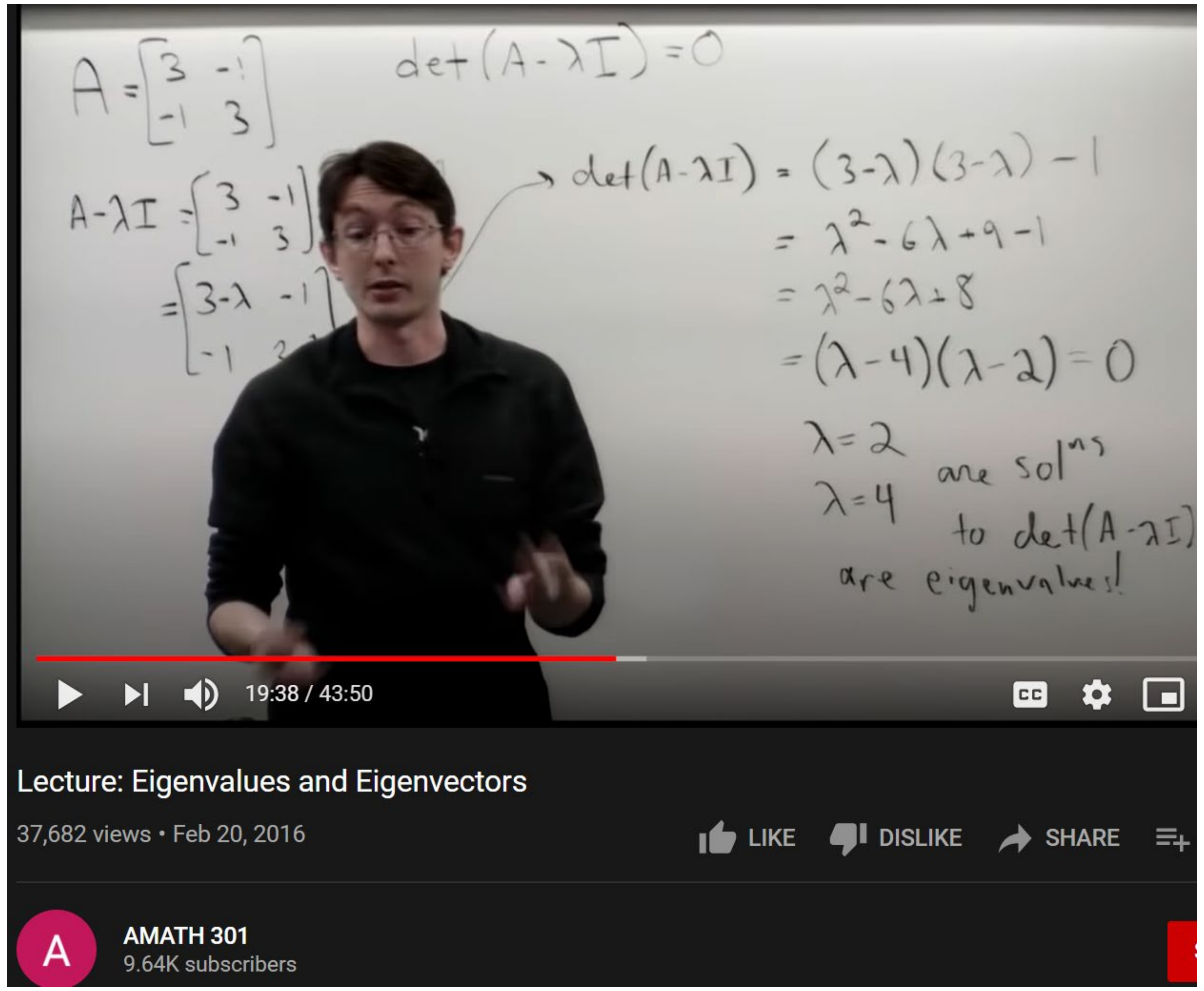
More examples

4. How to find eigenvalues/vectors (process):

a. Chasnov (L33): https://www.youtube.com/watch?v=29keVZGvqME&list=PLkZjai-2Jcxlg-Z1roB0pUwFU-P58tvOx&index=33



5. Steve Brunton's lecture for eigenvalues
https://www.youtube.com/watch?v=OELTJdaU8aA



$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **8.1.3**

Lecture : **Eigen and Singular Values**

Topic : **Similarity and Diagonalization**

Concept : **When can we represent** $\boldsymbol{A = PDP^{-1}}$

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Last updated: 7Nov 2022

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Similarity and Diagonalization

Given two nxn matrix A and P, and P is invertible (means it has n-independent column), then we can have a matrix B generated as follows:

$$P^{-1}AP = B$$
$$AP = PB$$
$$A = PBP^{-1}$$

then we say that A and B are similar matrixes,
and the transformation from A to $\mathrm{B} = P^{-1}AP$ is called similarity transformation.

In the special case that B is a diagonal matrix, then we also say that A is diagonalizable!

Wiki: insight

Similar matrices represent the same linear map under two (possibly) different bases, with P being the change of basis matrix.[1][2]

See Lecture 8.1.5B

Similarity and Diagonalization

Similar Matrix is an important concept because Similar matrixes share certain characteristics:

In general, any property that is preserved by a similarity transformation is called a ***similarity invariant*** and is said to be ***invariant under similarity***. Table 1 lists the most important similarity invariants. The proofs of some of these are given as exercises.

Table 1 Similarity Invariants

Property	Description
Determinant	A and $P^{-1}AP$ have the same determinant.
Invertibility	A is invertible if and only if $P^{-1}AP$ is invertible.
Rank	A and $P^{-1}AP$ have the same rank.
Nullity	A and $P^{-1}AP$ have the same nullity.
Trace	A and $P^{-1}AP$ have the same trace.
Characteristic polynomial	A and $P^{-1}AP$ have the same characteristic polynomial.
Eigenvalues	A and $P^{-1}AP$ have the same eigenvalues.
Eigenspace dimension	If λ is an eigenvalue of A (and hence of $P^{-1}AP$) then the eigenspace of A corresponding to λ and the eigenspace of $P^{-1}AP$ corresponding to λ have the same dimension.

We like a matrix to be similar to a diagonal matrix

Why we like diagonal matrix?

Diagonal matrixes have nice properties:

1) Eigenvalues of diagonal matrixes are its diagonal element
2) Determinant == product of diagonal entries
3) Rank == number of non-zero entries in the diagonal
4) Multiplication: given A and diagonal matrix D (AD and DA):

   - when we pre-multiply A by a diagonal matrix D, the rows of A are multiplied by the diagonal elements of D;
   - when we post-multiply A by D, the columns of A are multiplied by the diagonal elements of D.

5) A diagonal matrix's inverse is reciprocal of diagonal elements
6) Product of diagonal matrixes are easy to compute.

Ref:
1) https://www.statlect.com/matrix-algebra/diagonal-matrix
2) http://www.robertosmathnotes.com/uploads/8/2/3/9/8239617/la10-3_diagonal_matrices.pdf

When is A Diagonalizable?

Special case of similarity, B is D (a diagonal matrix).

THEOREM 5 **The Diagonalization Theorem**

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A. In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P.

In other words, A is diagonalizable if and only if there are enough eigenvectors to form a basis of $\mathbb{R}^n$. We call such a basis an **eigenvector basis** of $\mathbb{R}^n$.

Lay, pg 282, Ch 5.3

Ref:https://textbooks.math.gatech.edu/ila/similarity.html

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When is A Diagonalizable? Depends on P (the matrix containing the eigenvectors)

Since P is invertible, its columns $\mathbf{v}_1, \ldots, \mathbf{v}_n$ must be linearly independent. Also, since these columns are nonzero, the equations in (4) show that $\lambda_1, \ldots, \lambda_n$ are eigenvalues and $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are corresponding eigenvectors. This argument proves the "only if" parts of the first and second statements, along with the third statement, of the theorem.

Finally, given any n eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$, use them to construct the columns of P and use corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$ to construct D. By equations (1)–(3), $AP = PD$. This is true without any condition on the eigenvectors. If, in fact, the eigenvectors are linearly independent, then P is invertible (by the Invertible Matrix Theorem), and $AP = PD$ implies that $A = PDP^{-1}$. ■

Proof: When is A Diagonalizable?

THEOREM 5

In other words, A is diagonalizable if and only if there are enough eigenvectors to form a basis of $\mathbb{R}^n$. We call such a basis an **eigenvector basis** of $\mathbb{R}^n$.

PROOF First, observe that if P is any $n \times n$ matrix with columns $\mathbf{v}_1, \dots, \mathbf{v}_n$, and if D is any diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$, then

$$AP = A[\,\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n\,] = [\,A\mathbf{v}_1 \quad A\mathbf{v}_2 \quad \cdots \quad A\mathbf{v}_n\,] \tag{1}$$

while

$$PD = P\begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} = [\,\lambda_1\mathbf{v}_1 \quad \lambda_2\mathbf{v}_2 \quad \cdots \quad \lambda_n\mathbf{v}_n\,] \tag{2}$$

Now suppose A is diagonalizable and $A = PDP^{-1}$. Then right-multiplying this relation by P, we have $AP = PD$. In this case, equations (1) and (2) imply that

$$[\,A\mathbf{v}_1 \quad A\mathbf{v}_2 \quad \cdots \quad A\mathbf{v}_n\,] = [\,\lambda_1\mathbf{v}_1 \quad \lambda_2\mathbf{v}_2 \quad \cdots \quad \lambda_n\mathbf{v}_n\,] \tag{3}$$

Equating columns, we find that

$$A\mathbf{v}_1 = \lambda_1\mathbf{v}_1, \quad A\mathbf{v}_2 = \lambda_2\mathbf{v}_2, \quad \dots, \quad A\mathbf{v}_n = \lambda_n\mathbf{v}_n \tag{4}$$

Examine

$$AP = PD$$

For columns of P being eigenvectors of A, and D the eigenval of A

Practice Problems 2

2. Let $A = \begin{bmatrix} -3 & 12 \\ -2 & 7 \end{bmatrix}$, $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Suppose you are told that $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors of A. Use this information to diagonalize A.

Sol:

2. Compute $A\mathbf{v}_1 = \begin{bmatrix} -3 & 12 \\ -2 & 7 \end{bmatrix}\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 1 \cdot \mathbf{v}_1$, and

$$A\mathbf{v}_2 = \begin{bmatrix} -3 & 12 \\ -2 & 7 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = 3 \cdot \mathbf{v}_2$$

So, $\mathbf{v}_1$ and $\mathbf{v}_2$ are eigenvectors for the eigenvalues 1 and 3, respectively. Thus

$$A = PDP^{-1}, \quad \text{where} \quad P = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

```
% Example in Slide 8.1.3 (diagonalization)
A = [-3 12; -2 7];
[P,D] = eig(A);

P1 = [3 2; 1 1]
D1 = [ 1 0; 0 3]
P1*D1*inv(P1)
```

```
A =

    -3    12
    -2     7


P =

   -0.9487   -0.8944
   -0.3162   -0.4472


D =

    1.0000         0
         0    3.0000
```

```
P1 =

     3     2
     1     1


D1 =

     1     0
     0     3


ans =

   -3.0000   12.0000
   -2.0000    7.0000
```

Recap: Steps to diagonalize a Matrix

Given a matrix *A* size NxN, to diagonalize it to D, perform:

1) Find the eigenvalues of A.
2) For each eigenvalue, find the eigenvectors of corresponding λ_i
3) If there are N independent eigenvectors v_i, then the matrix A can be represented as:

$$
\begin{aligned}
AP &= PD \\
A &= PDP^{-1} \\
P^{-1}AP &= D
\end{aligned}
$$

Where *D* = diagonal matrix with eigenvalues λ_i
And *P* is a matrix with columns that are corresponding eigenvectors v_i.

When is A diagonalizable? Sufficient condition: If A has Distinct EigenValues -> Diagonalizable

THEOREM 6 Lay, 4thEd, pg 284, Ch 5.3

An $n \times n$ matrix with n distinct eigenvalues is diagonalizable.

PROOF Let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be eigenvectors corresponding to the n distinct eigenvalues of a matrix A. Then $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is linearly independent, by Theorem 2 in Section 5.1. Hence A is diagonalizable, by Theorem 5. ■

It is not *necessary* for an $n \times n$ matrix to have n distinct eigenvalues in order to be diagonalizable. The 3×3 matrix in Example 3 is diagonalizable even though it has only two distinct eigenvalues.

Theorem & proof: distinct eigenvalues means distinct eigenvectors

THEOREM 2 If $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \ldots, \lambda_r$ of an $n \times n$ matrix A, then the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is linearly independent.

Lay 4th, pg 270, Ch 5.1

PROOF Suppose $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ is linearly dependent. Since $\mathbf{v}_1$ is nonzero, Theorem 7 in Section 1.7 says that one of the vectors in the set is a linear combination of the preceding vectors. Let p be the least index such that $\mathbf{v}_{p+1}$ is a linear combination of the preceding (linearly independent) vectors. Then there exist scalars $c_1, \ldots, c_p$ such that

$$c_1\mathbf{v}_1 + \cdots + c_p\mathbf{v}_p = \mathbf{v}_{p+1} \tag{5}$$

Multiplying both sides of (5) by A and using the fact that $A\mathbf{v}_k = \lambda_k\mathbf{v}_k$ for each k, we obtain

$$\begin{aligned} c_1A\mathbf{v}_1 + \cdots + c_pA\mathbf{v}_p &= A\mathbf{v}_{p+1} \\ c_1\lambda_1\mathbf{v}_1 + \cdots + c_p\lambda_p\mathbf{v}_p &= \lambda_{p+1}\mathbf{v}_{p+1} \end{aligned} \tag{6}$$

Multiplying both sides of (5) by λ_{p+1} and subtracting the result from (6), we have

$$c_1(\lambda_1 - \lambda_{p+1})\mathbf{v}_1 + \cdots + c_p(\lambda_p - \lambda_{p+1})\mathbf{v}_p = \mathbf{0} \tag{7}$$

Since $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is linearly independent, the weights in (7) are all zero. But none of the factors $\lambda_i - \lambda_{p+1}$ are zero, because the eigenvalues are distinct. Hence $c_i = 0$ for $i = 1, \ldots, p$. But then (5) says that $\mathbf{v}_{p+1} = \mathbf{0}$, which is impossible. Hence $\{\mathbf{v}_1, \ldots, \mathbf{v}_r\}$ cannot be linearly dependent and therefore must be linearly independent. ■

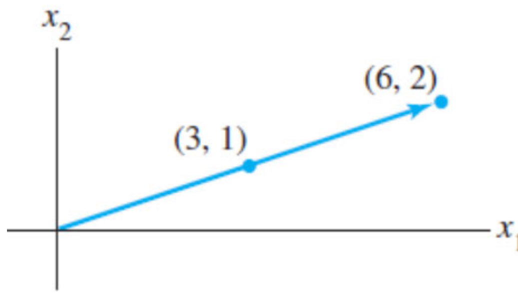
Recap: linear independence

Lay 4th Ed, ch1.7) Revision linear independence

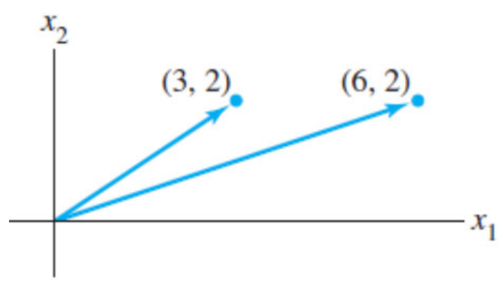
THEOREM 7 **Characterization of Linearly Dependent Sets**

An indexed set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ of two or more vectors is linearly dependent if and only if at least one of the vectors in S is a linear combination of the others. In fact, if S is linearly dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then some $\mathbf{v}_j$ (with $j > 1$) is a linear combination of the preceding vectors, $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$.

A set of two vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent if at least one of the vectors is a multiple of the other. The set is linearly independent if and only if neither of the vectors is a multiple of the other.



Linearly dependent



Linearly independent

Linear Independence of Matrix Columns

Suppose that we begin with a matrix $A = [\,\mathbf{a}_1 \ \cdots \ \mathbf{a}_n\,]$ instead of a set of vectors. The matrix equation $A\mathbf{x} = \mathbf{0}$ can be written as

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$$

Each linear dependence relation among the columns of A corresponds to a nontrivial solution of $A\mathbf{x} = \mathbf{0}$. Thus we have the following important fact.

The columns of a matrix A are linearly independent if and only if the equation $A\mathbf{x} = \mathbf{0}$ has *only* the trivial solution. (3)

Example: Distinct EigenValues -> Diagonalizable

EXAMPLE 5 Determine if the following matrix is diagonalizable.

$$A = \begin{bmatrix} 5 & -8 & 1 \\ 0 & 0 & 7 \\ 0 & 0 & -2 \end{bmatrix}$$

SOLUTION This is easy! Since the matrix is triangular, its eigenvalues are obviously 5, 0, and -2. Since A is a 3×3 matrix with three distinct eigenvalues, A is diagonalizable. ■

See Slide 8.1.2 (pg 8) To see slide "EigenValues of Triangular Matrixes"

THEOREM 1 The eigenvalues of a triangular matrix are the entries on its main diagonal.

Distinct eigenvalue is a sufficient BUT not necessary condition to have linearly independent eigenvectors.

Example 3 (later) shows that eigenvalues are repeated, but it is still diagonalizable.
And Example 4 shows counter-example.

When is a matrix with repeated roots diagonalizable? Introducing algebraic and geometrix multiplicity.

Algebraic
multiplicity = multiplicity of eigenvalues λ_k

Geometric
multiplicity = Dimension of eigenspace corresponding to eigenvalue λ_k

Ref: https://people.math.carleton.ca/~kcheung/math/notes/MATH1107/wk10/10_algebraic_and_geometric_multiplicities.html

Introducing terminology: Algebraic and Geometric Multiplicity

Eigenspaces

Let λ be an eigenvalue of A. Recall that the eigenvectors of A for λ are the nonzero vectors in the nullspace of $A - \lambda I$. We call the nullspace $A - \lambda I$ the **eigenspace** of A for λ denoted by $\mathcal{E}_A(\lambda)$. In other words, $\mathcal{E}_A(\lambda)$ consists of all the eigenvectors of A for λ and the zero vector.

Example

Let $A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$. Note that -1 is an eigenvalue of A. Then $A - (-1)I_2 = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}$. The nullspace of this matrix is spanned by the single vector $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Hence, $\mathcal{E}_A(-1)$ is the span of $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Sanity check:

```
% example in sli
A = [1 2; 1 0]
[P,D] = eig(A)
```

```
A =

     1     2
     1     0


P =

    0.8944   -0.7071
    0.4472    0.7071


D =

     2     0
     0    -1
```

Ref:https://people.math.carleton.ca/~kcheung/math/notes/MATH1107/wk10/10_algebraic_and_geometric_multiplicities.html

Geometric Multiplicity (is the dimension of Eigen Space)
Algebraic Multiplicity (is the number of repeated roots)

Algebraic multiplicity vs geometric multiplicity

The **geometric multiplicity** of an eigenvalue λ of A is the dimension of $\mathcal{E}_A(\lambda)$. In the example above, the geometric multiplicity of -1 is 1 as the eigenspace is spanned by one nonzero vector.

In general, determining the geometric multiplicity of an eigenvalue requires no new technique because one is simply looking for the dimension of the nullspace of $A - \lambda I$.

The **algebraic multiplicity** of an eigenvalue λ of A is the number of times λ appears as a root of p_A. For the example above, one can check that -1 appears only once as a root. Let us now look at an example in which an eigenvalue has multiplicity higher than 1.

In mathematics, the **dimension** of a vector space *V* is the cardinality (i.e. the number of vectors) of a basis of *V* over its base field.[1]

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$.

```
% Example:
% repeated roots lambda=1,
% BUT eigenspace ==1
A = [1 2; 0 1]
[P,D] = eig(A)
```

```
A =

     1     2
     0     1


P =

    1.0000   -1.0000
         0    0.0000


D =

     1     0
     0     1
```

Ref:https://people.math.carleton.ca/~kcheung/math/notes/MATH1107/wk10/10_algebraic_and_geometric_multiplicities.html

When is a matrix with repeated roots diagonalizable? Introducing algebraic and geometrix multiplicity.

In general, **the algebraic multiplicity and geometric multiplicity of an eigenvalue can differ. However, the geometric multiplicity can never exceed the algebraic multiplicity**.

It is a fact that summing up the algebraic multiplicities of all the eigenvalues of an $n \times n$ matrix A gives exactly n. **If for every eigenvalue of A, the geometric multiplicity equals the algebraic multiplicity, then A is said to be *diagonalizable***.

See also (Theorem 7) in Lay, 4thEd, pg 285, Ch 5.3

Ref: https://people.math.carleton.ca/~kcheung/math/notes/MATH1107/wk10/10_algebraic_and_geometric_multiplicities.html

Example 3: Diagonalizable A with repeated eigenValue

EXAMPLE 3 Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

That is, find an invertible matrix P and a diagonal matrix D such that $A = PDP^{-1}$.

SOLUTION There are four steps to implement the description in Theorem 5.

Step 1. ***Find the eigenvalues of A.*** As mentioned in Section 5.2, the mechanics of this step are appropriate for a computer when the matrix is larger than 2×2. To avoid unnecessary distractions, the text will usually supply information needed for this step. In the present case, the characteristic equation turns out to involve a cubic polynomial that can be factored:

$$\begin{aligned} 0 = \det(A - \lambda I) &= -\lambda^3 - 3\lambda^2 + 4 \\ &= -(\lambda - 1)(\lambda + 2)^2 \end{aligned}$$

The eigenvalues are $\lambda = 1$ and $\lambda = -2$.

Note: In this example, the eigenvalues are NOT distinct ($\lambda = -2$), i.e repeated, But this matrix is diagonalizable.

Algebraic multiplicity of eigenvalue = -2 is 2
AND
Geometric multiplicity of eigenvalue = -2 is ALSO 2.

HENCE there is a complete set of linearly independent eigenvectors for A, allowing A to be diagonalizable.

Lay4th, pg 283, Ch 5.2

Example 3: Diagonalizable A with repeated eigenValue

Step 2. ***Find three linearly independent eigenvectors of A.*** *Three* vectors are needed because A is a 3×3 matrix. This is the critical step. If it fails, then Theorem 5 says that A cannot be diagonalized. The method in Section 5.1 produces a basis for each eigenspace:

$$\text{Basis for } \lambda = 1: \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{Basis for } \lambda = -2: \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

You can check that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set.

Step 3. ***Construct P from the vectors in step 2.*** The order of the vectors is unimportant. Using the order chosen in step 2, form

$$P = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Sanity check:

```
% Example: slide 5.1.3, pg 17
A = [1 3 3; -3 -5 -3; 3 3 1]
[P,D] = eig(A)
```

```
A =

     1     3     3
    -3    -5    -3
     3     3     1


P =

   -0.5774   -0.7876    0.4206
    0.5774    0.2074   -0.8164
   -0.5774    0.5802    0.3957


D =

    1.0000         0         0
         0   -2.0000         0
         0         0   -2.0000
```

```
P23 = P(:,2:3);
v2  = [-1 1 0]'
c2 = pinv(P23)*v2
v2_est = P23*c2
```

```
c2 =

    0.7121
   -1.0440


v2_est =

   -1.0000
    1.0000
   -0.0000
```

Note that matlab produces col 2,3 of P that does not resemble v_2 *and* v_3.
But you can check that v_2 *and* v_3 can be formed by appropriate linear combinations of col2,3, of P!

Example 3: Diagonalizable A with repeated eigenValue

Step 4. Construct D from the corresponding eigenvalues. In this step, it is essential that the order of the eigenvalues matches the order chosen for the columns of P. Use the eigenvalue $\lambda = -2$ twice, once for each of the eigenvectors corresponding to $\lambda = -2$:

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

It is a good idea to check that P and D really work. To avoid computing P^{-1}, simply verify that $AP = PD$. This is equivalent to $A = PDP^{-1}$ when P is invertible. (However, be sure that P is invertible!) Compute

$$AP = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix}$$

$$PD = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix} \quad \blacksquare$$

Example 4: NOT Diagonalizable A with repeated eigenValue

EXAMPLE 4 Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

SOLUTION The characteristic equation of A turns out to be exactly the same as that in Example 3:

$$0 = \det(A - \lambda I) = -\lambda^3 - 3\lambda^2 + 4 = -(\lambda - 1)(\lambda + 2)^2$$

The eigenvalues are $\lambda = 1$ and $\lambda = -2$. However, it is easy to verify that each eigenspace is only one-dimensional:

$$\text{Basis for } \lambda = 1: \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\text{Basis for } \lambda = -2: \quad \mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

There are no other eigenvalues, and every eigenvector of A is a multiple of either $\mathbf{v}_1$ or $\mathbf{v}_2$. Hence it is impossible to construct a basis of $\mathbb{R}^3$ using eigenvectors of A. By Theorem 5, A is *not* diagonalizable. ■

Note: In this example, the eigenvalues are NOT distinct
($\lambda = -2$), i.e repeated,
But this matrix is NOT diagonalizable.

Algebraic multiplicity of eigenvalue = -2 is 2
BUT
Geometric multiplicity of eigenvalue = -2 is ONLY 1.

Hence incomplete basis of eigenvectors for A =>
A is NOT diagonalizable.

Example 4: NOT Diagonalizable A with repeated eigenValue

It is NOT possible to diagonalize as A does
NOT have a full set of independent eigenvectors.
Sanity check: what does Matlab produce?

```
A =

     2     4     3
    -4    -6    -3
     3     3     1

>> [P,D] = eig(A)

P =

   0.5774 + 0.0000i   0.7071 + 0.0000i   0.7071 - 0.0000i
  -0.5774 + 0.0000i  -0.7071 + 0.0000i  -0.7071 + 0.0000i
   0.5774 + 0.0000i   0.0000 - 0.0000i   0.0000 + 0.0000i


D =

   1.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 + 0.0000i
   0.0000 + 0.0000i  -2.0000 + 0.0000i   0.0000 + 0.0000i
   0.0000 + 0.0000i   0.0000 + 0.0000i  -2.0000 - 0.0000i
```

Note that columns of P are the eigen vectors.
Column 2 == Column 3.

Compare Column 2 to v_2 .
We see Direction is the same,
BUT scaled differently.
Matlab gives eigenVectors
with norm == 1

References

Similar Matrixes
A) Trefor Bazett: Similar matrices have similar properties
Link: https://www.youtube.com/watch?v=jNtiENbAcFM

B) MIT Strang: “Similar Matrixes”
https://www.youtube.com/watch?v=TSdXJw83kyA
https://www.youtube.com/watch?v=LKMGo8G7-vk
https://www.youtube.com/watch?v=KUuxdk_V7To

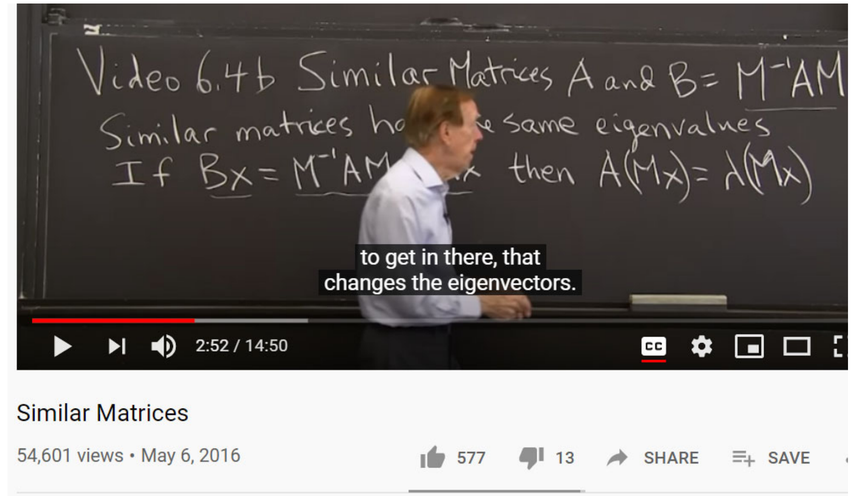
C) Technion 1M :
https://youtu.be/MJic8o5ph5M,
https://youtu.be/l2BEIOnG54k

D) Dan Yasaki https://youtu.be/ObTVoBQDBx8

E) Mathaholic:
https://www.youtube.com/watch?v=H2pv1Rug0RQ

Diagonalization:
A) MIT Strang: “Diagonalizing a matrix”
https://www.youtube.com/watch?v=U8R54zOTVLw

B) MIT Strang: “Diagonalization and Powers of A”
https://www.youtube.com/watch?v=13r9QY6cmjc
Example: Time == 22:10 (triangular matrix and eigenvalues)
Time == 24:00 (algebraic multiplicity==2, geometric multiplicity=1) -> not diagonalizable.



Prof. Strang shows similar matrices A and B have the same eigenvalues in the first 3 minutes and works out some examples of similar matrices later!

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

CX1104: **Linear Algebra for Computing**

Chap. No : **8.1.4**

Lecture : **Eigen and Singular Values**

Topic : **Powers of A**

Concept : **Efficiently compute** $\boldsymbol{A^K}$

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Rev: 3rd July 2020

Power of A: Example 1

In many cases, the eigenvalue–eigenvector information contained within a matrix A can be displayed in a useful factorization of the form $A = PDP^{-1}$ where D is a diagonal matrix. In this section, the factorization enables us to compute A^k quickly for large values of k, a fundamental idea in several applications of linear algebra. Later, in Sections 5.6 and 5.7, the factorization will be used to analyze (and *decouple*) dynamical systems.

The following example illustrates that powers of a diagonal matrix are easy to compute.

EXAMPLE 1 If $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$, then $D^2 = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}$

and

$$D^3 = DD^2 = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix} = \begin{bmatrix} 5^3 & 0 \\ 0 & 3^3 \end{bmatrix}$$

In general,

$$D^k = \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix} \quad \text{for } k \geq 1$$

Lay, pg 281

Example 2: Finding A^k from $A = PDP^{-1}$

EXAMPLE 2 Let $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$. Find a formula for A^k, given that $A = PDP^{-1}$, where

$$P = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

SOLUTION The standard formula for the inverse of a 2×2 matrix yields

$$P^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

Then, by associativity of matrix multiplication,

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD\underbrace{(P^{-1}P)}_{I}DP^{-1} = PDDP^{-1}$$

$$= PD^2P^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix}\begin{bmatrix} 5^2 & 0 \\ 0 & 3^2 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

Example 2: Finding A^k from $A = PDP^{-1}$

Again,

$$A^3 = (PDP^{-1})A^2 = (PD\underbrace{P^{-1})P}_{I}D^2P^{-1} = PDD^2P^{-1} = PD^3P^{-1}$$

In general, for $k \geq 1$,

$$A^k = PD^kP^{-1} = \begin{bmatrix} 1 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5^k & 0 \\ 0 & 3^k \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \cdot 5^k - 3^k & 5^k - 3^k \\ 2 \cdot 3^k - 2 \cdot 5^k & 2 \cdot 3^k - 5^k \end{bmatrix} \quad \blacksquare$$

A square matrix A is said to be **diagonalizable** if A is similar to a diagonal matrix, that is, if $A = PDP^{-1}$ for some invertible matrix P and some diagonal matrix D.

Impt: when A is diagonalizable to $A = PDP^{-1}, then$

$$A^k = PD^kP^{-1}$$

Practice Problems 1

PRACTICE PROBLEMS

1. Compute A^8, where $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$.

SOLUTIONS TO PRACTICE PROBLEMS

1. $\det(A - \lambda I) = \lambda^2 - 3\lambda + 2 = (\lambda - 2)(\lambda - 1)$. The eigenvalues are 2 and 1, and the corresponding eigenvectors are $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Next, form

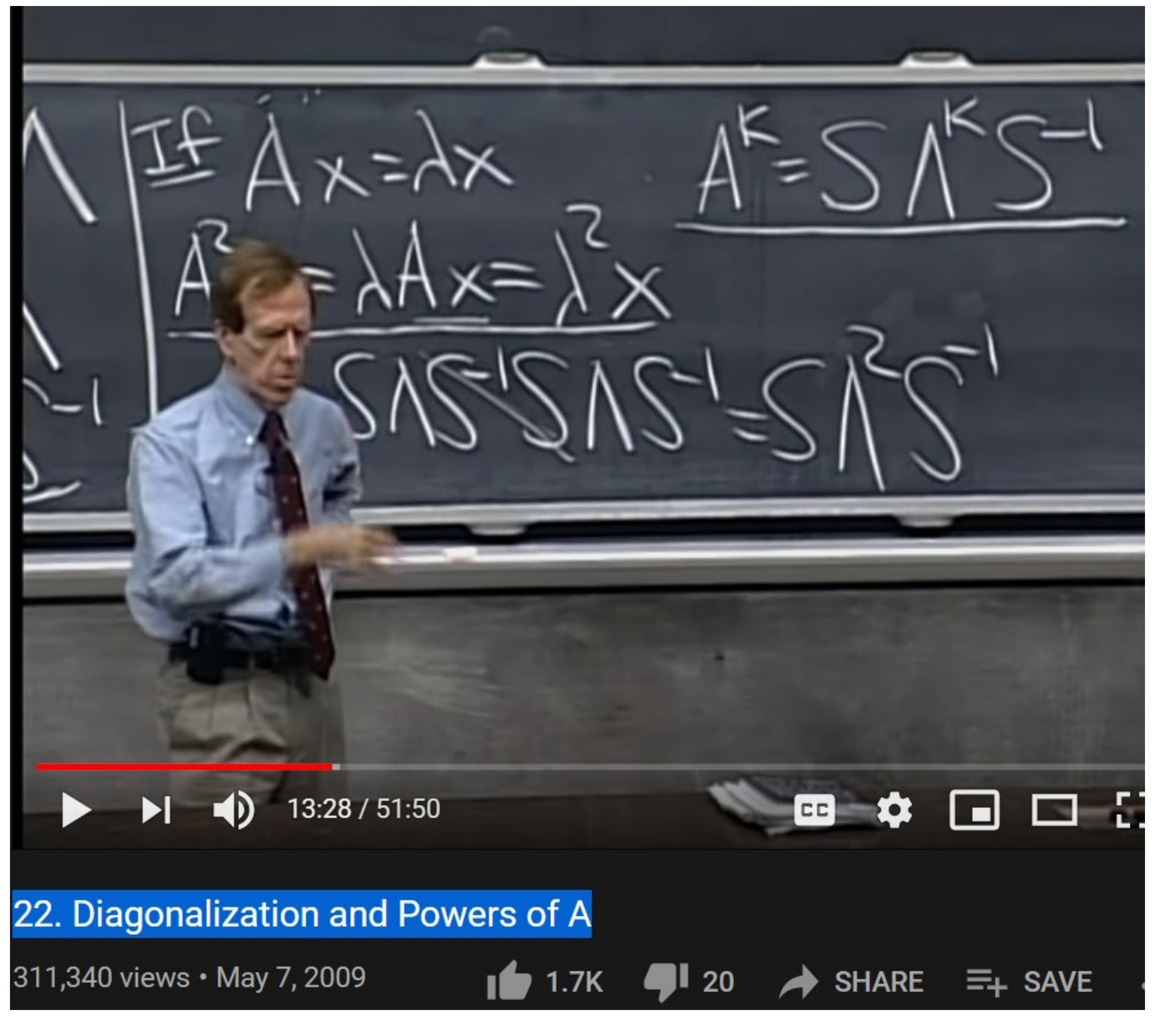
$$P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}, \qquad D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{and} \quad P^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

Since $A = PDP^{-1}$,

$$\begin{aligned} A^8 = PD^8P^{-1} &= \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2^8 & 0 \\ 0 & 1^8 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 256 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 766 & -765 \\ 510 & -509 \end{bmatrix} \end{aligned}$$

References

1) MIT Strang : Lect 22. Diagonalization and Powers of A
https://www.youtube.com/watch?v=13r9QY6cmjc



CX1104: **Linear Algebra for Computing**

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m\times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m\times 1}}_{b}$$

Chap. No : **8.1.5A**

Topic : **Revision – coordinate systems**

Concept : **Coordinate systems and Change of basis**

Note: This is a quick revision on coordinates and change of basis

Instructor: **A/P Chng Eng Siong**

TAs: **Zhang Su, Vishal Choudhari**

Basis

Let H be a subspace of a vector space V. An indexed set of vectors $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ in V is a **basis** for H if

(i) $\mathcal{B}$ is a linearly independent set, and

(ii) the subspace spanned by $\mathcal{B}$ coincides with H; that is,

$$H = \text{Span}\,\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$$

Lay,4th,pg209,ch4.3

Two Views of a Basis

When the Spanning Set Theorem is used, the deletion of vectors from a spanning set must stop when the set becomes linearly independent. If an additional vector is deleted, it will not be a linear combination of the remaining vectors, and hence the smaller set will no longer span V. Thus a basis is a spanning set that is as small as possible.

A basis is also a linearly independent set that is as large as possible. If S is a basis for V, and if S is enlarged by one vector—say, $\mathbf{w}$—from V, then the new set cannot be linearly independent, because S spans V, and $\mathbf{w}$ is therefore a linear combination of the elements in S.

Lay,4th,pg212,ch4.3

Basis

EXAMPLE 10 The following three sets in $\mathbb{R}^3$ show how a linearly independent set can be enlarged to a basis and how further enlargement destroys the linear independence of the set. Also, a spanning set can be shrunk to a basis, but further shrinking destroys the spanning property.

$$\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix}\right\} \qquad \left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix},\begin{bmatrix}4\\5\\6\end{bmatrix}\right\} \qquad \left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}2\\3\\0\end{bmatrix},\begin{bmatrix}4\\5\\6\end{bmatrix},\begin{bmatrix}7\\8\\9\end{bmatrix}\right\}$$

Linearly independent but does not span $\mathbb{R}^3$ | A basis for $\mathbb{R}^3$ | Spans $\mathbb{R}^3$ but is linearly dependent ■

Lay,4th,pg212,ch4.3

Revision: Coordinate system

THEOREM 7 **The Unique Representation Theorem**

Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for a vector space V. Then for each $\mathbf{x}$ in V, there exists a unique set of scalars $c_1, \ldots, c_n$ such that

$$\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n \tag{1}$$

PROOF Since $\mathcal{B}$ spans V, there exist scalars such that (1) holds. Suppose $\mathbf{x}$ also has the representation

$$\mathbf{x} = d_1\mathbf{b}_1 + \cdots + d_n\mathbf{b}_n$$

for scalars $d_1, \ldots, d_n$. Then, subtracting, we have

$$\mathbf{0} = \mathbf{x} - \mathbf{x} = (c_1 - d_1)\mathbf{b}_1 + \cdots + (c_n - d_n)\mathbf{b}_n \tag{2}$$

Since $\mathcal{B}$ is linearly independent, the weights in (2) must all be zero. That is, $c_j = d_j$ for $1 \le j \le n$. ■

Lay4th, Sec 4.4, pg 216

Another way to look at this:
e.g if $x \in R^n$, $c \in R^n$, and
$P_B = [b_1 \ldots b_n] \in R^{nxn}$,
P_B is invertible since cols
are independent, and relating
x and c by
$x = P_B c$

Then
$$c = P_B^{-1} x$$

Since P_B^{-1} $exist, then$
c is unique. $[x]_b = c$ is a
vector $\in R^n$ representing x
In the B basis.

Revision: Coordinate Systems

DEFINITION

Suppose $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is a basis for V and $\mathbf{x}$ is in V. The **coordinates of x relative to the basis $\mathcal{B}$** (or the **$\mathcal{B}$-coordinates of x**) are the weights $c_1, \ldots, c_n$ such that $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$.

If $c_1, \ldots, c_n$ are the $\mathcal{B}$-coordinates of $\mathbf{x}$, then the vector in $\mathbb{R}^n$

$$[\,\mathbf{x}\,]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

is the **coordinate vector of x (relative to $\mathcal{B}$)**, or the **$\mathcal{B}$-coordinate vector of x**. The mapping $\mathbf{x} \mapsto [\,\mathbf{x}\,]_{\mathcal{B}}$ is the **coordinate mapping (determined by $\mathcal{B}$)**.[1]

---

[1] The concept of a coordinate mapping assumes that the basis $\mathcal{B}$ is an indexed set whose vectors are listed in some fixed preassigned order. This property makes the definition of $[\,\mathbf{x}\,]_{\mathcal{B}}$ unambiguous.

Lay,4th,pg216,ch4.4

Revision: Coordinate Systems

EXAMPLE 1 Consider a basis $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ for $\mathbb{R}^2$, where $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{b}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Suppose an $\mathbf{x}$ in $\mathbb{R}^2$ has the coordinate vector $[\,\mathbf{x}\,]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$. Find $\mathbf{x}$.

SOLUTION The $\mathcal{B}$-coordinates of $\mathbf{x}$ tell how to build $\mathbf{x}$ from the vectors in $\mathcal{B}$. That is,

$$\mathbf{x} = (-2)\mathbf{b}_1 + 3\mathbf{b}_2 = (-2)\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$$

■

EXAMPLE 2 The entries in the vector $\mathbf{x} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ are the coordinates of $\mathbf{x}$ relative to the *standard basis* $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$, since

$$\begin{bmatrix} 1 \\ 6 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 6 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1 \cdot \mathbf{e}_1 + 6 \cdot \mathbf{e}_2$$

If $\mathcal{E} = \{\mathbf{e}_1, \mathbf{e}_2\}$, then $[\,\mathbf{x}\,]_{\mathcal{E}} = \mathbf{x}$. ■

Revision: coordinate system

EXAMPLE 4 Let $\mathbf{b}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, and $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$. Find the coordinate vector $[\,\mathbf{x}\,]_\mathcal{B}$ of $\mathbf{x}$ relative to $\mathcal{B}$.

SOLUTION The $\mathcal{B}$-coordinates c_1, c_2 of $\mathbf{x}$ satisfy

$$c_1 \underset{\mathbf{b}_1}{\begin{bmatrix} 2 \\ 1 \end{bmatrix}} + c_2 \underset{\mathbf{b}_2}{\begin{bmatrix} -1 \\ 1 \end{bmatrix}} = \underset{\mathbf{x}}{\begin{bmatrix} 4 \\ 5 \end{bmatrix}}$$

or

$$\underset{\mathbf{b}_1 \quad \mathbf{b}_2}{\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \underset{\mathbf{x}}{\begin{bmatrix} 4 \\ 5 \end{bmatrix}} \tag{3}$$

This equation can be solved by row operations on an augmented matrix or by using the inverse of the matrix on the left. In any case, the solution is $c_1 = 3$, $c_2 = 2$. Thus $\mathbf{x} = 3\mathbf{b}_1 + 2\mathbf{b}_2$, and

$$[\,\mathbf{x}\,]_\mathcal{B} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

See Fig. 4.

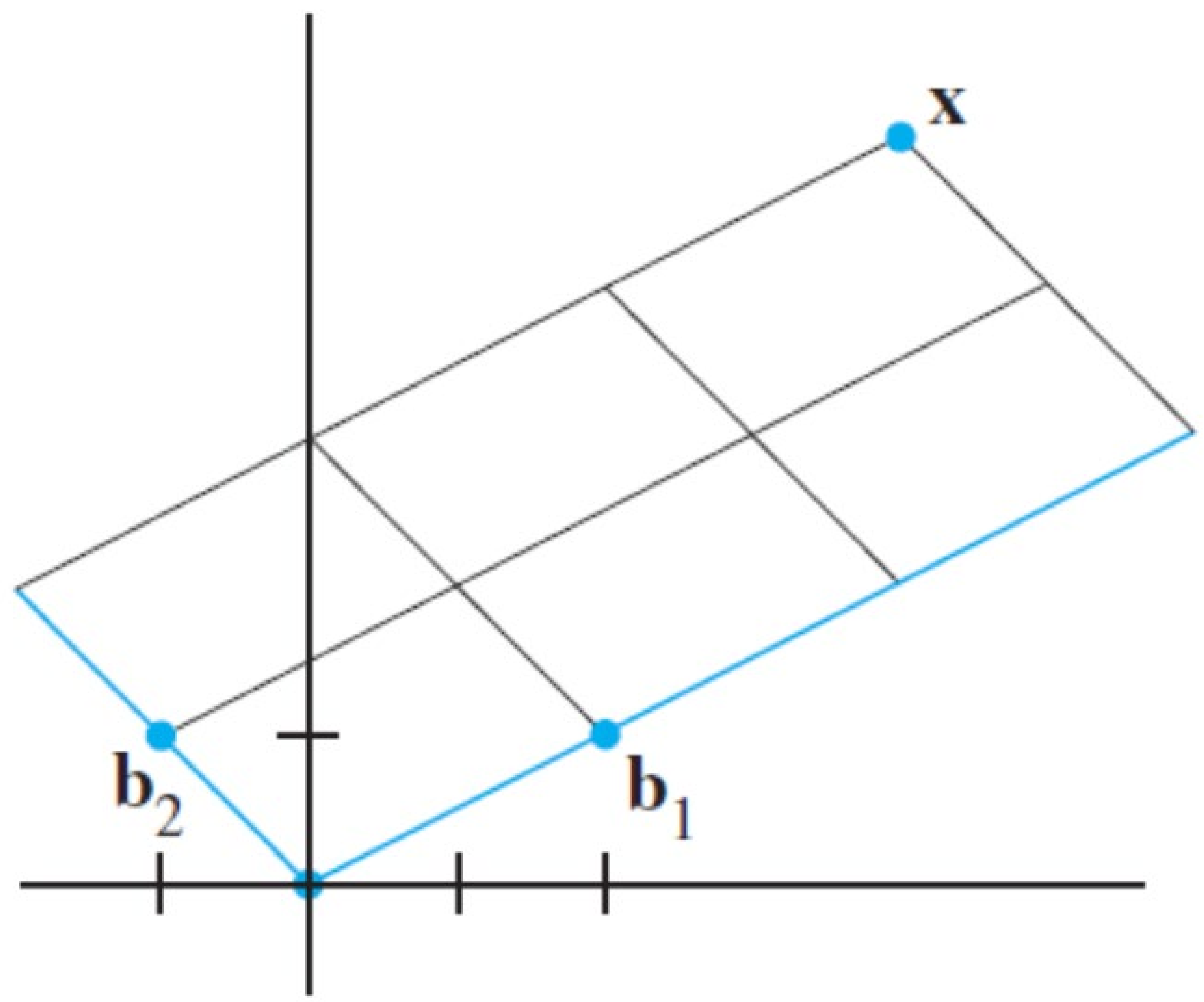


FIGURE 4
The $\mathcal{B}$-coordinate vector of $\mathbf{x}$ is (3, 2).

Lay,4th,pg218,ch4.4

Revision: change of coordinate matrix

$$\begin{array}{ccc} P_B & [x]_B & = \quad x \\ \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} & \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} & = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\ \mathbf{b}_1 \quad \mathbf{b}_2 & & \mathbf{x} \end{array} \tag{3}$$

The matrix in (3) changes the $\mathcal{B}$-coordinates of a vector $\mathbf{x}$ into the standard coordinates for $\mathbf{x}$. An analogous change of coordinates can be carried out in $\mathbb{R}^n$ for a basis $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$. Let

$$P_{\mathcal{B}} = [\,\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_n\,]$$

Then the vector equation

$$\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_n\mathbf{b}_n$$

is equivalent to

$$\boxed{\mathbf{x} = P_{\mathcal{B}}[\,\mathbf{x}\,]_{\mathcal{B}}} \tag{4}$$

We call $P_{\mathcal{B}}$ the **change-of-coordinates matrix** from $\mathcal{B}$ to the standard basis in $\mathbb{R}^n$. Left-multiplication by $P_{\mathcal{B}}$ transforms the coordinate vector $[\,\mathbf{x}\,]_{\mathcal{B}}$ into $\mathbf{x}$. The change-of-coordinates equation (4) is important and will be needed at several points in Chapters 5 and 7.

Lay,4th,pg219,ch4.4

Revision: change of coordinate matrix

Since the columns of $P_{\mathcal{B}}$ form a basis for $\mathbb{R}^n$, $P_{\mathcal{B}}$ is invertible (by the Invertible Matrix Theorem). Left-multiplication by $P_{\mathcal{B}}^{-1}$ converts $\mathbf{x}$ into its $\mathcal{B}$-coordinate vector:

$$P_{\mathcal{B}}^{-1}\mathbf{x} = [\,\mathbf{x}\,]_{\mathcal{B}}$$

The correspondence $\mathbf{x} \mapsto [\,\mathbf{x}\,]_{\mathcal{B}}$, produced here by $P_{\mathcal{B}}^{-1}$, is the coordinate mapping mentioned earlier. Since $P_{\mathcal{B}}^{-1}$ is an invertible matrix, the coordinate mapping is a one-to-one linear transformation from $\mathbb{R}^n$ onto $\mathbb{R}^n$, by the Invertible Matrix Theorem.

Working:

$$P_B[x]_B = x$$

$$(P_B)^{-1}\ P_B[x]_B = (P_B)^{-1}x$$

$$[x]_B = (P_B)^{-1}x$$

Revision: Coordinate System

THEOREM 8

Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for a vector space V. Then the coordinate mapping $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

The Coordinate Mapping

Choosing a basis $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ for a vector space V introduces a coordinate system in V. The coordinate mapping $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ connects the possibly unfamiliar space V to the familiar space $\mathbb{R}^n$. See Fig. 5. Points in V can now be identified by their new "names."

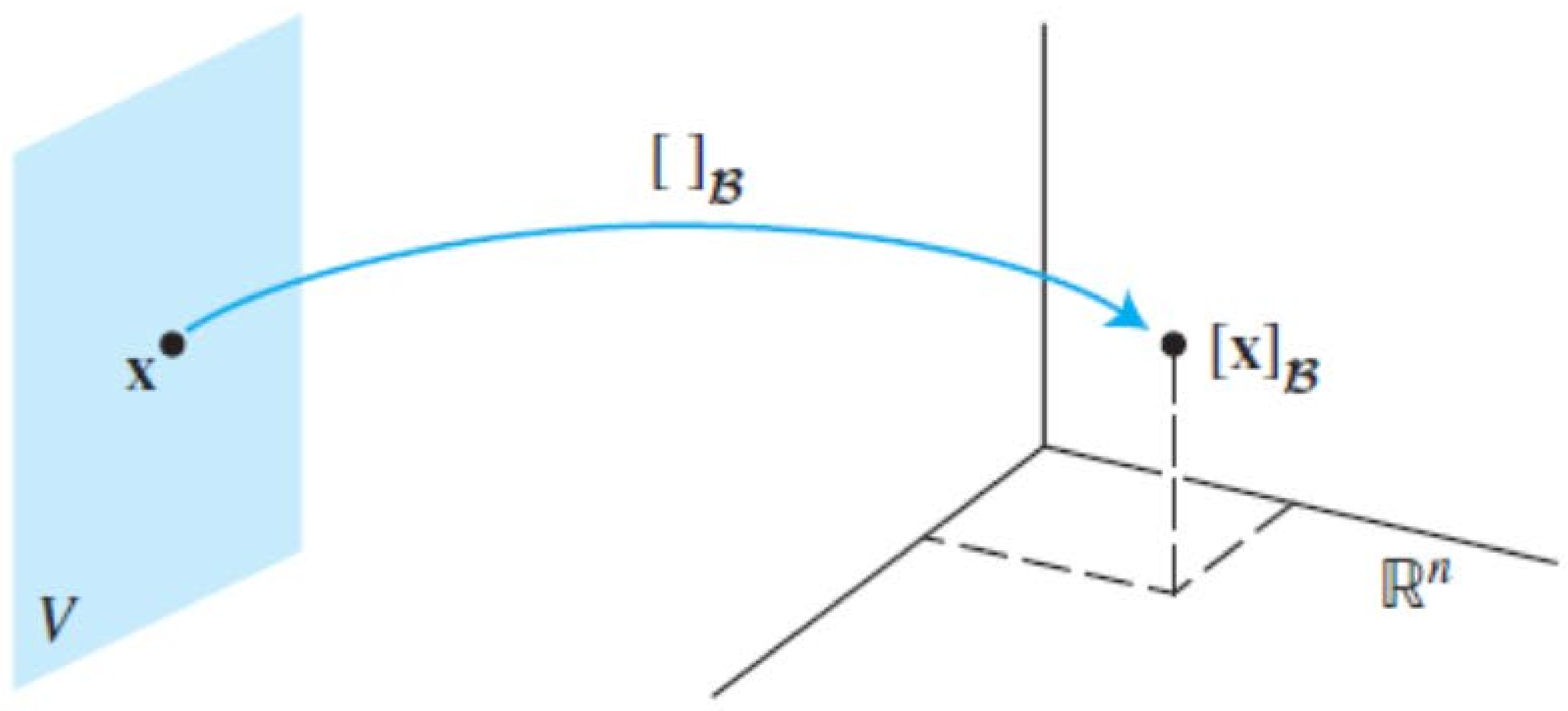


FIGURE 5 The coordinate mapping from V onto $\mathbb{R}^n$.

Lay,4th,pg219,ch4.3

Terminology:
$[x]_B$ is isomorphic to x and
x is isomorphic to $[x]_B$

"isomorphism"
is not part of syllabus.

8 Why are Isomorphisms useful? – Jonathan Dewein Jul 12 '13 at 3:33

55 Lots of reasons; one of the primary ones, however, is that it allows you to replace an object that you need to deal with with another object which has the same structure, but you are more familar with. For instance, The vector space $P_2(\mathbb{R}) = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\}$ of polynomials of degree at most 2 is isomorphic to $\mathbb{R}^3$; which one do you have more intuition about when it comes to vector space properties? – Nick Peterson Jul 12 '13 at 3:37

https://math.stackexchange.com/questions/441758/what-does-isomorphic-mean-in-linear-algebra

Revision: Coordinate System

THEOREM 8 Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis for a vector space V. Then the coordinate mapping $\mathbf{x} \mapsto [\,\mathbf{x}\,]_{\mathcal{B}}$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

PROOF Take two typical vectors in V, say,

$$\mathbf{u} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$$
$$\mathbf{w} = d_1\mathbf{b}_1 + \cdots + d_n\mathbf{b}_n$$

Then, using vector operations,

$$\mathbf{u} + \mathbf{w} = (c_1 + d_1)\mathbf{b}_1 + \cdots + (c_n + d_n)\mathbf{b}_n$$

It follows that

$$[\,\mathbf{u} + \mathbf{w}\,]_{\mathcal{B}} = \begin{bmatrix} c_1 + d_1 \\ \vdots \\ c_n + d_n \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} + \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} = [\,\mathbf{u}\,]_{\mathcal{B}} + [\,\mathbf{w}\,]_{\mathcal{B}}$$

So the coordinate mapping preserves addition. If r is any scalar, then

$$r\mathbf{u} = r(c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n) = (rc_1)\mathbf{b}_1 + \cdots + (rc_n)\mathbf{b}_n$$

So

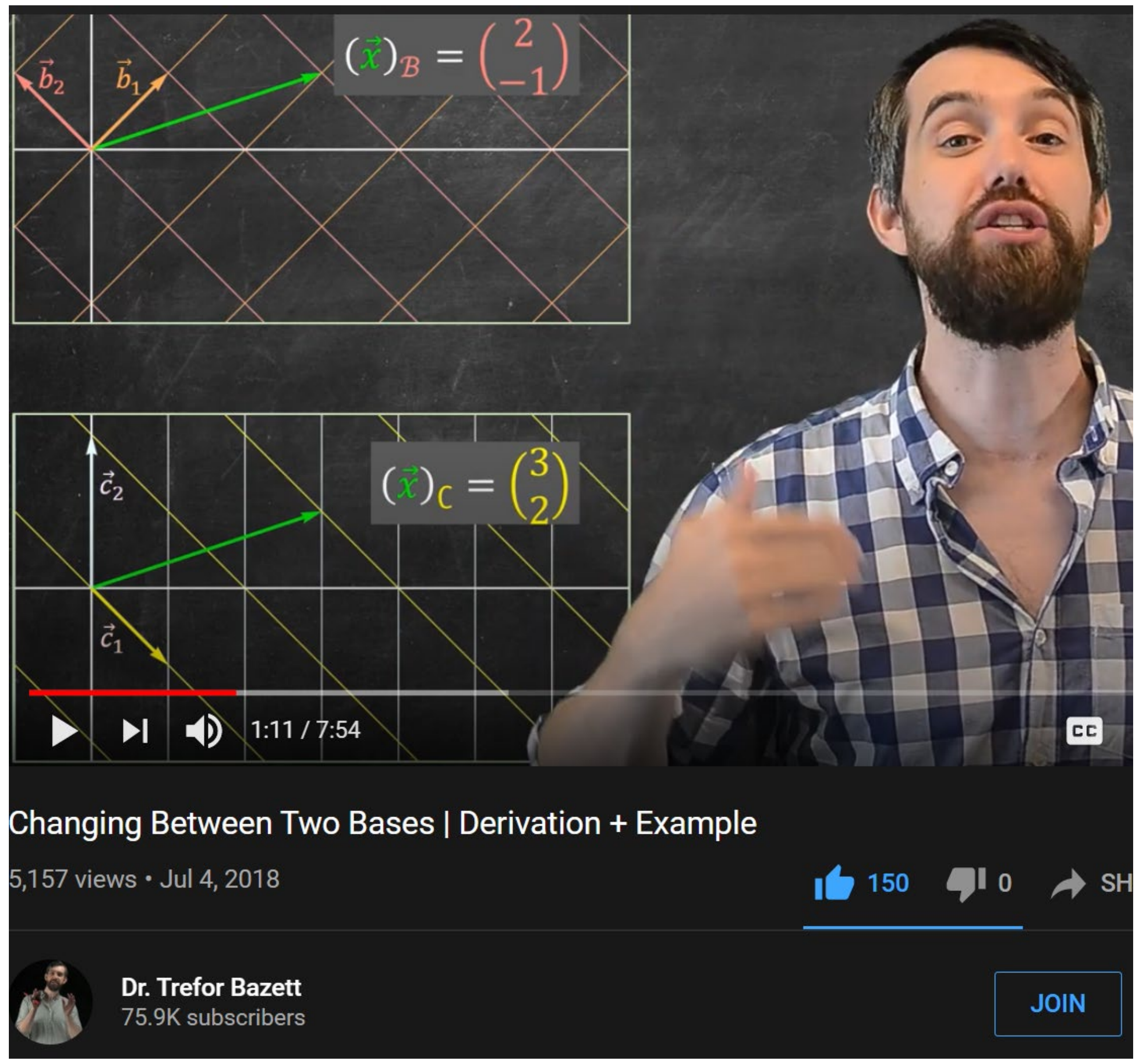
$$[\,r\mathbf{u}\,]_{\mathcal{B}} = \begin{bmatrix} rc_1 \\ \vdots \\ rc_n \end{bmatrix} = r\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = r[\,\mathbf{u}\,]_{\mathcal{B}}$$

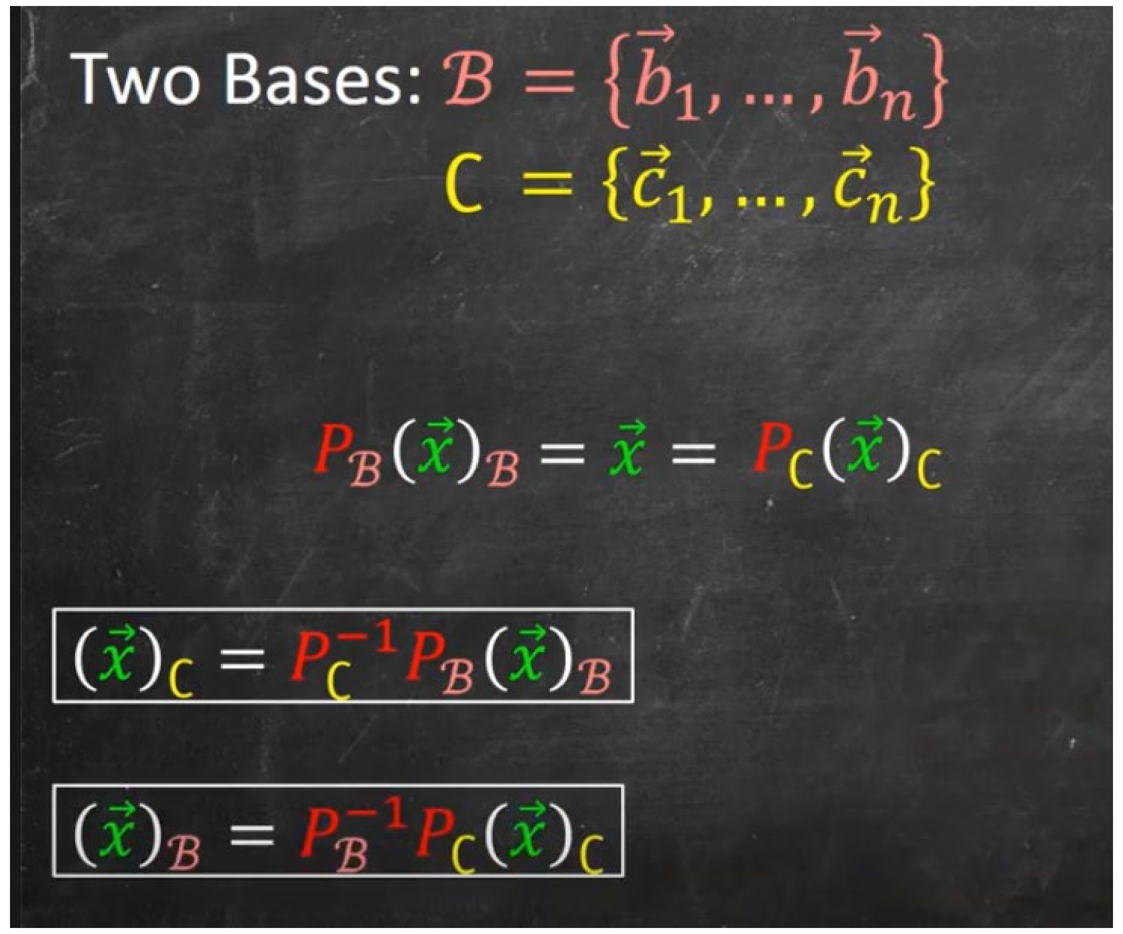
Thus the coordinate mapping also preserves scalar multiplication and hence is a linear transformation.

Lay,4th,pg219,ch4.3

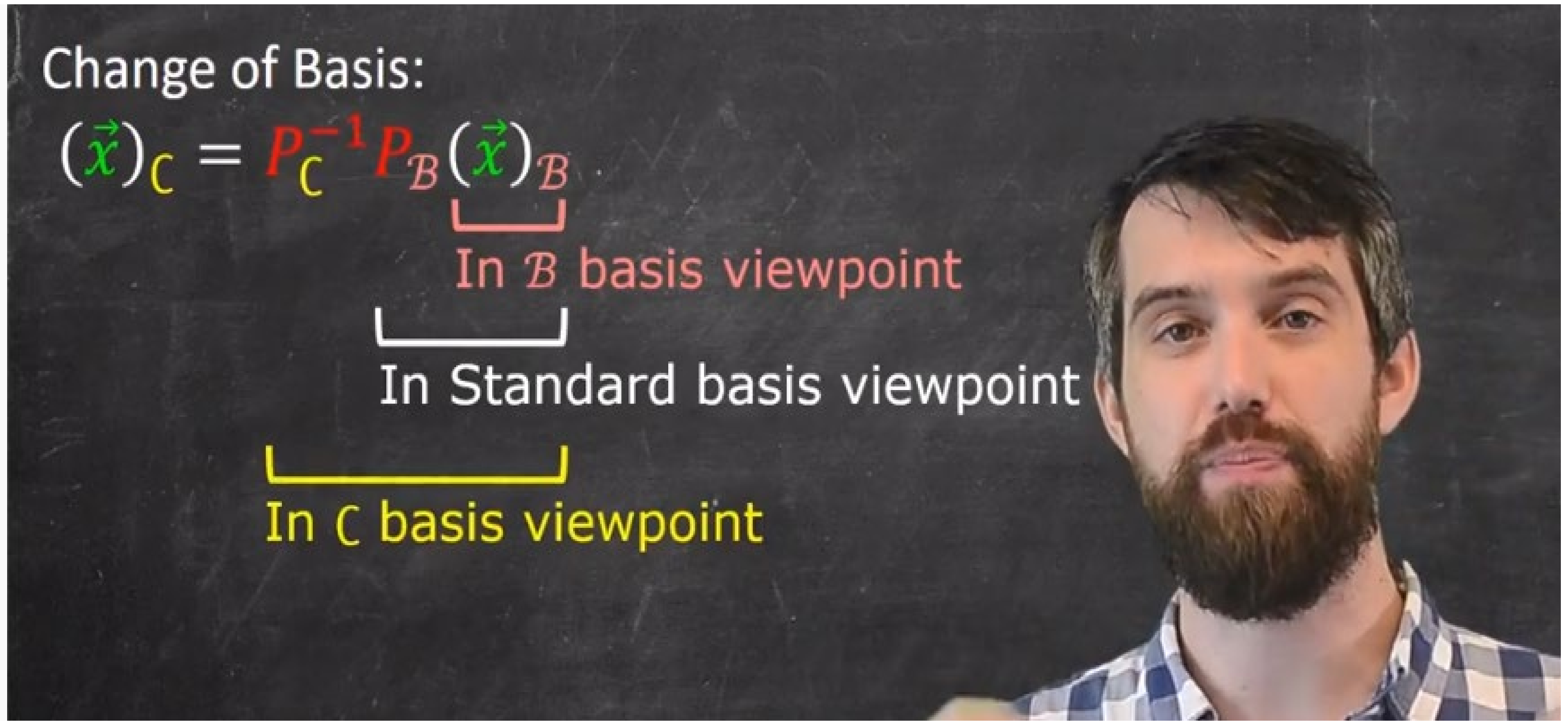
Ref: Trefor Bazett explaining Change of Basis

Trefor Bazett: Changing between 2 bases
https://www.youtube.com/watch?v=KjlTOLhaI9s





- Trefor Bazett:Visualizing Change Of Basis Dynamically
- https://www.youtube.com/watch?v=s4GC6zoULi0



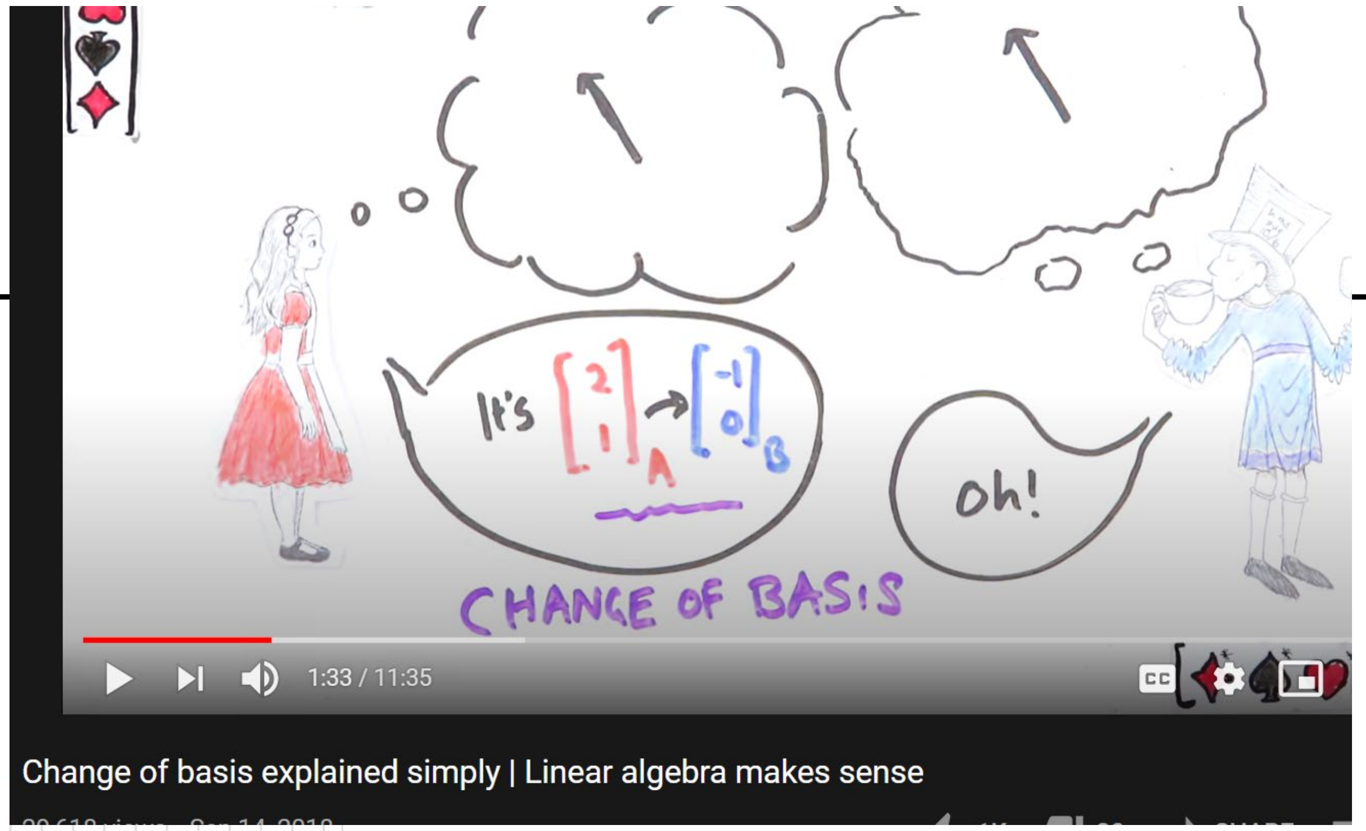
Ref: Other notable explanations of change of coordinate

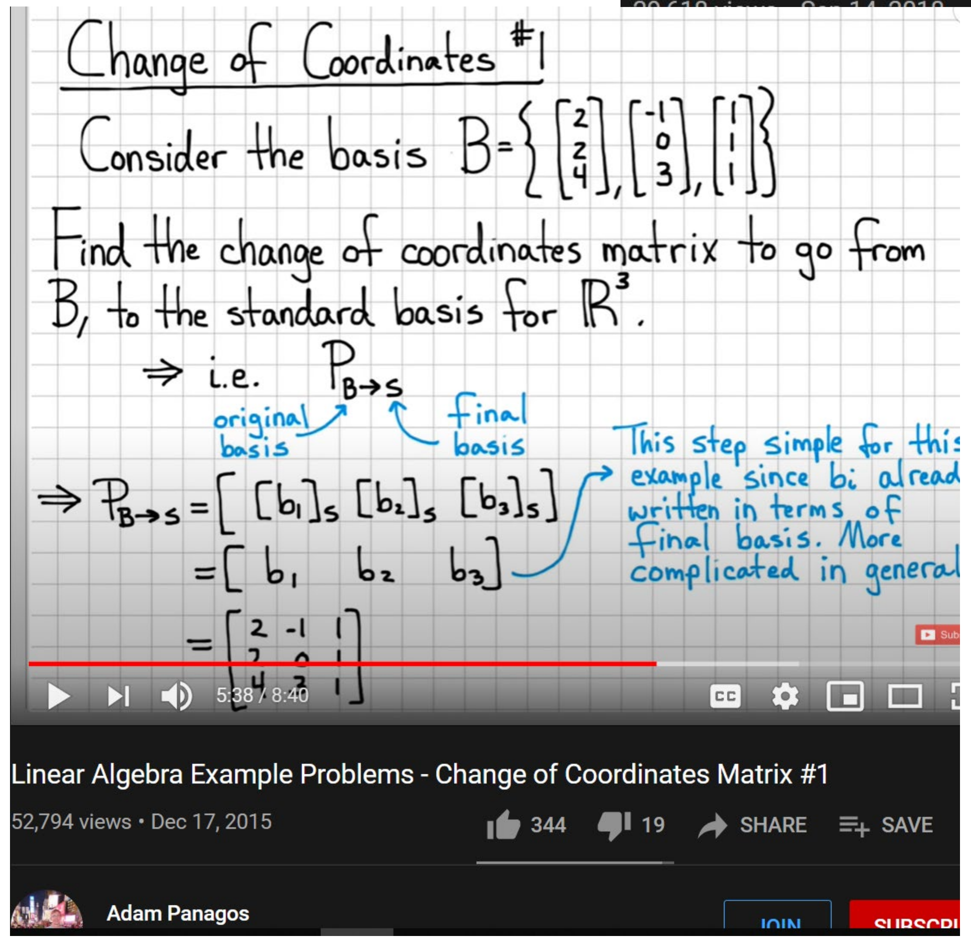
1) Looking Glass Universe (Change of Basis)
- https://youtu.be/Qp96zg5YZ_8

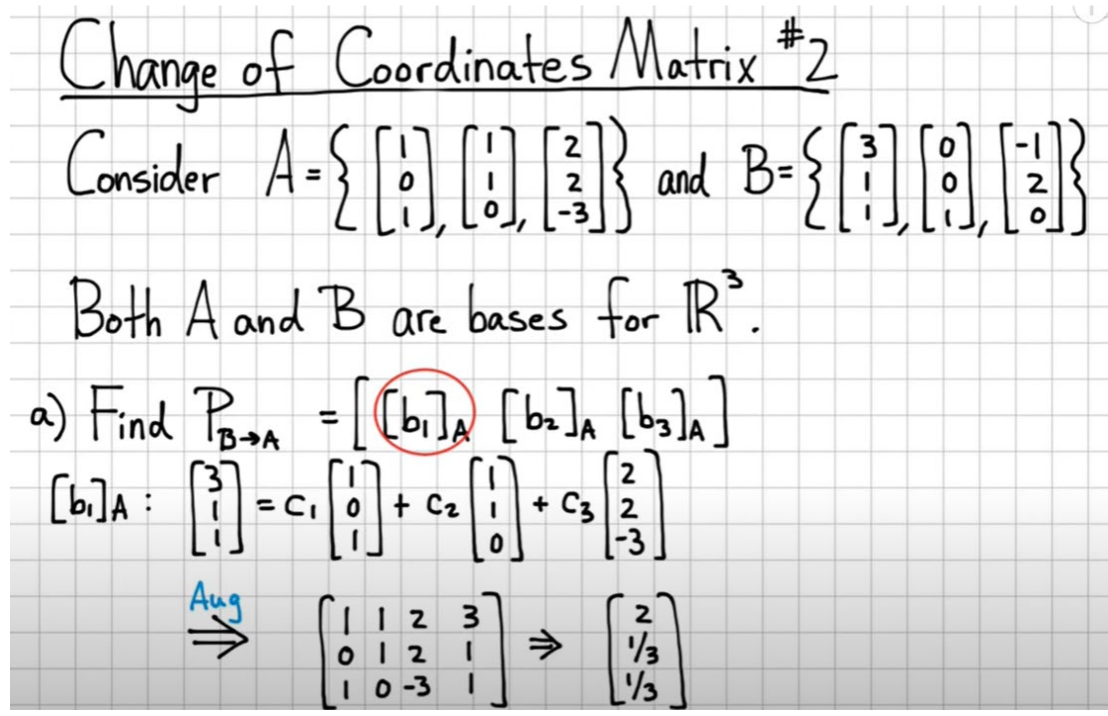
2) Adam Panagos – 2examples change of coordinates
- https://www.youtube.com/watch?v=VG4-8yW3Ce8
- https://youtu.be/2K6ipONMIgg

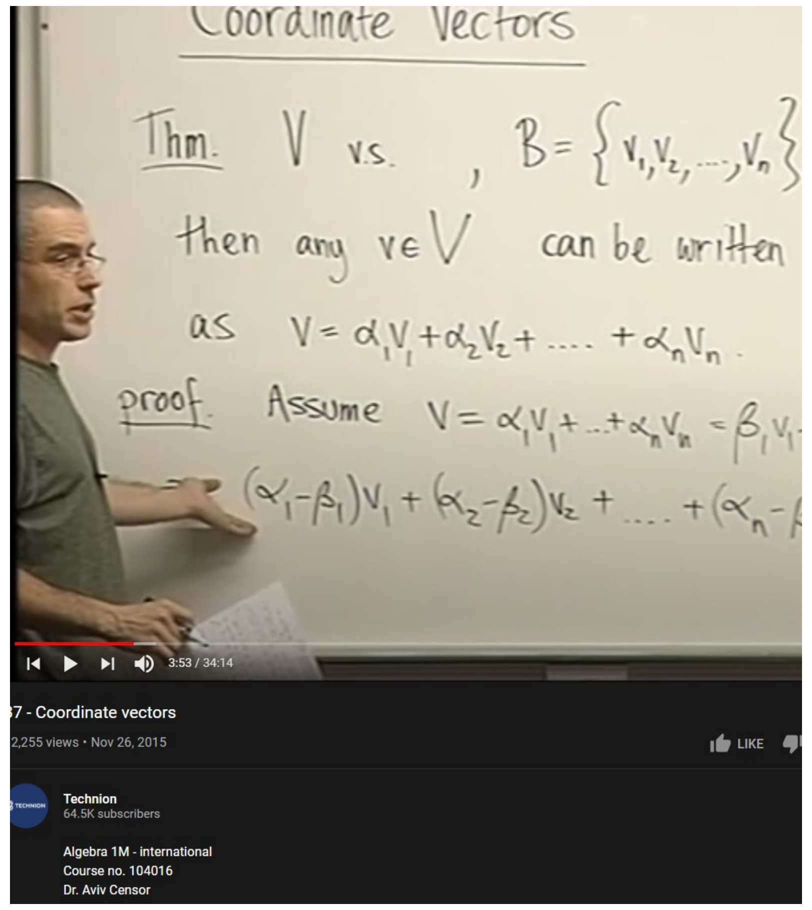
3) 3Blue1Brown: change of basis
- https://www.youtube.com/watch?v=P2LTAUO1TdA

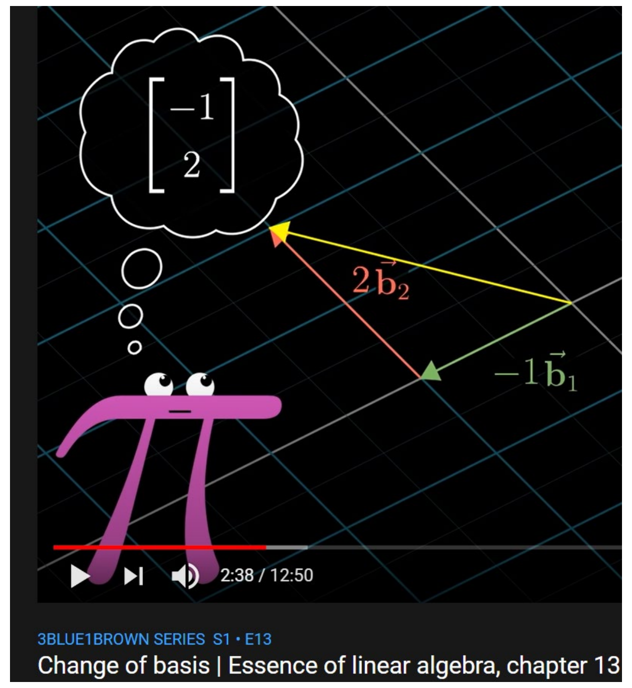
4) Technion, Dr Aviv Censor (lecture 37 – coordinate vectors)
https://youtu.be/XxRwKd9qPxw











Appendix: Isomorphism (not in syllabus)

1.3 Definition An *isomorphism* between two vector spaces V and W is a map $f\colon V \to W$ that

(1) is a correspondence: f is one-to-one and onto;*

(2) *preserves structure:* if $\vec{v}_1, \vec{v}_2 \in V$ then

$$f(\vec{v}_1 + \vec{v}_2) = f(\vec{v}_1) + f(\vec{v}_2)$$

and if $\vec{v} \in V$ and $r \in \mathbb{R}$ then

$$f(r\vec{v}) = r\, f(\vec{v})$$

(we write $V \cong W$, read "V is isomorphic to W", when such a map exists).

("Morphism" means map, so "isomorphism" means a map expressing sameness.)

1.2 Example Another two spaces we can think of as "the same" are $\mathcal{P}_2$, the space of quadratic polynomials, and $\mathbb{R}^3$. A natural correspondence is this.

$$a_0 + a_1x + a_2x^2 \quad\longleftrightarrow\quad \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} \qquad (\text{e.g., } 1 + 2x + 3x^2 \longleftrightarrow \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix})$$

The structure is preserved: corresponding elements add in a corresponding way

$$\begin{array}{r} a_0 + a_1x + a_2x^2 \\ +\; b_0 + b_1x + b_2x^2 \\ \hline (a_0+b_0) + (a_1+b_1)x + (a_2+b_2)x^2 \end{array} \quad\longleftrightarrow\quad \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_0+b_0 \\ a_1+b_1 \\ a_2+b_2 \end{pmatrix}$$

and scalar multiplication corresponds also.

$$r \cdot (a_0 + a_1x + a_2x^2) = (ra_0) + (ra_1)x + (ra_2)x^2 \quad\longleftrightarrow\quad r \cdot \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} ra_0 \\ ra_1 \\ ra_2 \end{pmatrix}$$

An isomorphism of vector spaces is a linear map, which has a two sided inverse **linear** map.

Ref:
1) Isa Jubran - https://web.cortland.edu/jubrani/272ch3.pdf
2) https://math.stackexchange.com/questions/2535063/if-t-is-invertible-prove-it-is-isomorphic
3) http://pi.math.cornell.edu/~andreim/Lec25.pdf

CX1104: **Linear Algebra for Computing**

Chap. No : **8.1.5B**

Lecture : **Linear Transformation and EigenVectors**

Concept : **Linear Transformation and Change of Basis using eigen Basis**

Note: Revise 8.1.5A first if unfamiliar with coordinate system and change of basis.

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m \times n}}_{A} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}}_{x} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m \times 1}}_{b}$$

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1) Overview: $Ax = y$, Least Squares vs Linear Transformation

1) **In Gaussian Elimination for system of equations and Least Squares**, the equation $Ax = y$ is presented.
   In this case, A and y represent the given equation values and target respectively
   - The problem there is to find x that minimize the error to approximate y.

2) **In Linear Transformation**, the problem also has the same equation $Ax = y$,
   in this case, the interpretation is different to (1). Here,
   - x is the input vector to be transformed to desired vector y through the linear transformation $\mathrm{y} = \mathrm{T}(x)$, evaluated as $\mathrm{y} = \mathrm{A}x$

   - In some machine learning problems, x represents the input, y the target and A the model. The machine learning algorithm strive to find the best A.
   - Here, the ML community do not use linear models A, but nonlinear models.
   - and the number of examples (x_i, y_i) typically number into the millions.

1.1) Linear Transformation $y = T(x) = Ax$

Given input vector $x \in R^n$, with coordinates in the standard basis and linear transformation $T: R^n \to R^m$ represented by matrix A, then $y = T(x) = Ax \in R^m$,
The columns of A are:

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix}$$

and $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n$ *denote the standard basis vectors for* $\mathbb{R}^n$.
This A *is called the* **matrix of** T.

$$\begin{tikzcd}
\text{abstract} & x \arrow[r, "T"] \arrow[d] & y = T(x) \arrow[d] \\
\text{Std basis} & x \in R^n \arrow[r, "A"] & y = Ax \in R^m
\end{tikzcd}$$

$$A \in R^{mxn}$$

Fig: representation of a linear transformation

Ref:

1) https://math.libretexts.org/Bookshelves/Linear_Algebra/Book%3A_A_First_Course_in_Linear_Algebra_(Kuttler)/05%3A_Linear_Transformations (Ch5.1, 5.2)

1.2) Proof that $T(x)$ can be represented as a matrix product Ax

Theorem. Consider a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$. The transformation T can be represented as a matrix product $\mathbf{x} \mapsto A\mathbf{x}$, for some matrix $A \in \mathbb{R}^{n\times n}$.

Proof. Consider a matrix $\mathbf{x} \in \mathbb{R}^n$ given by

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

We will construct a matrix $A \in \mathbb{R}^{n\times n}$ such that $T(\mathbf{x}) = A\mathbf{x}$.

The vector $\mathbf{x}$ can also be written as

$$\begin{aligned} \mathbf{x} &= x_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \cdots + x_n \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \\ &= x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n \\ &= \sum_{i=1}^{n} x_i\mathbf{e}_i, \end{aligned}$$

where $\mathbf{e}_i$ are the standard basis vectors in $\mathbb{R}^n$.

Consider the transformation $T(\mathbf{x})$. Rewriting $\mathbf{x}$ as above, we have

$$\begin{aligned} T(\mathbf{x}) &= T\left(\sum_{i=1}^{n} x_i\mathbf{e}_i\right) \\ &= \sum_{i=1}^{n} T(x_i\mathbf{e}_i) \\ T(\mathbf{x}) &= \sum_{i=1}^{n} x_i T(\mathbf{e}_i). \qquad (1) \end{aligned}$$

Let the matrix $A \in \mathbb{R}^{n\times n}$ be defined by

$$\begin{aligned} A &= [\, T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n) \quad] \\ &= \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}, \end{aligned}$$

where each $T(\mathbf{e}_i)$ is a column of A, and each $a_{ij} = T(\mathbf{e}_i) \cdot \mathbf{e}_j$ is the jth component of $T(\mathbf{e}_i)$. Then, by the definition of matrix-vector multiplication, we have

$$\begin{aligned} A\mathbf{x} &= \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= \begin{bmatrix} x_1 a_{11} + \cdots + x_n a_{1n} \\ \vdots \\ x_1 a_{n1} + \cdots + x_n a_{nn} \end{bmatrix} \\ &= x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{n1} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{n1} \\ \vdots \\ a_{nn} \end{bmatrix} \\ &= x_1 T(\mathbf{e}_1) + \cdots + x_n T(\mathbf{e}_n) \\ A\mathbf{x} &= \sum_{i=1}^{n} x_i T(\mathbf{e}_i). \qquad (2) \end{aligned}$$

Therefore, by eqs. (1) and (2), we have that

$$T(\mathbf{x}) = \sum_{i=1}^{n} x_i T(\mathbf{e}_i) \qquad A\mathbf{x} = \sum_{i=1}^{n} x_i T(\mathbf{e}_i),$$

and we reach $T(\mathbf{x}) = A\mathbf{x}$, as was to be shown.

The crux of the proof is that using the standard basis and by linearity, $\mathbf{x} = \sum \mathbf{e}_i x_i \implies T(\mathbf{x}) = \sum T(\mathbf{e}_i) x_i = \sum \mathbf{a}_i x_i$ where the $\mathbf{a}_i$ can be arranged as the columns of the matrix. – user65203

Ref:
1) https://math.stackexchange.com/questions/916192/proving-any-linear-transformation-can-be-represented-as-a-matrix

2.1) If A is a dense matrix, computing Ax can sometimes be simplified by converting A and x into another basis.

$$
\begin{tikzcd}
\text{abstract} & x \arrow[r, "T(x)"] \arrow[d] & y = T(x) \arrow[d] \\
\text{Std basis} & x \in R^n \arrow[r, "A"] \arrow[d, "P^{-1}"] & \begin{array}{l} y = Ax = P[y]_B \\ \quad = PDP^{-1}x \end{array} \\
\text{B-basis} & {[x]_B = P^{-1}x} \arrow[r, "D"] & {[y]_B = D[x]_B} \arrow[u, "P"']
\end{tikzcd}
$$

Motivation: why change basis?
We wish to perform change of basis on x to another basis $[x]_B$.
In basis B, the transformation A will be a very sparse diagonal matrix D, hence calculating $\mathrm{A}x$ is cheap.

P is a matrix with columns containing a basis,

$$P = \{b_1, b_2, \dots, b_n\}$$

How is standard basis related to B-basis:

$$x = P[x]_B$$
$$P^{-1}x = [x]_B$$

$$y = \mathrm{T(x)}$$
$$y = Ax$$
$$= P(D\ \underbrace{(P^{-1}x)}\ )$$

Changing from B-basis to Std basis

Transformation T in B-basis

Changing from Std to B-basis, this is $[x]_B$

2.2) Linear Transformation from V into V

THEOREM 8

Diagonal Matrix Representation

Suppose $A = PDP^{-1}$, where D is a diagonal $n \times n$ matrix. If $\mathcal{B}$ is the basis for $\mathbb{R}^n$ formed from the columns of P, then D is the $\mathcal{B}$-matrix for the transformation $\mathbf{x} \mapsto A\mathbf{x}$.

Lay,4th,pg291,ch4.4

Note: P contains the set of eigenvectors of A,
and for P^{-1} to exist, it means that P contains n number of independent eigenvectors, and
D is a diagonal matrix containing the eigenvalues.
Not all matrix have this characteristics. When matrix have this characteristic, we say these matrixes are diagonalizable.

See Tut8, Q7A youTube: https://youtu.be/zaSqGGmNokw

2.3) D is the transformation matrix in B basis.

See pg 7 of slide 8.1.5A

$$\boxed{\mathbf{x} = P_{\mathcal{B}}[\,\mathbf{x}\,]_{\mathcal{B}}} \qquad \boxed{P_{\mathcal{B}}^{-1}\mathbf{x} = [\,\mathbf{x}\,]_{\mathcal{B}}} \qquad (4)$$

We call $P_{\mathcal{B}}$ the **change-of-coordinates matrix** from $\mathcal{B}$ to the standard basis in $\mathbb{R}^n$.

$$y = \ Ax$$

if A can be decomposed into $A \ = PDP^{-1}$, therefore

$$y \ = PDP^{-1}x,$$

We can view $P^{-1}x$ as changing coordinate to B, i.e $[x]_B$

$$\begin{aligned} y \ &= P\,D(P^{-1}x) \\ &= P\,D[x]_B \\ &= P\,[y]_B \end{aligned}$$

We can view $D[x]_B$ as transformation matrix D applied on $[x]_B$, $(Theorem\ 8)$
i.e D is the transformation matrix in B basis,
and $[y]_B = D[x]_B$ (y in the B-basis). Lastly

$$\text{y} = \ P\,[y]_B$$

can be viewed as converting y in B-basis back to standard basis.

2.4) Linear Transformation wrt B-basis pictorially

$$\begin{array}{ccc} x & \xrightarrow{A = PDP^{-1}} & y \;= Ax = PDP^{-1}x \\ P^{-1}\big\downarrow \big\uparrow P & & P^{-1}\big\downarrow \big\uparrow P \\ [x]_B & \xrightarrow{[T]_B = D} & [y]_B = [Ax]_B = [T]_B\,[x]_B \end{array}$$

$$x = P[x]_B$$
$$P^{-1}x = [x]_B$$

$$y = P[y]_B$$
$$P^{-1}y = [y]_B$$

8

Lets start from y:

$$y = P\ [y]_B$$
$$y = P\ D\ [x]_B$$
$$y = P\ D\ P^{-1}x$$

Therefore

$$A = PDP^{-1}$$
$$P^{-1}AP = D$$

Lay,4th,pg291,ch4.4

THEOREM 8

Diagonal Matrix Representation

Suppose $A = PDP^{-1}$, where D is a diagonal $n \times n$ matrix. If $\mathcal{B}$ is the basis for $\mathbb{R}^n$ formed from the columns of P, then D is the $\mathcal{B}$-matrix for the transformation $\mathbf{x} \mapsto A\mathbf{x}$.

PROOF Denote the columns of P by $\mathbf{b}_1, \dots, \mathbf{b}_n$, so that $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $P = [\,\mathbf{b}_1 \ \cdots \ \mathbf{b}_n\,]$. In this case, P is the change-of-coordinates matrix $P_{\mathcal{B}}$ discussed in Section 4.4, where

$$P[\mathbf{x}]_{\mathcal{B}} = \mathbf{x} \quad \text{and} \quad [\mathbf{x}]_{\mathcal{B}} = P^{-1}\mathbf{x}$$

If $T(\mathbf{x}) = A\mathbf{x}$ for $\mathbf{x}$ in $\mathbb{R}^n$, then

$$\begin{aligned} [T]_{\mathcal{B}} &= [\,[T(\mathbf{b}_1)]_{\mathcal{B}} \ \cdots \ [T(\mathbf{b}_n)]_{\mathcal{B}}\,] && \text{Definition of } [T]_{\mathcal{B}} \\ &= [\,[A\mathbf{b}_1]_{\mathcal{B}} \ \cdots \ [A\mathbf{b}_n]_{\mathcal{B}}\,] && \text{Since } T(\mathbf{x}) = A\mathbf{x} \\ &= [\,P^{-1}A\mathbf{b}_1 \ \cdots \ P^{-1}A\mathbf{b}_n\,] && \text{Change of coordinates} \\ &= P^{-1}A[\,\mathbf{b}_1 \ \cdots \ \mathbf{b}_n\,] && \text{Matrix multiplication} \\ &= P^{-1}AP && (6) \end{aligned}$$

Since $A = PDP^{-1}$, we have $[T]_{\mathcal{B}} = P^{-1}AP = D$. ■

2.5) Important point: if we are asked to find x[k] , for k very large $x[n+1] = Ax[n]$ and A is diagonalizable, it is computationally efficient if we convert $x \rightarrow [x]_B$ to work on the problem!

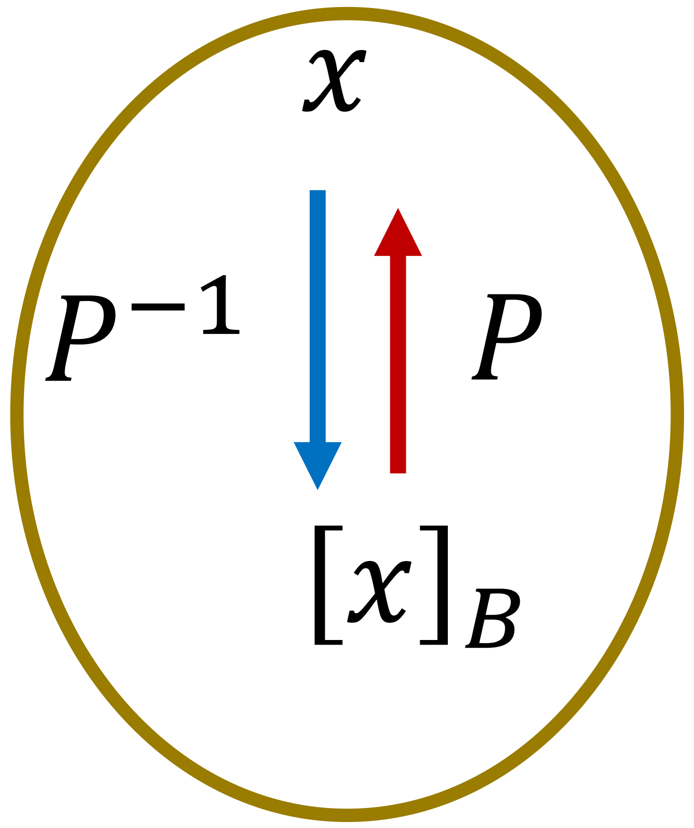
Given A and $x[0]$, and $x[n+1] = Ax[n]$,
we convert problem to B-basis (eigenvectors of A if exists) since its computationally cheaper.

Proof:
$x[1] = Ax[0]$
$x[2] = Ax[1] = A(Ax[0]) = A^2x[0]$
Therefore, $x[k] = A^k x[0]$

if $A = PDP^{-1}$,
then $A^2 = (PDP^{-1})(PDP^{-1}) \quad = PD^2P^{-1}$ and
$A^3 \;= (PDP^{-1})(PD^2P^{-1}) = PD^3P^{-1}$
Therefore $A^k = PD^kP^{-1}$

$\Rightarrow x[k] = A^k x[0] = PD^kP^{-1}x[0]$
$\Rightarrow \; x[k] = P\,D^k\;\, x[0]_B$

Note:
Computing A^k is expensive if A is a dense matrix.
Computing D^k is cheap since it is a diagonal matrix.



Change of Basis

$$x \quad = P[x]_B$$
$$P^{-1}x = \quad [x]_B$$

3) Lay's notes: General Equation of Matrix of Linear Transformation T: $R^n \rightarrow R^m$

The Matrix of a Linear Transformation

Let V be an n-dimensional vector space, let W be an m-dimensional vector space, and let T be any linear transformation from V to W. To associate a matrix with T, choose (ordered) bases $\mathcal{B}$ and $\mathcal{C}$ for V and W, respectively.

Given any $\mathbf{x}$ in V, the coordinate vector $[\,\mathbf{x}\,]_{\mathcal{B}}$ is in $\mathbb{R}^n$ and the coordinate vector of its image, $[\,T(\mathbf{x})\,]_{\mathcal{C}}$, is in $\mathbb{R}^m$, as shown in Figure 1.

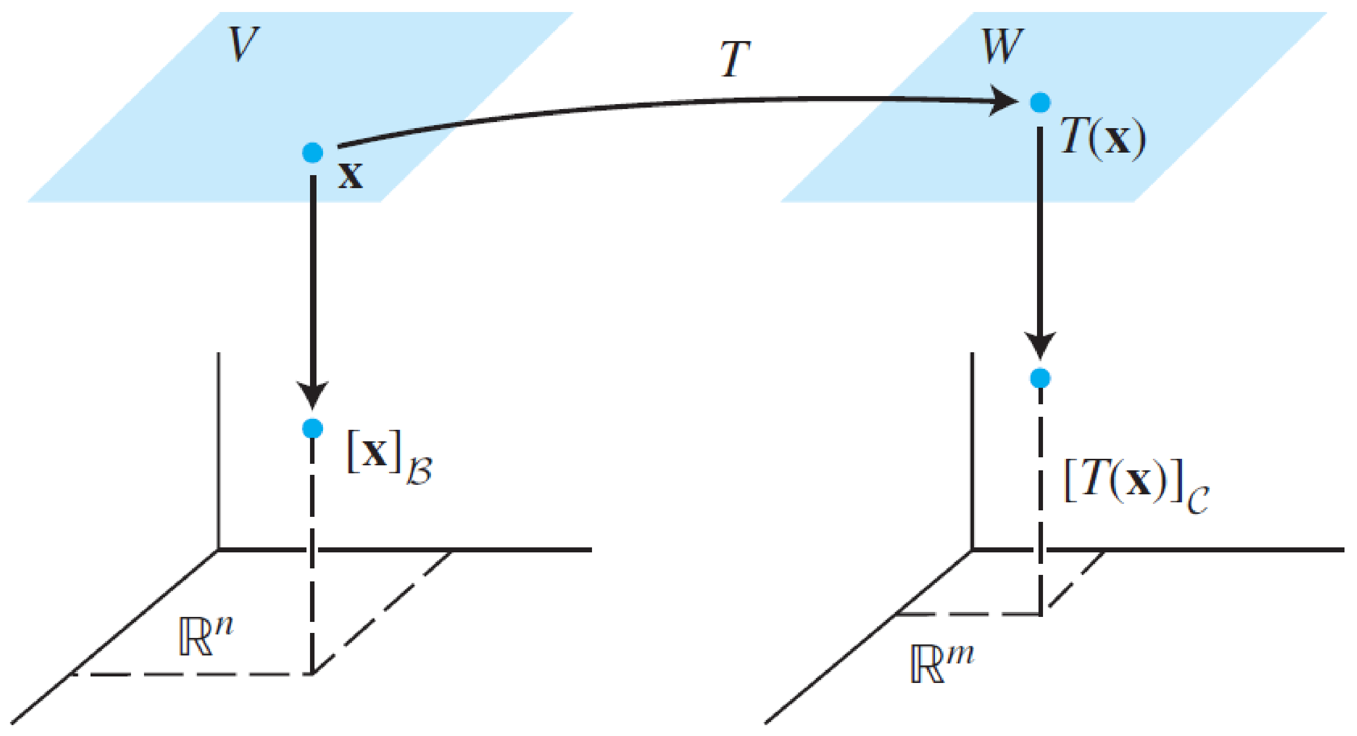


FIGURE 1 A linear transformation from V to W.

$$\begin{tikzcd}
\mathbf{x} \arrow[r, "T"] \arrow[d] & T(\mathbf{x}) \arrow[d] \\
[\mathbf{x}]_{\mathcal{B}} \arrow[r, "\text{Multiplication}", "\text{by } M"'] & [T(\mathbf{x})]_{\mathcal{C}}
\end{tikzcd}$$

FIGURE 2

Lay 5e, pg 290, Ch5.4

The connection between $[\,\mathbf{x}\,]_{\mathcal{B}}$ and $[\,T(\mathbf{x})\,]_{\mathcal{C}}$ is easy to find. Let $\{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be the basis $\mathcal{B}$ for V. If $\mathbf{x} = r_1\mathbf{b}_1 + \cdots + r_n\mathbf{b}_n$, then

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} r_1 \\ \vdots \\ r_n \end{bmatrix}$$

and

$$T(\mathbf{x}) = T(r_1\mathbf{b}_1 + \cdots + r_n\mathbf{b}_n) = r_1T(\mathbf{b}_1) + \cdots + r_nT(\mathbf{b}_n) \tag{1}$$

because T is linear. Now, since the coordinate mapping from W to $\mathbb{R}^m$ is linear (Theorem 8 in Section 4.4), equation (1) leads to

$$[\,T(\mathbf{x})\,]_{\mathcal{C}} = r_1[\,T(\mathbf{b}_1)\,]_{\mathcal{C}} + \cdots + r_n[\,T(\mathbf{b}_n)\,]_{\mathcal{C}} \tag{2}$$

Since $\mathcal{C}$-coordinate vectors are in $\mathbb{R}^m$, the vector equation (2) can be written as a matrix equation, namely,

$$[\,T(\mathbf{x})\,]_{\mathcal{C}} = M[\,\mathbf{x}\,]_{\mathcal{B}} \tag{3}$$

where

$$M = \big[\,[\,T(\mathbf{b}_1)\,]_{\mathcal{C}} \quad [\,T(\mathbf{b}_2)\,]_{\mathcal{C}} \quad \cdots \quad [\,T(\mathbf{b}_n)\,]_{\mathcal{C}}\,\big] \tag{4}$$

The matrix M is a matrix representation of T, called the **matrix for T relative to the bases $\mathcal{B}$ and $\mathcal{C}$**. See Figure 2.

Equation (3) says that, so far as coordinate vectors are concerned, the action of T on $\mathbf{x}$ may be viewed as left-multiplication by M.

3.1) Example: Linear Transformation

EXAMPLE 1 Suppose $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ is a basis for V and $\mathcal{C} = \{\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3\}$ is a basis for W. Let $T : V \to W$ be a linear transformation with the property that

$$T(\mathbf{b}_1) = 3\mathbf{c}_1 - 2\mathbf{c}_2 + 5\mathbf{c}_3 \quad \text{and} \quad T(\mathbf{b}_2) = 4\mathbf{c}_1 + 7\mathbf{c}_2 - \mathbf{c}_3$$

Find the matrix M for T relative to $\mathcal{B}$ and $\mathcal{C}$.

SOLUTION The $\mathcal{C}$-coordinate vectors of the *images* of $\mathbf{b}_1$ and $\mathbf{b}_2$ are

$$[\,T(\mathbf{b}_1)\,]_{\mathcal{C}} = \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix} \quad \text{and} \quad [\,T(\mathbf{b}_2)\,]_{\mathcal{C}} = \begin{bmatrix} 4 \\ 7 \\ -1 \end{bmatrix}$$

Hence

$$M = \begin{bmatrix} 3 & 4 \\ -2 & 7 \\ 5 & -1 \end{bmatrix}$$

If $\mathcal{B}$ and $\mathcal{C}$ are bases for the same space V and if T is the identity transformation $T(\mathbf{x}) = \mathbf{x}$ for $\mathbf{x}$ in V, then matrix M in (4) is just a change-of-coordinates matrix (see Section 4.7).

Lay 5e, pg 291, Ch5.4

Revision:

Let $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ be bases of a vector space V. Then there is a unique $n \times n$ matrix $\underset{\mathcal{C}\leftarrow\mathcal{B}}{P}$ such that

$$[\,\mathbf{x}\,]_{\mathcal{C}} = \underset{\mathcal{C}\leftarrow\mathcal{B}}{P}[\,\mathbf{x}\,]_{\mathcal{B}} \tag{4}$$

The columns of $\underset{\mathcal{C}\leftarrow\mathcal{B}}{P}$ are the $\mathcal{C}$-coordinate vectors of the vectors in the basis $\mathcal{B}$. That is,

$$\underset{\mathcal{C}\leftarrow\mathcal{B}}{P} = \begin{bmatrix} [\mathbf{b}_1]_{\mathcal{C}} & [\mathbf{b}_2]_{\mathcal{C}} & \cdots & [\mathbf{b}_n]_{\mathcal{C}} \end{bmatrix} \tag{5}$$

The matrix $\underset{\mathcal{C}\leftarrow\mathcal{B}}{P}$ in Theorem 15 is called the **change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$**. Multiplication by $\underset{\mathcal{C}\leftarrow\mathcal{B}}{P}$ converts $\mathcal{B}$-coordinates into $\mathcal{C}$-coordinates.[2] Figure 2 illustrates the change-of-coordinates equation (4).

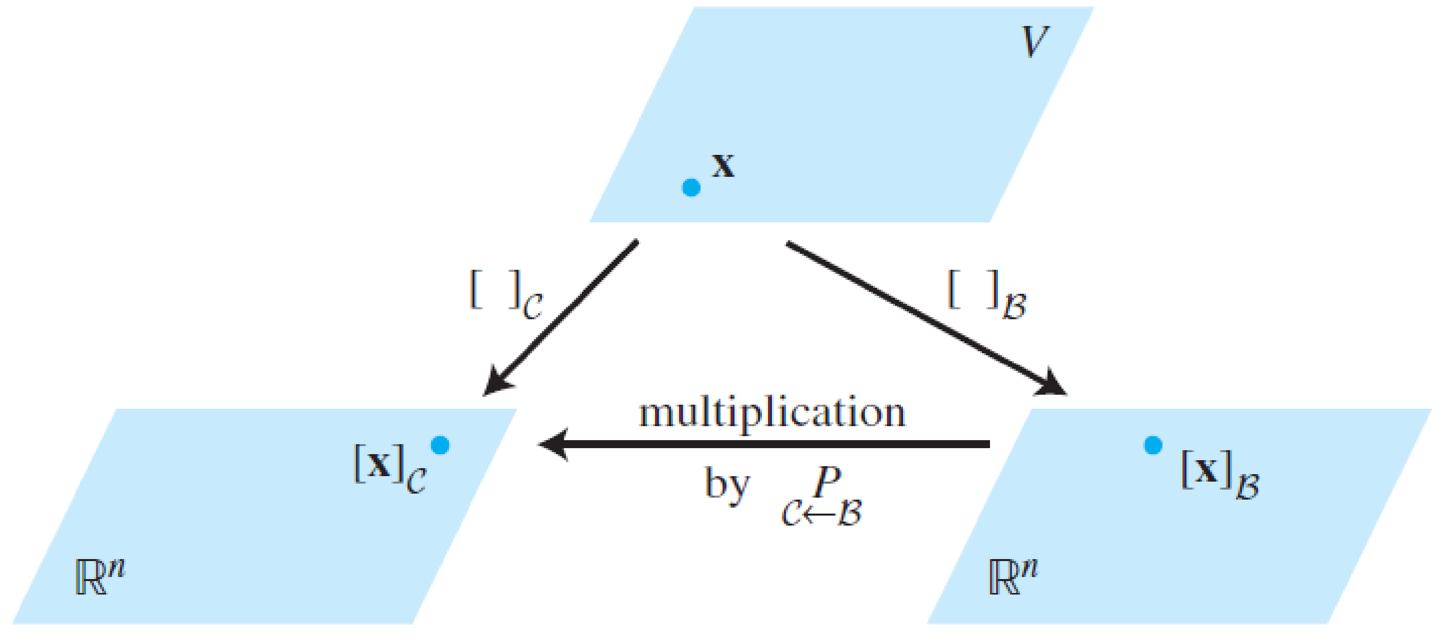


FIGURE 2 Two coordinate systems for V.

Lay 5e, pg 242, Ch 4.7

Appendix: Some useful information

References:

1)Youtube:

a)3Blue1Brown: Linear Transformation (Ch3) https://www.youtube.com/watch?v=kYB8IZa5AuE

b) Strang Lect 30, 18.06 (2005) https://www.youtube.com/watch?v=Ts3o2I8_Mxc

c) Adams Panagos: "Finding A" https://www.youtube.com/watch?v=61knWwBM3gQ

d) Technion: L54 Matrix Representation of Linear Map : https://www.youtube.com/watch?v=tRbXrnoVJI8

2) Problems in Yutsumura.com:

a) https://yutsumura.com/find-matrix-representation-of-linear-transformation-from-r2-to-r2/

b) https://yutsumura.com/linear-transformation-tr2-to-r2-given-in-figure/

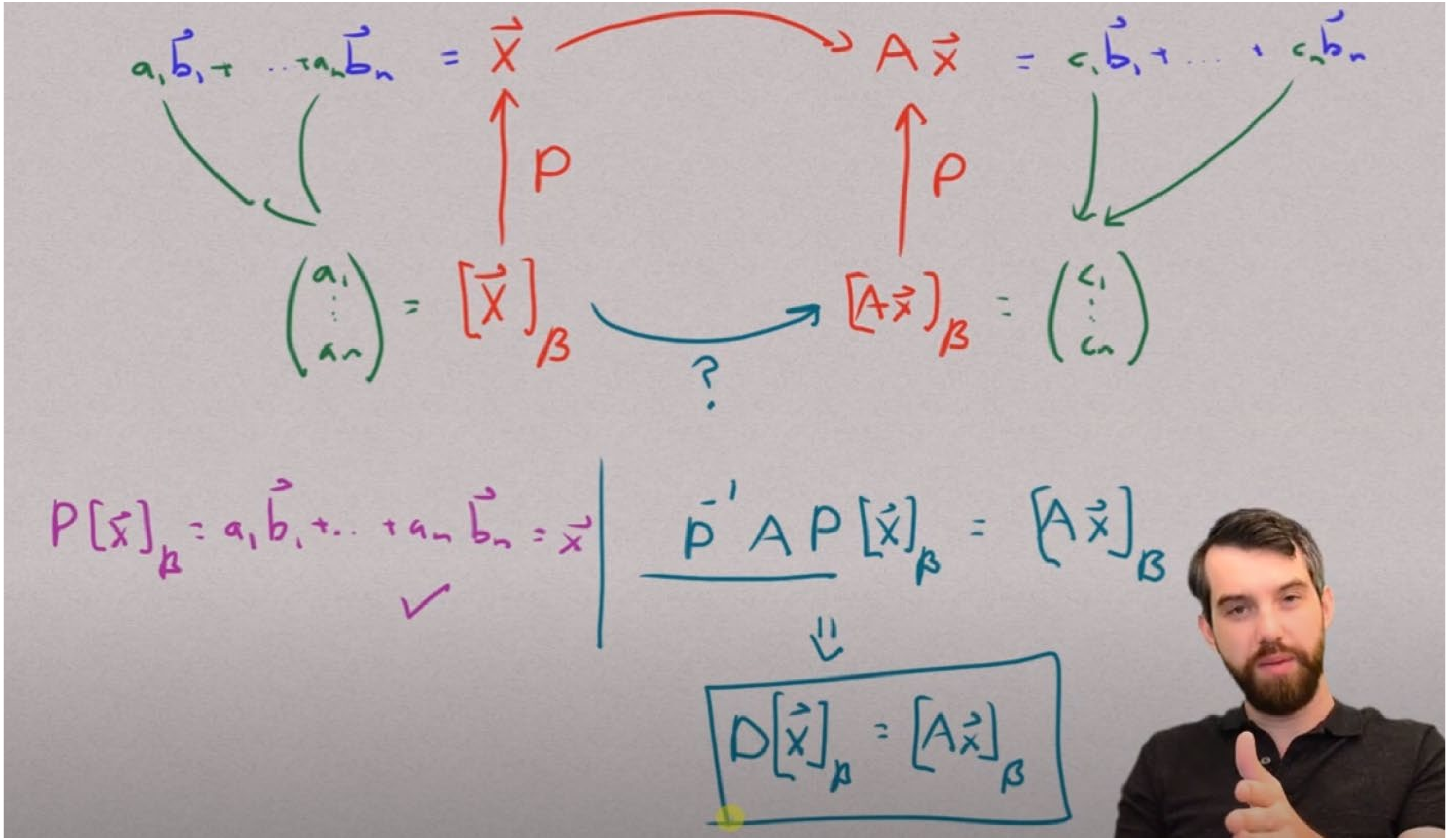
b) https://yutsumura.com/find-a-general-formula-of-a-linear-transformation-from-r2-to-r3/#more-2526

3) Reference from others:

a) Upenn: https://www2.math.upenn.edu/~moose/240S2013/slides7-23.pdf

b) Abbasi notes on deriving the transformation matrix: https://www.12000.org/my_notes/similarity_transformation_and_SVD/ind

Trefor Bazett: The Similarity Relationship Represents a Change of Basis
Youtube: https://www.youtube.com/watch?v=s4c5LQ5a4ek



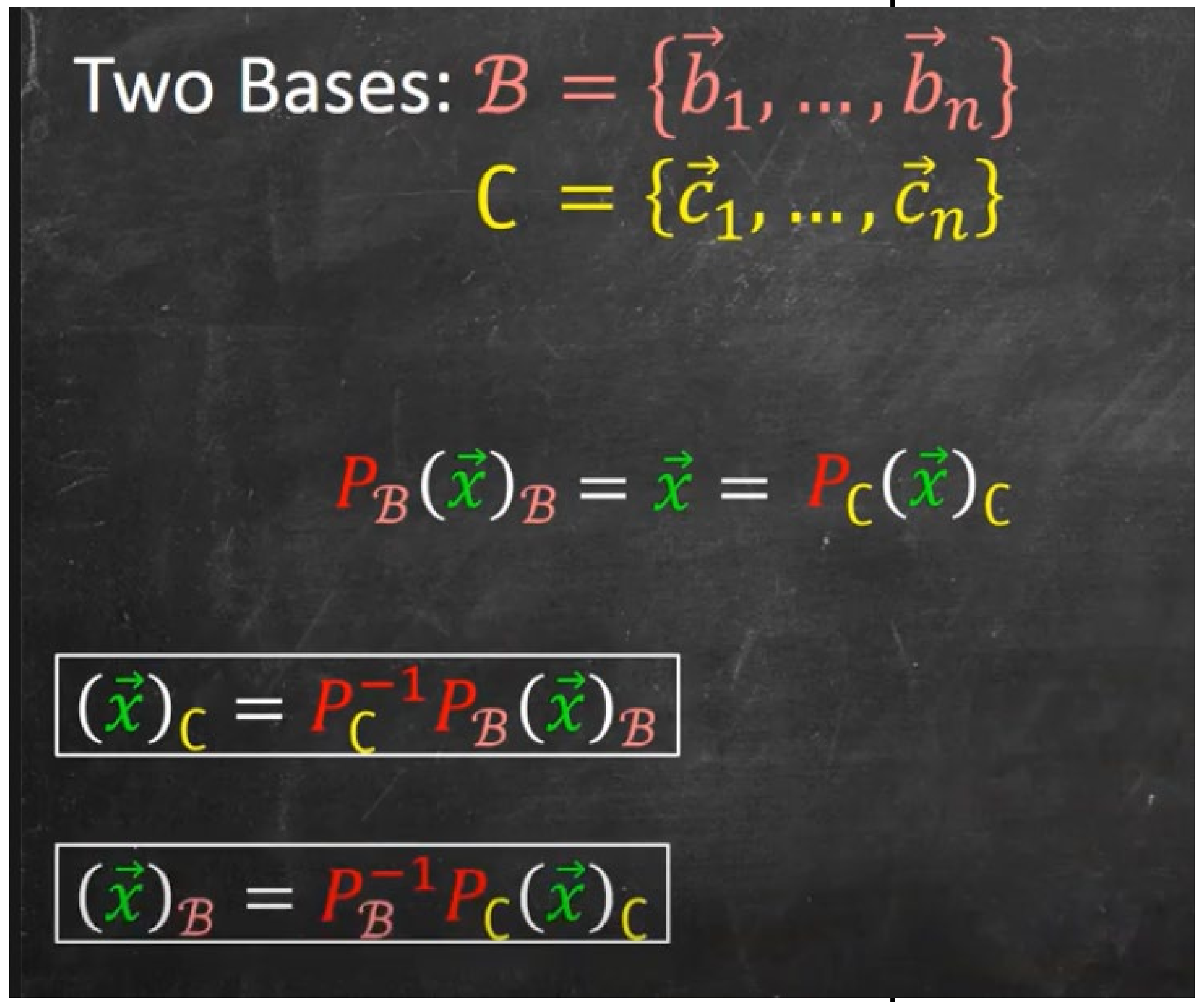
Change of Basis from Trefor Bazett

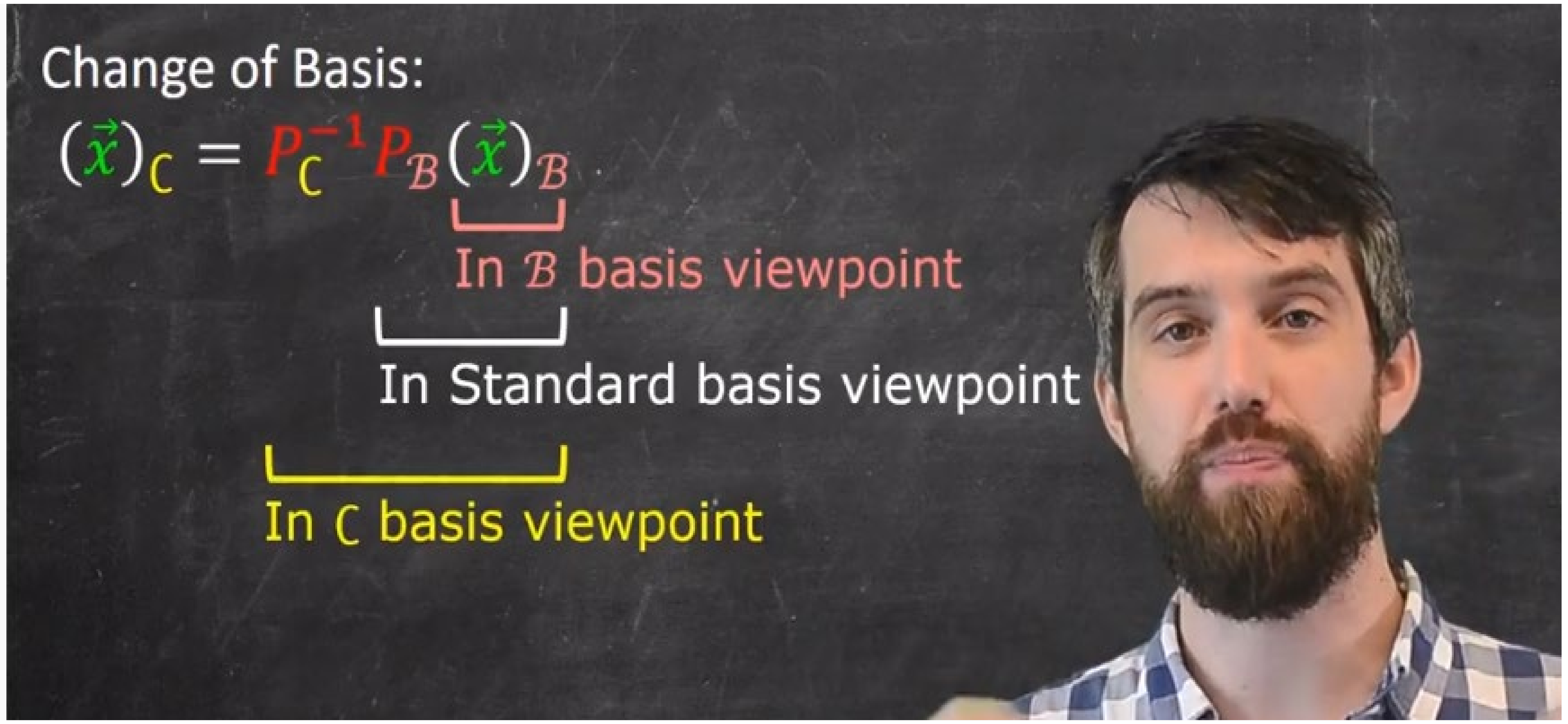
$$\begin{tikzcd}[column sep=huge]
\mathbf{x} \arrow[r, "T"] \arrow[d] & T(\mathbf{x}) \arrow[d] \\
{[\mathbf{x}]_{\mathcal{B}}} \arrow[r, "\text{Multiplication}", "\text{by } M"'] & {[T(\mathbf{x})]_{\mathcal{C}}}
\end{tikzcd}$$

FIGURE 2

Trefor Bazett: Changing between 2 bases
https://www.youtube.com/watch?v=KjlTOLhaI9s

Remember: $x = P_B[x]_B = P_C\ [x]_C$
The vector x represented in basis B and C





Lay4th, sec5.4,pg289

2021/ 5 NOV 2021.

Topic : Change of Basis

LA/ Ch 8 week 12/13.

and Representation of Linear Transformation in different basis.

References :

A) Interpreting x, $[x]_B$, $[x]_C$ and their relationship.

a) Lay 5e, pg 241.

B) $T(x)$ linear transformation of Ax ($R^n \to R^n$) and its representation in another basis.

b) pg 292 ~293 Lay 5e.

c) $T(x)$ $R^n \to R^m$ linear transformation in different basis.

c) Lay 5e, pg 291. Matrix for T relative to the bases B and C.

2A

A) Interpreting x, $[x]_B$, $[x]_C$ and their relationship.

a) A basis (set of independent vectors that span a vector space) for vector space V allow us to represent $x \in V$ with a coordinate system, and it will make V act like R^n.

b) different basis allow us to view V differently.

c) Given a basis $B = \{\underline{b}_1, \underline{b}_2, \dots, \underline{b}_n\}$ for V, then for each $\underline{x} \in V$, there exist a unique set of scaler $c_1, c_2, \dots c_n$, such that

$$\underset{\in V}{x} \rightarrow [x]_{std} = \beta_1 \underline{b}_1 + \beta_2 \underline{b}_2 + \dots \beta_n \underline{b}_n .$$

represented in std basis $\in R^n$

$$= \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ b_1 & b_2 & \cdots & b_n \\ \downarrow & \downarrow & & \downarrow \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}$$

$$[x]_{std} = B \; [x]_B$$

d) The coordinate of x with respect to the B-basis are the weight $\alpha_1, \alpha_2 \dots \alpha_n$

$$[x]_B = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} \in R^n$$

and $[x]_{std}$ in std coordinate means wrt std basis

$$\text{std basis} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & & 0 \\ 0 & 0 & 1 & & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & & 1 \end{bmatrix} \Rightarrow \text{each col } e_i = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in R^n$$

$e_1 \quad e_2 \quad e_n$

i^{th} entry.

$$\underset{\in V}{x} \rightarrow \underset{\in R^n}{[x]_{\text{std basis}}}$$

Typically we do not indicate std-basis $[\;]_{\text{std basis}}$ just

In most cases, we simply write $[x]_{std}$ as x when it is clear from context.

e) $\therefore \; [x]_{std} = B[x]_B = C[x]_C$

B basis. C basis

Note B, C are basis of R^n

$\Rightarrow$ there will be square invertible matrix $(n \times n)$ since col of B are independent and span R^n.

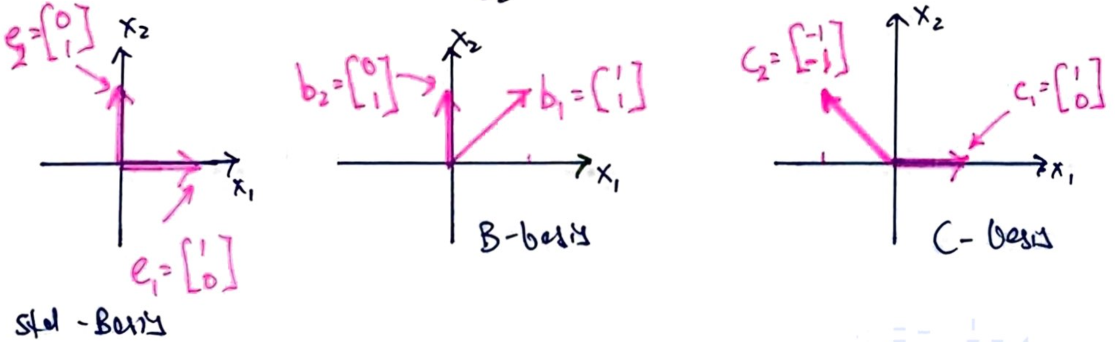
A) Interpreting x, $[x]_B$, $[x]_C$: example 1

From B basis $\rightarrow$ std basis $\rightarrow$ C basis.

Q: Given $\begin{bmatrix} 2 \\ 3 \end{bmatrix}_{\substack{\text{std} \\ \text{basis}}}$, find $[x]_B$, $[x]_C$

given $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $C = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

b_1 b_2 c_1 c_2

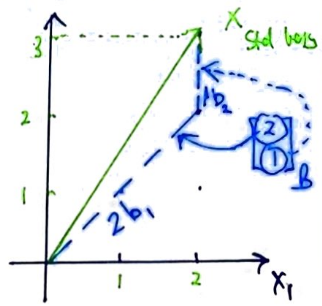


1a) To find $[x]_B$ given $[x]_{std}$ and B, note

$$x = [x]_{std} = B[x]_B$$

$$\therefore [x]_B = B^{-1}[x]_{std}$$

$$\underbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}_B}_{x \text{ in } B \text{ basis}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}}_{B^{-1}} \underbrace{\begin{bmatrix} 2 \\ 3 \end{bmatrix}_{std}}_{x \text{ std basis}}$$



Interpretation

i) $[x]_B = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}_B \Rightarrow x = \begin{bmatrix} \uparrow & \uparrow \\ b_1 & b_2 \\ \downarrow & \downarrow \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}_B$

$= \beta_1 b_1 + \beta_2 b_2$

ii) Graphically, we can see

$x = 2b_1 + 1b_2$ = travel along $\begin{bmatrix} 1 \\ 1 \end{bmatrix} = b_1$ by 2 units

and along $\begin{bmatrix} 0 \\ 1 \end{bmatrix} = b_2$ by 1 unit.

to reach the same pt x via B-basis

1b) Find $[x]_C$ given $x = \begin{bmatrix} 2 \\ 3 \end{bmatrix}_{\text{std basis}}$, $C = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

$$\therefore [x]_{std} = C[x]_C$$

$$\therefore [x]_C = C^{-1}[x]_{std}.$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_C = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_C = \begin{bmatrix} 5 \\ 3 \end{bmatrix}_C$$

Sanity check : $x = 5\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}_{std}$

x_1 in C-basis, c_1, x_2 in C-basis, c_2.

A) Interpreting x, $[x]_B$, $[x]_C$: summary & from $[x]_B$ to $[x]_C$.

Summary:

$$[x]_{std} = B[x]_B$$
$$[x]_B = B^{-1}[x]_{std}.$$
$$[x]_{std} = C[x]_C$$
$$[x]_C = C^{-1}[x]_{std}.$$

$$\therefore B[x]_B = [x]_{std} = C[x]_C$$

$$\Rightarrow [x]_B = B^{-1}C[x]_C$$
$$\Rightarrow [x]_C = C^{-1}B[x]_B . \blacksquare$$

changing to std basis

changing std to C Basis.

Note: Pre-multiply with basis B with $[x]_B$

a) $\Rightarrow$ changing to $[x]_{std}$.

b) Pre-multiply with B^{-1} with $[x]_{std}$

$\Rightarrow$ changing to $[x]_B$.

Ex 2: $[x]_C$ from $[x]_B = \begin{bmatrix}2\\1\end{bmatrix}_B$ given $B = \begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}$, $C = \begin{bmatrix}1 & -1\\0 & 1\end{bmatrix}$

$$\therefore [x]_C = C^{-1}B\begin{bmatrix}2\\1\end{bmatrix}_B$$
$$= \begin{bmatrix}1 & -1\\0 & 1\end{bmatrix}^{-1}\begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}\begin{bmatrix}2\\1\end{bmatrix}$$
$$= \begin{bmatrix}5\\3\end{bmatrix}_C \blacksquare$$

// same results as pg 3

Lay's notation.

let $\underset{C \leftarrow B}{P} = C^{-1}B = C^{-1}[b_1 \; b_2]$ — eq^n 1

$\therefore [x]_C = C^{-1}B[x]_B$

$= P[x]_B$.

in Lay's book.

$\underset{C \leftarrow B}{P} = [(b_1)_C \; (b_2)_C]$ $\longrightarrow$ eq^n 2

comparing col of eq^n 1 to col of eq^n 2.

$\therefore (b_1)_C = C^{-1} b_1$ // i.e col b_1 of B basis changed to C basis.

$= \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 2 \\ 1 \end{bmatrix}_C$

$(b_2)_C = C^{-1} b_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}_C$

2D

$\therefore \underset{C \leftarrow B}{P} = C^{-1}B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ ☒

Sanity check.

$[x]_C = \underset{C \leftarrow B}{P} [x]_B = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$ ☒

Note: Lay's notation $(b_1)_C$

= take col 1 from Basis B and convert to C basis by pre-multiply with C^{-1}. ☒

3) $T(x)$ linear transformation $R^n \to R^n$
represented by Ax.

and $A = PDP^{-1}$

where P = col^n contains a basis.

D = diagonal. matrix

$\therefore \; AP = PD \Rightarrow$ diagonalisation of A

P = col^n = eigenvectors

D = diagonal elements = eigen values.

then $y = Ax$

$$[y]_{std} = A[x]_{std}$$

$$= (PD\underbrace{P^{-1})[x]_{std}}_{\text{changing to } P\text{-basis.}}$$

$$= P\underbrace{D[x]_P}_{\text{transformation using } D \text{ in } P\text{-basis.}}$$

$$= \underbrace{P[y]_P}_{\text{converting to std basis.}}$$

$$= [y]_{std}.$$

$$\begin{tikzcd}
x \arrow[rr, bend left, "T"] \arrow[d] & & y \arrow[d] \\
{[x]_{std}} \arrow[rr, "A"] \arrow[d, "P^{-1}"'] & & {[y]_{std}} = A[x]_{std} = P[y]_P = PDP^{-1}[x]_{std}. \arrow[d, shift right, "P^{-1}"'] \\
P^{-1}[x]_{std} = [x]_P \arrow[rr, "D"] & & {[y]_P} = D[x]_P = DP^{-1}[x]_{std}. \arrow[u, shift right, "P"']
\end{tikzcd}$$

$\therefore$ A is T in std basis = std matrix of T.

D is T in P basis.

// see Lay 5e section 1.9 (pg 71) Matrix of linear transformation. (Theorem 10).

$T: R^n \to R^m$ be linear transformation.

$T(x) = Ax$; $A \in R^{m \times n}$, $x \in R^n$

where $A = [T(e_1), T(e_2), \ldots, T(e_n)]$

where e_j is col^n j of identity matrix.

Example : $A = \begin{bmatrix} 2 & 0 \\ 2.5 & -0.5 \end{bmatrix}$.

can be diagonalize to

$P = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & -\frac{1}{2} \end{bmatrix}$

eigen vector   eigenvalue

a) Write the close form sol^n of $x(n)$, given $x(n) = A\,x(n-1)$; $n \geq 1$

$x(0) = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

$$\therefore \quad x(1) = A\,x(0)$$
$$x(2) = A\,x(1) = AA\,x(0) = A^2 x(0)$$
$$\vdots$$
$$x(n) = A^n x(0).$$
$$= (PDP^{-1})^n\, x(0).$$

// since A can be diagonalize to P, D

means $AP = PD$

$\therefore A = PDP^{-1}$.

3B

$$\therefore \quad x(n) = \underbrace{(PDP^{-1})(PDP^{-1})\ldots}_{n \text{ times}}\, x(0)$$

$$= P \underbrace{\underbrace{D^n \underbrace{P^{-1}\, x(0)}_{\text{convert to P basis}}}_{\text{transformation in P basis}}}_{\text{convert to std basis}}$$

$$\therefore \quad x(n) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2^n & 0 \\ 0 & -\frac{1}{2}^n \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 4 \end{bmatrix}}_{[x(0)]_P = \begin{bmatrix} 1 \\ 3 \end{bmatrix}_P}$$

$$= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2^n & 0 \\ 0 & (-\frac{1}{2})^n \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \left(2^n + 3(-\tfrac{1}{2})^n\right).$$

$$x(n) = 2^n \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 3(-\tfrac{1}{2})^n \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad ※$$

ex:

$$\therefore \quad x(10) = 2^{10} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 3(-\tfrac{1}{2})^{10} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad ※$$

$R^2 \to R^3$: Lay 5e, pg 291

4A

$$[T(x)]_C = M[x]_B.$$

$$M = \Big[[T(b_1)]_C \;\; [T(b_2)]_C \cdots [T(b_n)]_C \Big]$$

matrix for T relative to B and C.

interpreting col of M.

$T(b_1) = Ab_1$

$[T(b_1)]_C = C^{-1}[\;]_{\text{std basis}}$ = converting to C-basis.

$$\underbrace{C^{-1}\underbrace{AB}_{T \text{ acting on } B \text{ basis}}}_{\text{converting to } C\text{-basis.}}$$

$$\begin{array}{ccc}
x & \xrightarrow{T} & y = T(x) \\
[x]_{std} & \xrightarrow{A} & [y]_{std} = A[x]_{std}. \\
B^{-1} \downarrow \;\; \uparrow B & & C \uparrow \;\; \downarrow C^{-1} \\
[x]_B & \xrightarrow{M} & [y]_C = C^{-1}[y]_{std}
\end{array}$$

Graphically.

$$[y]_{std} = A[x]_{std}$$

$$[y]_C = M[x]_B.$$

c basis to std

$$[y]_{std} = C[y]_C = CM[x]_B.$$

B basis to std.

$$[y]_{std} = \underbrace{CMB^{-1}}_{=A}[x]_{std}.$$

$$\therefore A = CMB^{-1}$$

$$\Rightarrow C^{-1}AB = M.$$

converting to C-basis

A acting on col of B

Transformation from $R^2 \to R^3$ / Lay 5e pg 291

4B

$y = Ax$; $x \in R^2$, $y \in R^3$

$A \in 3\times 2$ matrix.

Given B matrix representing a basis $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

C matrix representing a basis for R^3 $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1/2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

find $[y]_c = M[x]_B$; M represent A from B basis to C basis.

given $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$, $[x]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

Ans:

$\therefore [x]_{std} = B[x]_B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

$\therefore [y]_{std} = A[x]_{std} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix}_{std}$.

$\therefore [y]_{std} = A[x]_{std}$.

$= A \underbrace{B[x]_B}$. from $[x]_B \to [x]_{std}$

convert to C basis from std basis.

$[y]_c = C^{-1}[y]_{std} = \underbrace{C^{-1}AB}_{M}[x]_B$

$= \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1/2 & 1 \\ 0 & 0 & 1 \end{bmatrix}^{-1}\begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix}_B$

C^{-1} A B $[X]_B$

$[y]_c = \begin{bmatrix} 3 \\ -6 \\ 5 \end{bmatrix}$ $\underset{A:C\leftarrow B}{M} = \begin{bmatrix} 0 & 1 \\ 0 & -2 \\ 1 & 1 \end{bmatrix}$ #

$[y]_{std} = C[y]_c = AB[x]_B$

$= \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix}_{std}$.

Maths/LA/Tut8 EigenValue/Vectors

23 Oct 2021

Chng Eng Siong

Tutorial 8 Help links

Youtube link: playlist (Tut 8)

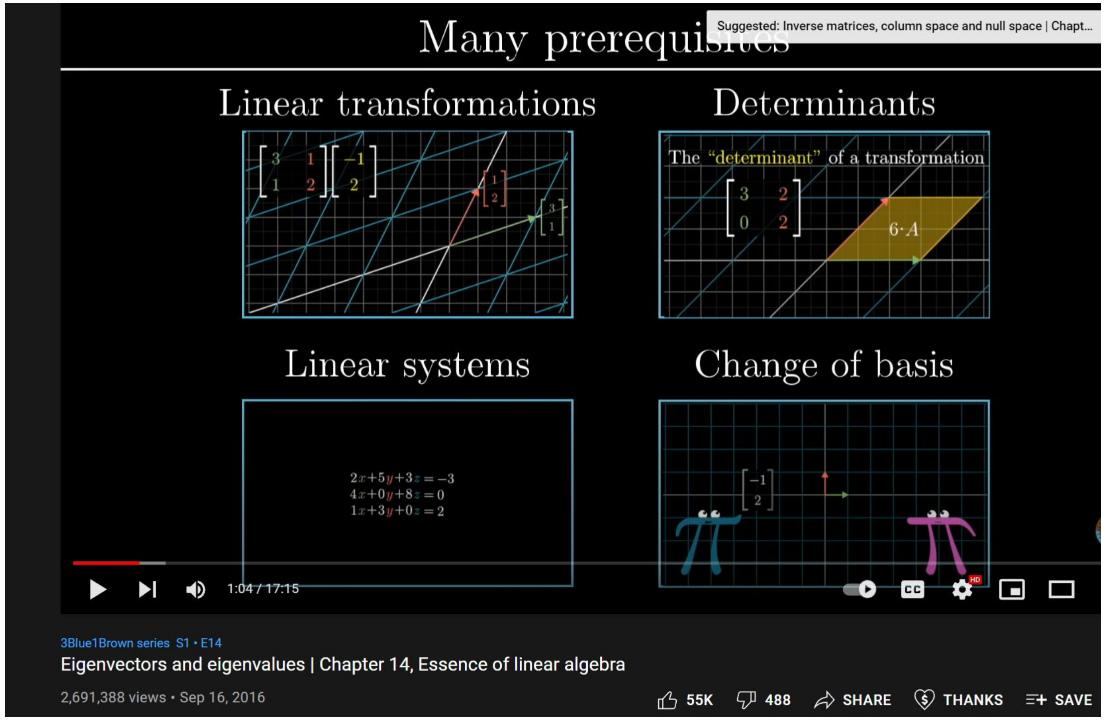
https://www.youtube.com/playlist?list=PLki3aFwg-9ewVB6R252cfd2QpJWLmtuGA

Solutions:

PDF:
https://www.dropbox.com/s/k1odexxe2jhs15r/Tut8revA_withSol.docx?dl=0

Insight to EigenValue/Vectors

- 3Blue1Bro Ch14
- https://www.youtube.com/watch?v=PFDu9oVAE-g



Q1) Intro to EigenValue/Vectors and examples how to calculate them

1) Prof Dave Explans (overview and 2x2 example)

https://www.youtube.com/watch?v=TQvxWaQnrqI

2) PatrickJMT (2x2 example)

https://www.youtube.com/watch?v=IdsV0RaC9jM

3) Adam Panagos: “Eigenvalue and Eigen Vector computation” 3x3 example:
https://www.youtube.com/watch?v=cHOsd2PhkqE&feature=youtu.be

https://www.adampanagos.org/courses/ala/eigenvaluesandeigenvectors

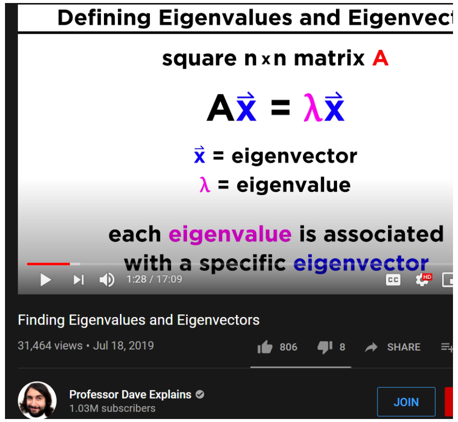
4) Scot Annin: EigenValue/Vector Intro

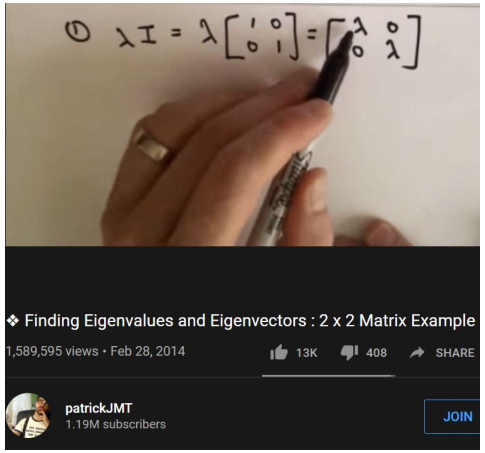
Part1: https://www.youtube.com/watch?v=Z3ULegjcArA

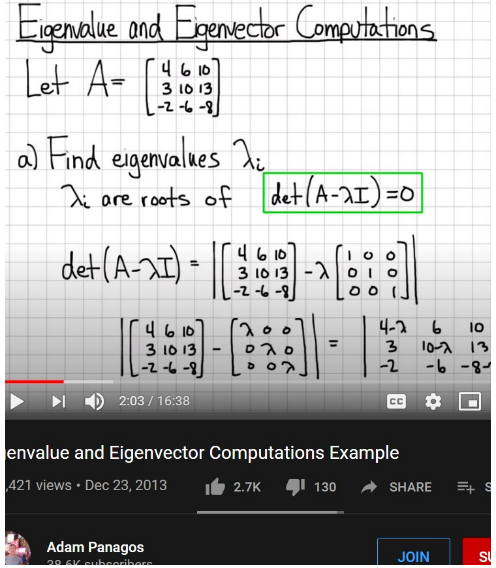
Part2: https://www.youtube.com/watch?v=o4nH1JgyfII

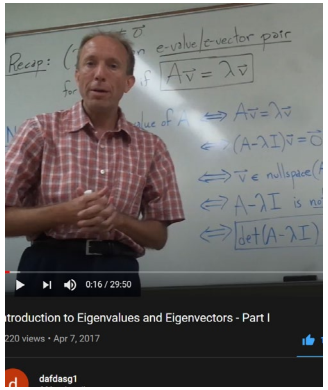
5) MIT Strang: Lecture 21 (18.06, Spring 2005)

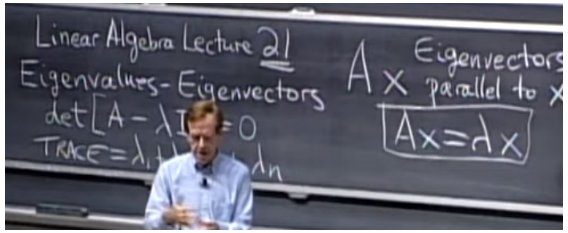
https://www.youtube.com/watch?v=cdZnhQjJu4I











Linear Algebra — Part 6: eigenvalues and eigenvectors

Sho Nakagome Following
Oct 14, 2018 · 5 min read

Q1) Characteristic Eqn/Polynomial

Characteristic Equation

The characteristic equation is the equation which is solved to find a matrix's eigenvalues, also called the characteristic polynomial. For a general $k \times k$ matrix $\mathbf{A}$, the characteristic equation in variable λ is defined by

$$\det(\mathbf{A} - \lambda\,\mathbf{I}) = 0,$$

Ref:

1)https://mathworld.wolfram.com/CharacteristicEquation.html

2)https://textbooks.math.gatech.edu/ila/characteristic-polynomial.html

Definition. Let A be an $n \times n$ matrix. The ***characteristic polynomial*** of A is the function $f(\lambda)$ given by

$$f(\lambda) = \det(A - \lambda I_n).$$

Find the characteristic polynomial of the matrix

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}.$$

Solution

We have

$$\begin{aligned} f(\lambda) = \det(A - \lambda I_2) &= \det\left(\begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right) \\ &= \det\begin{pmatrix} 5-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} \\ &= (5-\lambda)(1-\lambda) - 2\cdot 2 = \lambda^2 - 6\lambda + 1. \end{aligned}$$

The roots for $\lambda^2 - 6\lambda + 1$, are the eigen values of A.

Solution

In the above example we computed the characteristic polynomial of A to be $f(\lambda) = \lambda^2 - 6\lambda + 1$. We can solve the equation $\lambda^2 - 6\lambda + 1 = 0$ using the quadratic formula:

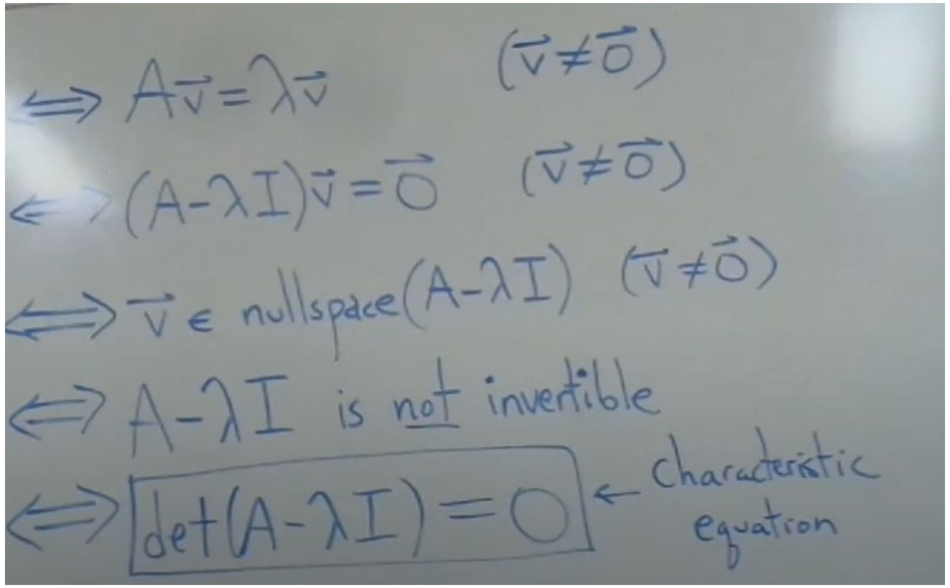
$$\lambda = \frac{6 \pm \sqrt{36-4}}{2} = 3 \pm 2\sqrt{2}.$$

Therefore, the eigenvalues are $3 + 2\sqrt{2}$ and $3 - 2\sqrt{2}$.

Q1) Why determinant $(A - \lambda I) = 0$? To find eigenvalues?

Scott Annin's board: last eqn showing det(A-λI) $= 0$

Part1: https://www.youtube.com/watch?v=Z3ULegjcArA



Ref:
https://math.stackexchange.com/questions/2619022/why-can-the-determinant-be-assumed-to-be-0

For a square matrix like $M = (A - \lambda I)$, the equation $Mx = 0$ will have a non-zero solution x if and only if M doesn't have an inverse, which is true if and only if the determinant of M is 0.

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answered Jan 24 '18 at 12:27
Ben Grossmann
172k 10 118 236

As for *why* you are interested in the values of λ that make the determinant equal to 0, remember that

$$rank(A - \lambda I) = n \iff det(A - \lambda I) \neq 0$$

So, if $det(A - \lambda I) \neq 0$, you will find that the *only* solution to $(A - \lambda I)x = 0$ is $x = 0$ (due to the fact that the rank of the matrix is full, hence the kernel only contains the 0 vector). This means that the *only* x such that $Ax = \lambda x$ is $x = 0$, which means that x is *not* an eigenvector.

So the only way to have eigenvectors is to have the determinant of $A - \lambda I$ be equal to zero, so that's why to find eigenvalues you look for the values of λ that make $det(A - \lambda I) = 0$

share cite edit follow

edited Jan 24 '18 at 14:41

answered Jan 24 '18 at 12:32
Ant
19.9k 3 33 91

Diagonalization

Ref:

1)Ref: http://www2.math.uconn.edu/~troby/math2210f16/LT/sec5_3.pdf

$$
\begin{aligned}
AP &= PD \\
APP^{-1} &= PDP^{-1} \\
A &= PDP^{-1}
\end{aligned}
$$

Where P = eigenvector,
D = diagonal matrix of eigen values

5.3 Diagonalization

The goal here is to develop a useful factorization $A = PDP^{-1}$, when A is $n \times n$. We can use this to compute A^k quickly for large k.

The matrix D is a *diagonal* matrix (i.e. entries off the main diagonal are all zeros).

D^k is trivial to compute as the following example illustrates.

EXAMPLE: Let $D = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}$. Compute D^2 and D^3. In general, what is D^k, where k is a positive integer?

Solution:

$$D^2 = \begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} & 0 \\ 0 & \end{bmatrix}$$

$$D^3 = D^2 D = \begin{bmatrix} 5^2 & 0 \\ 0 & 4^2 \end{bmatrix}\begin{bmatrix} 5 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} & 0 \\ 0 & \end{bmatrix}$$

and in general,

$$D^k = \begin{bmatrix} 5^k & 0 \\ 0 & 4^k \end{bmatrix}$$

THEOREM 5 The Diagonalization Theorem

An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.

In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A. In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P.

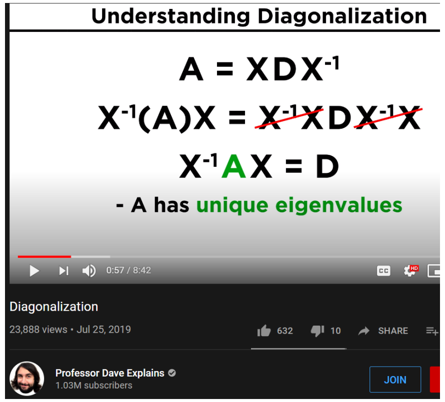
Q7) Diagonalization and Powers of A

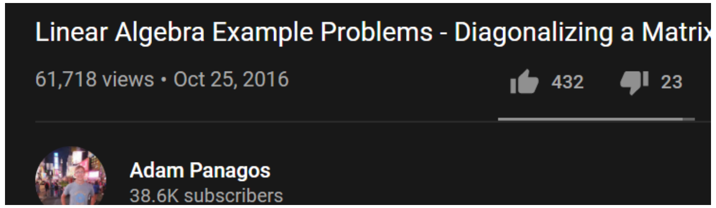
1) Prof Dave Expains: “Diagonalization”
https://www.youtube.com/watch?v=WTLl03D4TNA

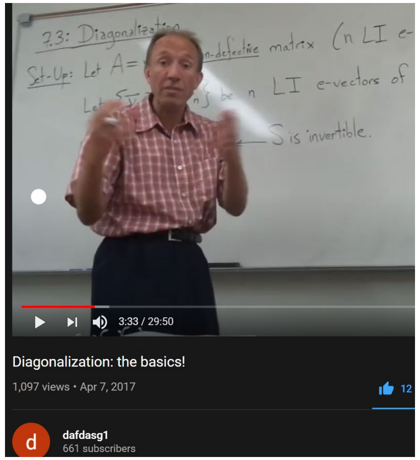
2) Adams Panagos:
https://www.youtube.com/watch?v=zEoHJfiQvt8

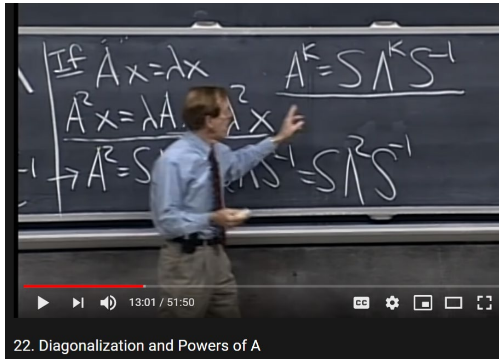
3) Prof Scott Annin, Cal State U Fullerton:
Dafdasg1 video: Diagonalization:
https://www.youtube.com/watch?v=6FknM_bPhUk

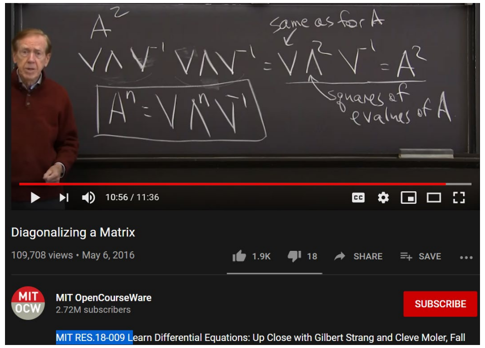
4) Strang:
   A) MIT Strang L22 (Spring 2005, 18.06):
   Diagonalization and Powers of A
   https://www.youtube.com/watch?v=13r9QY6cmjc
   B) MIT 2015 “Diagonalizing a Matrix”,
   https://www.youtube.com/watch?v=U8R54zOTVLw











Q7) Lay's example, 5th edition, pg 280
Application of diagonalization and power of A

Application to Dynamical Systems

Eigenvalues and eigenvectors hold the key to the discrete evolution of a dynamical system, as mentioned in the chapter introduction.

EXAMPLE 5 Let $A = \begin{bmatrix} .95 & .03 \\ .05 & .97 \end{bmatrix}$. Analyze the long-term behavior of the dynamical system defined by $\mathbf{x}_{k+1} = A\mathbf{x}_k$ ($k = 0, 1, 2, \ldots$), with $\mathbf{x}_0 = \begin{bmatrix} .6 \\ .4 \end{bmatrix}$.

SOLUTION The first step is to find the eigenvalues of A and a basis for each eigenspace. The characteristic equation for A is

$$0 = \det \begin{bmatrix} .95 - \lambda & .03 \\ .05 & .97 - \lambda \end{bmatrix} = (.95 - \lambda)(.97 - \lambda) - (.03)(.05)$$
$$= \lambda^2 - 1.92\lambda + .92$$

By the quadratic formula

$$\lambda = \frac{1.92 \pm \sqrt{(1.92)^2 - 4(.92)}}{2} = \frac{1.92 \pm \sqrt{.0064}}{2}$$
$$= \frac{1.92 \pm .08}{2} = 1 \quad \text{or} \quad .92$$

It is readily checked that eigenvectors corresponding to $\lambda = 1$ and $\lambda = .92$ are multiples of

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 5 \end{bmatrix} \quad \text{and} \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

respectively.

The next step is to write the given $\mathbf{x}_0$ in terms of $\mathbf{v}_1$ and $\mathbf{v}_2$. This can be done because $\{\mathbf{v}_1, \mathbf{v}_2\}$ is obviously a basis for $\mathbb{R}^2$. (Why?) So there exist weights c_1 and c_2 such that

$$\mathbf{x}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = [\,\mathbf{v}_1 \quad \mathbf{v}_2\,]\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \tag{3}$$

In fact,

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = [\,\mathbf{v}_1 \quad \mathbf{v}_2\,]^{-1}\mathbf{x}_0 = \begin{bmatrix} 3 & 1 \\ 5 & -1 \end{bmatrix}^{-1} \begin{bmatrix} .60 \\ .40 \end{bmatrix}$$
$$= \frac{1}{-8}\begin{bmatrix} -1 & -1 \\ -5 & 3 \end{bmatrix}\begin{bmatrix} .60 \\ .40 \end{bmatrix} = \begin{bmatrix} .125 \\ .225 \end{bmatrix} \tag{4}$$

Because $\mathbf{v}_1$ and $\mathbf{v}_2$ in (3) are eigenvectors of A, with $A\mathbf{v}_1 = \mathbf{v}_1$ and $A\mathbf{v}_2 = .92\mathbf{v}_2$, we easily compute each $\mathbf{x}_k$:

$$\mathbf{x}_1 = A\mathbf{x}_0 = c_1A\mathbf{v}_1 + c_2A\mathbf{v}_2 \qquad \text{Using linearity of } \mathbf{x} \mapsto A\mathbf{x}$$
$$= c_1\mathbf{v}_1 + c_2(.92)\mathbf{v}_2 \qquad \mathbf{v}_1 \text{ and } \mathbf{v}_2 \text{ are eigenvectors.}$$
$$\mathbf{x}_2 = A\mathbf{x}_1 = c_1A\mathbf{v}_1 + c_2(.92)A\mathbf{v}_2$$
$$= c_1\mathbf{v}_1 + c_2(.92)^2\mathbf{v}_2$$

and so on. In general,

$$\mathbf{x}_k = c_1\mathbf{v}_1 + c_2(.92)^k\mathbf{v}_2 \quad (k = 0, 1, 2, \ldots)$$

Using c_1 and c_2 from (4),

$$\mathbf{x}_k = .125\begin{bmatrix} 3 \\ 5 \end{bmatrix} + .225(.92)^k\begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (k = 0, 1, 2, \ldots) \tag{5}$$

This explicit formula for $\mathbf{x}_k$ gives the solution of the difference equation $\mathbf{x}_{k+1} = A\mathbf{x}_k$. As $k \to \infty$, $(.92)^k$ tends to zero and $\mathbf{x}_k$ tends to $\begin{bmatrix} .375 \\ .625 \end{bmatrix} = .125\mathbf{v}_1$. ■

Q8 (21a) You can generate your own $A = PDP^{-1}$

You can generate your own A, and use [P,D] = eig(A) (Matlab) to check.
As long as your P is invertible and your D is diagonal.

E.g,

```
P =

    -1     1     1
    -1    -1     1
    -1     0    -1

>> D

D =

   -3.0000         0         0
         0    2.0000         0
         0         0    1.0000

>> A=P*D*inv(P)

A =

    0.5000   -1.5000   -2.0000
   -1.5000    0.5000   -2.0000
   -1.0000   -1.0000   -1.0000
```

```
>> A*P

ans =

    3.0000    2.0000    1.0000
    3.0000   -2.0000    1.0000
    3.0000         0   -1.0000
```

```
>> [P1,D1]=eig(A)

P1 =

    0.5774    0.7071   -0.5774
    0.5774   -0.7071   -0.5774
    0.5774   -0.0000    0.5774


D1 =

   -3.0000         0         0
         0    2.0000         0
         0         0    1.0000
```

Note, Matlab produced normalized columns of P, and the sorting of D and P may be arbitrary (not sorted according to eigenvalues)

Additionally, the eigenvector can be –ve direction!
Eig P(:,1) is –P1(:,1)
And P(:,3) is – P1(:,3)

Q8 (21c) When is a matrix diagonalizable

When there is a basis of eigenvectors, we can diagonalize the matrix. Where there is not, we can't. We can come close, but that's another very complicated story. So the good case is when the geometric multiplicity of each eigenvalue equals its algebraic multiplicity because then there are just enough eigenvectors to make a basis.

Definition: the *algebraic multiplicity* of an eigenvalue e is the power to which $(\lambda - e)$ divides the characteristic polynomial.

Definition: the *geometric multiplicity* of an eigenvalue is the number of linearly independent eigenvectors associated with it. That is, it is the dimension of the nullspace of $A - eI$.

In the example above, 1 has algebraic multiplicity two and geometric multiplicity 1. It is always the case that the algebraic multiplicity is at least as large as the geometric:

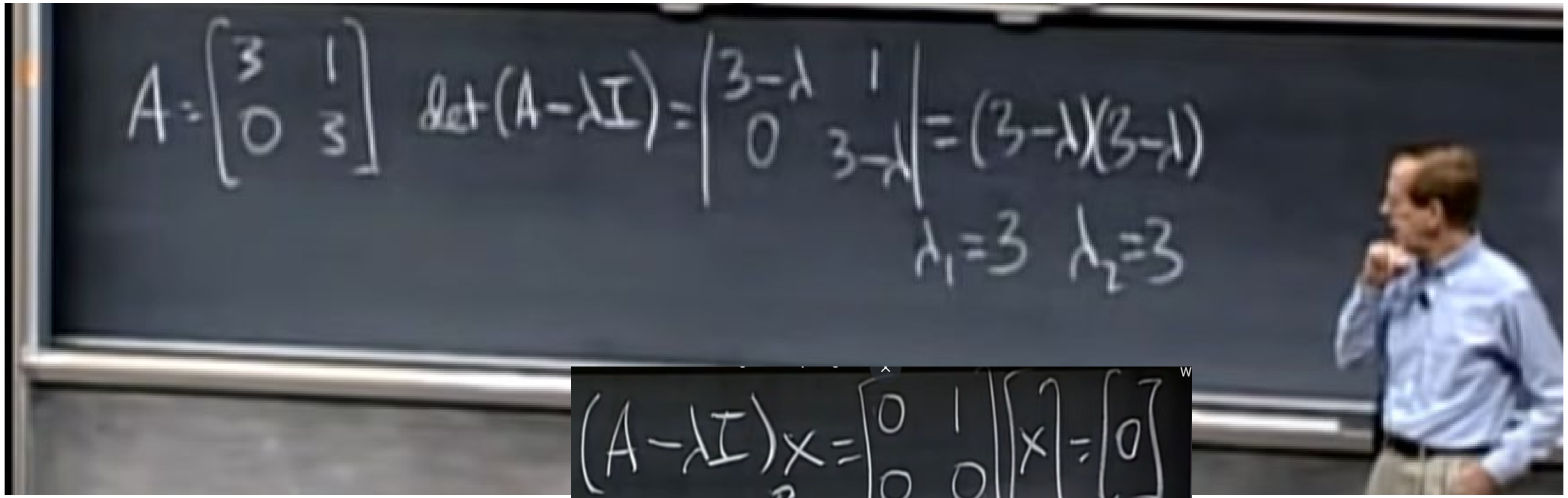
Theorem: if e is an eigenvalue of A then its algebraic multiplicity is at least as large as its geometric multiplicity.

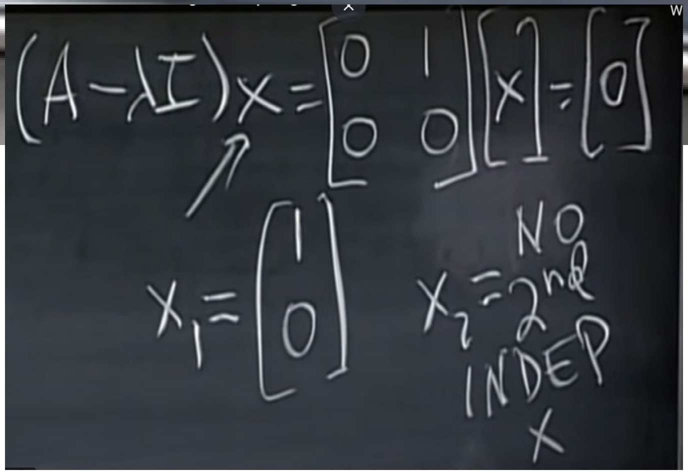
- https://staff.imsa.edu/~fogel/LinAlg/PDF/44%20Multiplicity%20of%20Eigenvalues.pdf

Q8 Degenerate Matrix – Strang
Not enough eigenvectors

- http://www.teachingtree.co/watch/degenerate-matrix-eigenvectors
- Strang: Lect **Degenerate Matrix Eigenvectors**
- Lec 21 | MIT 18.06 Linear Algebra, Spring 2005

Note, Matlab produced the same eigenvector! P(:,1) vs P(:,2)
P(:,2) = -ve P(:,1) which is dependent! Hence useless eigenvector






```
>> A = [3 1; 0 3]

A =

     3     1
     0     3

>> [P,D] = eig(A)

P =

    1.0000   -1.0000
         0    0.0000


D =

     3     0
     0     3
```

Q9,10,11) Eigenvectors and Linear Transformation

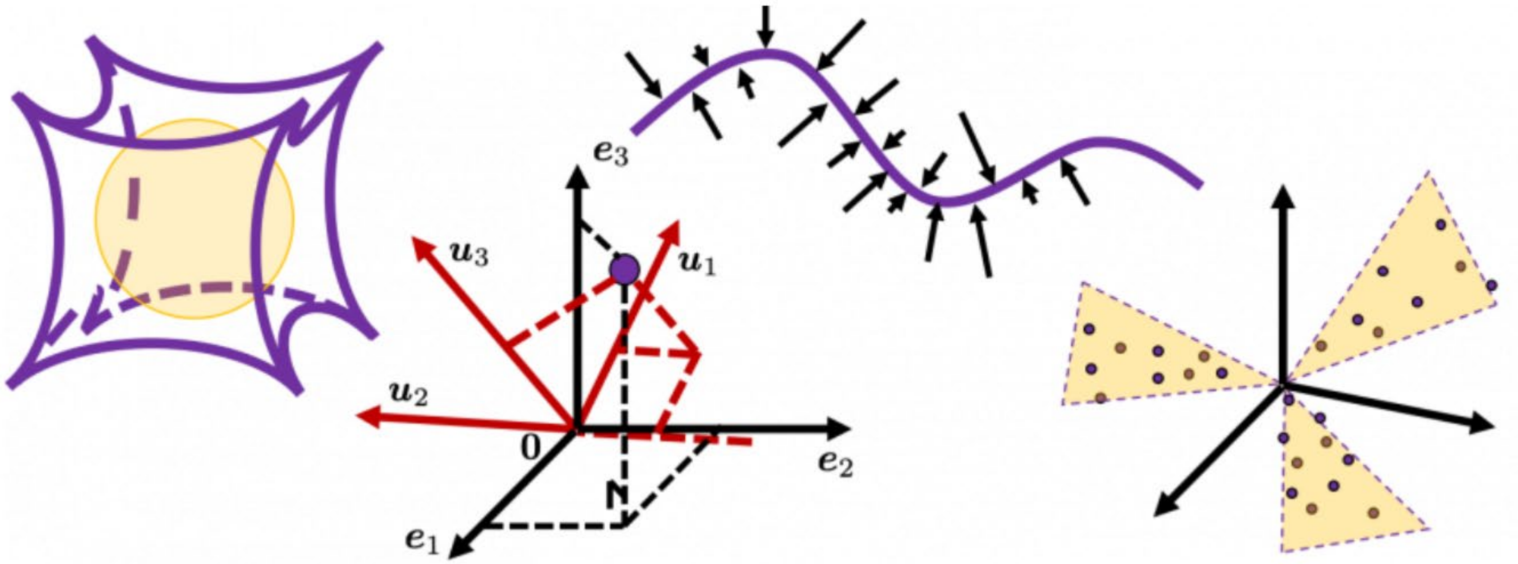
1) Paul Cartie: https://www.youtube.com/watch?v=DI6BU1sMaLs
2) Purdue Math:https://www.math.purdue.edu/~bkrummel/ma265_lecture5_4.pdf
3) U.of Michigan: http://www.math.lsa.umich.edu/~kesmith/CoordinateChange.pdf

4) Worked examples:
   a) https://math.stackexchange.com/questions/16386/eigenvalues-and-eigenvectors-of-linear-transformations
   b) https://math.stackexchange.com/questions/2560162/change-of-basis-for-linear-transformation-linear-algebra

   b) https://yutsumura.com/linear-algebra/eigenvalues-and-eigenvectors-of-linear-transformations/

5) CS Blog in DataScience by Yasuto Tamura:
https://data-science-blog.com/blog/2020/10/27/10360/

Q9,10,11) Eigenvectors and Linear Transformation



Rethinking linear algebra: visualizing linear transformations and eigenvectors

October 27, 2020 / in Data Mining, Data Science, Machine Learning, Main Category, Mathematics / by Yasuto Tamura

4) CS Blog in DataScience by Yasuto Tamura:
https://data-science-blog.com/blog/2020/10/27/10360/

Q9,10,11) Matrix of Linear Transformation with respect to a change of basis

Ref

1) Arnold Yim: Basis, coordinate system, and change of basis
   https://www.youtube.com/watch?v=IpKUPhNHFQA&list=PLUhmPGIKgSbIlTyKCvvZ6BLOcIgyHMLTr

2) Adams Panagos: https://www.youtube.com/watch?v=VG4-8yW3Ce8